FUNCTIONAL SPECIFICATION - TIRE CASING

DATE	***VER***	***REVISION***	
22 Apr 1998	*IR*	*RELEASED AC00-E-10682904-000*	***APPROVALS*:***
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10 Sep 1999	*IT*	*REVISED AND RETYPED: AC00-E-10842752-016*	*Maria Joandrea / Arne Heyden*
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5 Sep 2008	*1*	*REVISED, RE-TITLED TO FUNCTIONAL SPECIFICATION VERSION 1 (Details in Appendix A): PREPARED BY: D. KLEKAMP We will now store the Generic ES's which are now called Functional Specification (FS) in EDMS and include in SDS: TY-0117.*	
11 Jun 2012	*2*	*REVISED AND RETYPED (Details in Appendix A): PREPARED BY: D. KLEKAMP*	
1 Nov 2014	*3*	*REVISED, RETYPED, AND REFORMATTED (Details in Appendix A): Prepared by D. Klekamp, J. Vaillancourt & M. Knych*	
26 Aug 2019	*4*	*REVISED, RETYPED, AND REFORMATTED (Details in Appendix A): Prepared by Maria Joandrea and Mark Peschiutta*	
17 Nov 2021	*5*	*REVISED, RETYPED, AND REFORMATTED (Details in Appendix A): Prepared by Maria Joandrea and Mark Peschiutta*	
			REFERENCE TIRE ENGINEERING SPECS BEING REPLACED: *ES-F4DC-1508-AA* *ES-E8TA-1508-AA* *ES-E0TA-1508-BA* *S92VB-1508-AA* *S69AB-1508-AA* *ES-XU5A-1508-IZ* *Tire FS Version 1* *Tire FS Version 2* *Tire FS Version 3* *Tire FS Version 4*

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ABBREVIATIONS

Abbreviation	Word or Phrase	Abbreviation	Word or Phrase
AIAG	Automotive Industry Action Group	PPAP	Production Part Approval Process
AMS	Auto Motor and Sports (magazine)	PRAT	Plysteer Residual Aligning Torque
APQP	Advanced Product Quality Plan	PSW	Part Submission Warrant
BIS	Bureau of Indian Standards	PV	Production Validation
CAE	Computer-Aided Engineering	RHD	Right Hand Drive
CC	Critical Characteristic (PFMEA)	RMA	Rubber Manufacturers Association
CCC	China Compulsory Product Certification, 3C	RQT	Requirement
CDS	Construction Detail Sheet	RRC	Rolling Resistance Coefficient
CETP	Corporate Engineering Test Procedure	RWL	Raised White Letters
CUV	Crossover Utility Vehicle	SAE	Society of Automotive Engineers
D&R Engineer	Design & Release Engineer	SC	Significant Characteristic (PFMEA)
DFMEA	Design Failure Mode Effects Analysis	SDS	System Design Specification
DMA	Dynamic Mechanical Analysis	SREA	Supplier Request For Engineering Approval
DRW	Dual Rear Wheel	SRW	Single Rear Wheel
DV	Design Verification	STA	Supplier Technical Assistance
DVP&R	Design Verification Plan and Report	SUSE	Spare Unit Substitutive Equipment
ECE	Economic Commission of Europe	SUV	Sport Utility Vehicle
EPAS	Electric Power Assist Steering	T&RA	Tire and Rim Association
ETRTO	European Tyre and Rim Technical Organization	TIN	Tire Identification Number
F/CMVSS	Federal Motor Vehicle Safety Standards Canadian Motor Vehicle Safety Standards	Tire TA	Tire Target Agreement
FEA	Finite Element Analysis	TM	Test Method *[previously Corporate Engineering Test Procedure (CETP)]*
FNA	Ford North America	TSS	Tire Submission Summary
FoE	Ford of Europe	U.S. DOT	United States Department of Transportation
FS	Functional Specification	USTMA	United States Tire Manufacturers Association
Gage R&R	Gage Repeatability & Reproducibility	VER	Vehicle Evaluation Rating
GAT Team	Global Analytical Tire Team	WDK	Wirtschaftsverband Der Kautschukindustrie (Association of the Rubber Industry)
GAWR	Gross Axle Weight Rating	WLTP	Worldwide Harmonized Light Vehicle Test Procedures
GCC	Gulf Cooperation Council	WSW	White Sidewall
GVW	Gross Vehicle Weight	YC	Critical Characteristic (DFMEA)
GVWR	Gross Vehicle Weight Rating	YS	Significant Characteristic (DFMEA)
IRR	Initial Run at Rate		
ISO	International Organization for Standardization (*was International Standards Organization*)		
JATMA	Japan Automobile Tire Manufacturers' Association		
LHD	Left Hand Drive		
LSU	Low Speed Uniformity		
MAV	Multi Activity Vehicle		
NHTSA	National Highway Traffic Safety Administration		
NVH	Noise, Vibration, and Harshness		
OE	Original Equipment		
OWL	Outlined White Letters		
PD	Product Development		
PFMEA	Process Failure Mode Effects Analysis		

I. GENERAL

A. SCOPE:

1. This Functional Specification (FS) establishes the minimum engineering requirements for all tubeless tires used on passenger cars, light trucks, heavy trucks, vans, SUVs, MAVs, and CUV's. This includes the following tire types: P-Metric, Metric, T-Metric, LT-Metric, North American Truck/Bus, and European Commercial
    - Note – North American Truck/Bus and European Commercial refers to the standard used to classify the tire (e.g. T&RA, ETRTO, etc…). It is **NOT** meant to reference the region in which the tire is used.
2. The Tire FS applies to all tires in all markets for Ford Motor Company unless noted otherwise by Global Tire Manager
3. The Tire FS will be the document that a Tire Supplier will conform to in certifying a new tire design. All requirements in the Tire FS and the Tire DVP&R must be met unless otherwise directed by the Chassis Organization, with concurrence by Global Tire Manager. Minimum measures necessary for demonstrating compliance to these requirements are given in each section of this document.
4. The tests, sample sizes, and test frequencies contained within the Tire FS reflect the minimum requirements established to provide a consistent evaluation of conformance to design intent.
5. The Tire FS is intended to evaluate specific component characteristics as a supplement to normal Tire Supplier material inspections, dimensional checking, and in-process controls and should, in no way, adversely influence other inspections conducted by the Tire Supplier.

B. TIRE TARGET AGREEMENT (TIRE TA)

1. Specific targets and requirements for a new tire program, including the markets where the vehicle will be sold, will be defined by Ford Product Engineering via a program Tire TA. Where acceptance criteria are program specific, the specific tests and acceptance criteria beyond what is contained herein, will be defined in the TA as issued by Ford Purchasing.
2. Tire dimensions and marking requirements may be listed on the Tire TA.
3. All Tire TA's are to be signed by Ford's Tire Governance Board members. The signature can be electronic.
   - 3.1 Driving Dynamics Engineering Manager
   - 3.2 Vehicle Energy Management Manager
   - 3.3 Vehicle Engineering Manager
   - 3.4 Vehicle Engineering NVH Manager
   - 3.5 Global Tire Manager
   - 3.6 Global Tire Technical Specialist
   - 3.7 Tire D&R Supervisor
4. Tire Supplier should communicate any request for deviation from the Tire FS **before** sourcing.
5. Any change to program targets must be documented in the Tire TA. The new Tire TA must be approved and submitted to the Tire Supplier to determine if a re-quote is necessary, and a new timing assessment must be obtained.

C. TIRE DEVELOPMENT / PROCESS WORKFLOW

1. The document defined in Appendix D - Tire Submission Summary Sheet (TSS) will be the format and minimum information provided by a Tire Supplier for all tire submissions.
2. It is expected that one construction in the first submission will match what was provided in the bid feedback.
3. Tire Supplier should approach the design of each submission (including Submission 1) with the goal of providing at least one construction that meets all the TA targets (both lab and on-vehicle) and would be a candidate to select for production.
4. The following outlines the typical workflow for development, validation, and sign-off of a tire. These steps assume that the tire has already been formally sourced. This is intended to be a generic workflow, and some tires may not follow this process in its entirety.
    - Details of the process, as well as roles and responsibilities, can be found in the "Tire D&R Development Process & Checklist" file, which can be made available upon request.

    4.1 After a tire supplier has been formally sourced to a program, a Kick-Off meeting is held to review the overall project.

    4.2 When the supplier has prepared their proposal for the mold design, a meeting is held to review this proposal in detail.

    4.3 With an acceptable mold design, another meeting is held with the supplier to review their proposals for the submission tires. If the proposals are accepted, the submission tires can be built. If the proposals are not accepted, additional proposals and reviews will be necessary.

    4.4 Submission tires are supplied to Ford for lab testing and on-vehicle evaluation.

    4.5 In cases where a tire construction does not meet lab targets, the on-vehicle evaluation of that construction may not be completed. This will be at the discretion of the GAT Team (or equivalent) and Global Tire Manager

    4.6 The results of the lab testing and on-vehicle evaluation are consolidated onto the TSS. A review is held to discuss the results and plan for the next submission.

    4.7 Steps 2 thru 4 are repeated until a tire construction has been approved for production. At this point, the DVP&R process can begin.
    - The newest level **Generic Tire DVP&R** must always be used when preparing the sign-off documentation for a tire.
5. Tires which use a multi-compound strategy for the tread ("multi-compound tread") must be approved by the Global Tire Technical Specialist and Global Tire Manager. These tires must include a Tread Map Image with the TSS that objectively describes the location of each compound. A multi-compound tread is defined by the tread having more than one compound above the tread wear indicators.
6. Tire Supplier will provide a cut section of each construction submitted to Ford for testing. These sections will be cut and delivered according to the details contained in the "Shipping Details - Cut Sections" box found in the TSS Test Request tab.
    - If two constructions differ *only* by belt angle and / or tread compound, then only one cut section is required.
7. It is recommended that all tires used for DV testing are retained by the tire supplier until the DVP&R achieves Final sign-off, after which the test tires can be scrapped. Tire suppliers may scrap the test tires prior to Final sign-off with agreement from the Tire D&R and Global Tire Manager
8. Shipment of tires to Ford Building 4 for lab testing and vehicle evaluations:

    8.1 All development tires must be shipped on pallets.

    8.2 Tires must be shrink wrapped and no taller than 4.5 ft (1.37 meter)

    8.3 Loose tires will not be accepted

    8.4 Tire supplier is responsible for coordinating shipment of development tires to the Ford requested location, with no involvement from, or cost to Ford.

    8.5 Tire supplier is responsible for all shipping and logistics of development tires, including but not limited to, importation, clearing customs, use of customs brokers, etc…

    8.6 Failure to follow these rules may result in rejected shipments.
9. Reference documents used during the development process:

    9.1 Tire Functional Specification (Tire FS)

    9.2 Test Methods (TM's), previously Corporate Engineering Test Procedures (CETP's)

    9.3 Tire TSS

9.4 Requirements (RQT's)
9.5 Design Rules
9.6 Generic Tire DVP&R

D. DESIGN VALIDATION

1. It is the Tire Supplier's responsibility to prepare and submit an acceptable Design Failure Mode Effects Analysis (DFMEA). The Tire Supplier's DFMEA must identify Special / Critical Characteristics (SC / CC) as specified in Table 2 and must be reviewed and approved by Global Tire Manager.
2. The Tire DVP&R is the master sign-off document that validates all Tire FS and Tire SDS requirements have been met.
3. Each Tire Supplier, working with the Tire D&R Engineer, will develop a DVP&R "Plan" at the beginning of a program.
4. The Tire Supplier will present all final test data and results electronically to the responsible Tire D&R Engineer in the Generic DVP&R format.
5. Tire Suppliers will use the newest release of the Generic Tire DVP&R for all tire sign-off submissions; prior versions will not be accepted.
6. The following items will be stored electronically:
   6.1 A copy of the Tire TA
   6.2 The completed DVP&R file and its supporting evidence
7. The Tire DVP&R, which includes the signed Tire Construction Detail Sheet (CDS), will constitute the permanent sign-off record for the particular tire that is being approved. The DVP&R will be approved, and the Cover Sheet signed by:
   7.1 Tire D&R Engineer
   7.2 Tire Engineering Supervisor
   7.3 Global Tire Technical Specialist
   7.4 Tire Supplier
8. The format and data reporting elements of the Tire CDS are specified in Appendix B - Tire Construction Detail Sheet (CDS). The most recent Tire CDS revision level must be utilized for all tires; prior versions may not be submitted for sign-off.
9. One tire cut section obtained from the DV build must be reviewed by the Tire D&R Engineer to validate cut tire measurements on the Tire CDS.
10. Signatures required for the Tire CDS sign-off are:
    10.1 Tire D&R Engineer
    10.2 Tire Engineering Supervisor
    10.3 Global Tire Manager
    10.4 Tire Supplier
11. As part of the DVP&R the Tire Suppliers must supply:
    11.1 Written certification using the "Tire Supplier Regulatory Compliance Letter" (per the example included in the Generic DVP&R) that all tires conform to applicable regulatory requirements for markets indicated in the program market list. This letter should include applicable certification information for; US Federal Government Standards, Canadian Government Standards, European Government Standards, or any other government regulatory body standards (e.g. F/CMVSS 109, 119, 139, ECE, GCC, CCC, BIS, etc.) and any applicable state or country regulations [e.g. Gulf Coast Countries (GCC), ECE, INMETRO, etc...].
12. In addition, Tire Suppliers will comply with standards and recommendations of The European Tyre and Rim Technical Organization (ETRTO), AIAG B-11, The Tire & Rim Association (T&RA), The Japan Automobile Tire Manufacturers' Association, Inc. (JATMA), or other industry standards organizations as approved by Global Tire Manager. Tire Suppliers may also provide a letter, in lieu of test data, self-certifying to certain F/CMVSS standards to satisfy DVP&R requirements when approved by Global Tire Manager.
13. When applicable, the Tire Supplier must provide a certification letter of ISO RRC compliance, as per II.P.4 of this Tire FS, for all tires that are subject to Part 1037 in the Federal Register.
14. When applicable, the Tire Supplier shall provide a certification letter stating the provided measurements were done in accordance with ECE 117-02 and the rules of tire classification defined in the EG 1222/2009, required for vehicle WLTP homologation compliance, as per Section II.P.4.4 of this Tire FS, for all tires certified per European Government Standards.
15. The Tire Supplier must submit a letter for the DVP&R stating self-certification for Air Permeation.

16. Compliance to the Tire FS must be noted on the tire component drawing. In the event that an exception (deviation) to the Tire FS is approved by Global Tire Manager, the specific exception(s) must be noted on the component drawing. The component drawing for tires is defined by the Tire Drawing Template and is illustrated in Appendix C - Tire Drawing Template. Tire Suppliers are required to support requests from the Tire D&R Engineer for content to complete the drawing template when required per program milestones. Tire Suppliers will not submit alternative drawings to satisfy this request (e.g. 2-D drawings); only content consistent with the Tire Drawing Template is permitted.
17. All documents relating to the program (DVP&R and related evidence, DFMEA) must be kept by the Tire Supplier for a document retention period of Job Last + 5 years.
18. Tire Suppliers are to review and approve the latest released "Ford Tire Warranty Supplement" and add changes and/or sign-off when prompted by the Owner's Guide committee or Global Tire Manager via email notification. The tire warranty information provided must be reviewed and approved by Global Tire Manager.

E. TIRE IDENTIFICATION NUMBER CODE REQUIREMENTS

1. All tires must contain a molded-in Tire Identification Number (TIN) following the format of the National Highway Traffic Safety Administration (NHTSA) tire identification requirements. Tires marked with the letters **"DOT"** preceding the TIN would indicate compliance to U.S. Department of Transportation requirements.
   - 1.1 The requirement is defined in NHTSA 49 CFR Part 574 – Tire Identification and Recordkeeping
2. The tire type code will meet the following requirements:
   - 2.1 The combination of the TIN and construction number will be uniquely traceable to the released Ford Part number.
   - 2.2 The TIN and construction number will be documented on the Tire CDS. The Tire CDS will be provided to engineering as part of the DVP&R and be included as part of PSW documentation.
   - 2.3 A unique construction number is required for different sidewall treatments. For example, white sidewall, OWL and black sidewall versions of the same tire construction must have unique construction numbers.
   - 2.4 A unique combination of the TIN and construction number is mandatory for each tire end item supplied to Ford
   - 2.5 A new unique construction number may be required if a change in the tire company corporate specification (versus the local tire manufacturing plant specification) for the released tire occurs. The Tire Supplier will request a prior Ford Motor Engineering determination on the need for a construction number change in this circumstance.
3. Ford Motor Company mandates the TIN, including the date code, is present on the outboard sidewall when installed on a vehicle.
   Exception: For Dual Rear Wheel (DRW) vehicles, the TIN must face outboard when mounted on the vehicle for the front axle and inner rear DRW assemblies. The outboard rear DRW assemblies may have the TIN facing inboard when mounted on the vehicle.
   Exception: For directional tires that have the TIN, including the date code, on only one side of the tire, the date code may not be present on the outboard sidewall of all positions when installed on a vehicle. This exception must be reviewed and approved by the Global Tire Manager.
4. This applies to all Ford Motor Company tires worldwide. For markets which do not require DOT certification the "DOT" may be absent from in front of the TIN if approved by Global Tire Manager
5. Tires delivered to Ford Motor Company assembly plants or outside assemblers will have been manufactured within the prior ten (10) month time period. Tires used in the Ford assembly plant will have been manufactured within the prior twelve (12) month time period. No tires older than one (1) year will be used. The TIN date of the tire will constitute the date of manufacture for this requirement.
6. Submission Tires require a unique construction number, other than the TIN, permanently identified (molded /cured / branded) into the tire sidewall for identification purposes. The Submission Tire Construction Number requirement is optional for Production Tires, a TIN is always required.
7. For all tires globally, a maximum inflation pressure must be clearly indicated on at least one sidewall of the tire in both kPa and psi pressure units.

F. TIRE CHARACTERISTICS AND MARKINGS

1. Tread Depth
   1.1 Minimum tread depths are defined by many regulatory bodies and have a range of 1.6 - 4.0 mm [0.06 - 0.16 in]. It is the responsibility of the Tire Supplier to ensure that the minimum tread depth of the Tread Wear Indicators (TWI) satisfies all regulatory requirements for the markets defined in the Tire TA.
2. Tg – Glass Transition Temperature
   2.1 Tg is defined as the glass transition temperature point on the dynamic mechanical analysis (DMA) graph per TM-04.04-E-4335. The glass transition temperature point is defined as the x-axis point relating to the peak y-value of the DMA curve.
   2.2 Tires must meet any Tg requirement as defined on the Tire TA.
3. Snow/Winter Tires
   3.1 Definition: For the purposes of this Tire FS and all other Ford Motor Company documentation, the terms “snow tire” and “winter tire” are equivalent and define the same type of tire. The two terms can be used interchangeably but the default term used for definition purposes will be “winter tire”.
   3.2 Winter tires must be marketed by the tire supplier as such. Any exceptions for testing of winter tires noted through this Tire FS must be approved by Global Tire Manager.
   3.3 Specific conditions for usage are contained in;
      - TIRE: TIRE LOAD MARGIN [RQT-040400-017839]
      - TIRE: MINIMUM TIRE INFLATION & SPEED RATING/MARGIN [RQT-040400-017828]
      - TIRE: TIRE RELATED LABELS & PLACARD INFLATION PRESSURES [RQT-040400-017850]
4. All-Terrain (A/T) Tires
   4.1 Tires defined on the Tire TA as All-Terrain tires must be marked on the sidewall or marketed as All-Terrain tires by the tire supplier. This marketing can include, but is not limited to, print ads, websites, dealer information, television ads, etc…
   4.2 The tires must include nomenclature on the sidewall indicating that it is an All-Terrain tire (i.e. “All-Terrain”, “A/T”, etc…)
5. T-Type (mini-spare) Tires
   5.1 T-type (mini-spare) tires must have sidewalls marked with “Temporary Use Only”, as well as the maximum speed, per applicable regulatory requirements.
6. Tires should be able to mount using industry standard mounting equipment

G. PD PPAP REQUIREMENTS AND PRODUCTION VALIDATION

1. Tests described in this Tire FS are used to obtain an initial estimate of the process potential to produce parts that conform to engineering requirements, and to identify causal or predictive relationships between significant design and process characteristics (to be used for process control). These tests must be completed satisfactorily using initial parts from production tooling and processes before PSW approval and authorization of production parts can be issued. Table 3 summarizes the Tire FS requirements and tests and the acceptance parameters for each. They form the basis on which to develop a complete Control Plan for these and their related significant process characteristics.
2. Phased PPAP will apply for PPAP requirements. Global Phased PPAP Requirements Handbook Edition 4.0 is the current standard. Future PPAP Handbook editions will supersede the Edition 4.0 when issued.
3. Phased PPAP begins after the tire is signed off (DVP&R and Tire CDS sign-off complete)
4. One tire cut section obtained from the PPAP build must be reviewed by the Tire D&R Engineer to validate cut tire measurements are within supplier production tolerances.
5. Conditional DVP&R sign-off can occur per Global Tire Manager approval. This allows Phase 0 Run at Rate build.
6. Gage R&R should be conducted on a low speed uniformity (LSU) machine before PPAP samples are tested. Follow TM-04.04-E-414 for the LSU gage R&R to verify.
7. The Initial Run at Rate (IRR) of the capacity verification must be submitted to STA for approval.
8. Capacity verification form required per customer clarification & requirements
9. The PFMEA, Control Plan, and Tire Supplier APQP - PPAP Evidence Workbook must be reviewed and approved by Ford Supplier Technical Assistance (STA). The Tire Supplier will retain the original DFMEA, PFMEA and Control Plan. In addition, the Tire Supplier must demonstrate linkage between the DFMEA and PFMEA with a linkage document and have a Control Plan that ensures controls are in place at the supplier's plant to meet the Significant Characteristic (SC) and Critical Characteristic (CC) items. All documents relating to the program (DVP&R and related test reports, DFMEA, PFMEA, and Control Plans) must be submitted to Ford in PDF format in addition to being kept by the Tire Supplier for a document retention period of Job last + 5 years.
10. Tire weight must be assessed by the tire supplier at PPAP
11. Tire Rolling Resistance is not required for PPAP, since it cannot be completed at the Ford Building 4 facility.
    11.1 Tire Supplier must establish a nominal / baseline Rolling Resistance value, or define another means in the Control Plan, to demonstrate compliance to the In-Process Requirements I.H.2.2 and/or I.H.2.3, as approved by Global Tire Manager.

H. TIRE MANUFACTURING REQUIREMENTS

1. GENERAL REQUIREMENTS

1.1 Requirements are valid for all tires regardless of tire type, unless noted in the exceptions.

1.2 After Ford Vehicle Engineering has approved a tire construction, as part of the Engineering Sign-Off (DVP&R sign-off), the design cannot be changed without approval by Global Tire Manager, Vehicle Engineering, and the Tire D&R Engineer.

1.3 The tire manufacturing location may not be changed after Tire Supplier sourcing without approval by Global Tire Manager, Vehicle Engineering, the Tire D&R Engineer, and Ford Purchasing. All subsequent tires must be manufactured from this site (Submission tires, vehicle builds, etc.).

1.4 Tires of the same size and type with different load index, speed index, or construction supplied for production to a Ford manufacturing plant must not have the same sidewall appearance. Any deviation to this requirement must be approved by the Global Tire Manager.

1.5 No repairs to the tire are allowed unless approved by Global Tire Manager and STA Tire Technical Specialist. This includes, but is not limited to, repairs to the innerliner, beads, structural components or sidewall markings.

1.5.1 Green tire repairs are not allowed for innerliner, beads, or near any structural components (e.g. steel belts, body ply)

1.5.2 Cured tire repairs are not allowed for innerliner, beads, and sidewalls at locations of required government markings.

1.6 The bladder release agent used during the tire manufacturing process must not contribute to slippage between the wheel and tire.

2. IN-PROCESS REQUIREMENTS

2.1 Tire weight must be within ± 3% of the Tire Average Weight value on the approved Tire CDS.

2.2 SAE Rolling Resistance must be within ± 6% of the Rolling Resistance value on the approved Tire CDS.

2.3 Where applicable, the tire shall meet the DV sign off level per ISO 28580, in order to support vehicle manufacturer homologation per WLTP requirement.

3. TIRE APPEARANCE

3.1 Tires must be free from molding defects including; flash, bubbles, holes, color variation, surface texture variation, or other visual variations. The tolerances are defined by the boundary sample approved per section I.H.3.5.

3.2 The number and size of mold vents on a tire must be minimized.

3.3 Tires must not have vents or excessive flash in the bead seat if they have the potential to create air loss after mounting.

3.4 Bead barcodes, manufacturing marks, or stickers must not affect bead seat or air retention and must not be visible when the tire is mounted on the wheel.

3.4.1 Exception - Steel carcass tires, with approval from Global Tire Manager

3.5 Tire appearance boundary samples must be approved by the Tire D&R Engineer and Global Tire Manager as part of the DV process to ensure consistent appearance quality for all tires shipped to vehicle manufacturing locations. A minimum of one (1) sample per tire part number will be supplied for review and will be signed by the jury. The Tire Supplier must provide digital images of this master sample to Ford in order to support DV documentation. These images will capture key areas such as; mold lines, vents, overall tire, tread area, and jury signatures.

3.5.1 This will require physical appearance boundary samples to be submitted to Ford by the Tire Supplier to be evaluated by the jury identified above. This jury will establish acceptance criteria for the physical appearance of all production tires.

3.5.2 A document with images and descriptions clearly indicating unacceptable boundary conditions may be submitted in lieu of a physical sample if the content is incorporated into manufacturing process checks.

3.6 Normal tire plant process variation may require a cleanup operation to ensure that the quality appearance is maintained during production.

3.7 Tires are prohibited from being supplied with markings (color dots, writing, etc.) that were not approved on a physical appearance boundary sample or markings explicitly indicated as approved in this document.

3.8 The Tire Sidewall Protective Paint must protect against discoloring of white letters (WSW, OWL, and RWL) and not allow flaking or scaling of the protective paint from the initial application at the OE Tire Manufacturing Plant through our Assembly Plant operations; including the storage, shipping, handling, mounting, matching, inflation, load simulation, balance, roll test and wheel alignment. Afterwards the paint must be removable with Ford White Wall cleaner (w/o brushing) during new vehicle pre-delivery.

I. INSTRUCTIONS AND NOTES

1. Control Plans address all significant design and process characteristics, which include all FS tests and Control Item characteristics. They describe the process potential studies that will be performed for product validation (including PV tests) and the ongoing product and process evaluation for continuing improvement (including IP tests). They include acceptance criteria, sample sizes, frequencies, data analysis methods and reaction plans.
2. The Control Plan is developed, and updated as necessary, by the manufacturing source in conjunction with the design responsible Product Engineering activity and other appropriate functions such as STA. The Control Plan defines the management of the upstream production process and part variables (significant process characteristics) that affect the outcome of the FS tests or other significant design characteristics. The Control Plan also identifies the specific FS tests, with their sample sizes and frequencies that will be performed in order to:
   - 2.1 Confirm whether the process is being managed effectively.
   - 2.2 Further identify significant process characteristics.
   - 2.3 Evaluate performance of marginal processes.
   - 2.4 Better anticipate the customer effect of proposed process improvements
3. For any part on which FS tests have been specified, the manufacturing source must present the Control Plan and any revisions to the design responsible Product Design activity for review. This Product Engineering activity has flexibility to honor business relationships with suppliers having proprietary processes.

J. TIRE TRACEABILITY: TIN/VIN Tire Identification:
(All Vehicles Shipping to US/Canada, AND All North American Vehicle Manufacturing Sites)

1. Tire traceability process requires the Tire Supplier to either store the 4 digit DOT's (week/year) of the tires in each shipment for 10 years, or email (email ID – TIRETRAC@Ford.com) this information to Ford for data storage (storage location is in EDMS - https://www.edms.ford.com/webtop/drl/objectId/0b00cad9846513aa) In cases where Ford is storing tires in a warehouse as a capacity buffer, the warehouse will be required to email the data required for each Assembly Plant shipment.
2. The data stored or emailed to Ford for each shipment will be:
   2.1 Shipment number
   2.2 Ford Part number
   2.3 Date delivered to Assembly Plant
   2.4 Total number of tires in the shipment
   2.5 List of the DOT's (week/year) included in the shipment
3. If the above data (from I.J.2) is emailed, the data can be contained in a PDF file attached to the email.
4. Email Format: The email subject name and the PDF file name will follow the naming convention/format described below.
   4.1 Email Subject and PDF file naming structure - Use the 3-character Plant code in **Table 1 - Ford Plant Codes for DOT Week Identification** followed by the Ford Part number prefix and suffix and the date the shipment is delivered to the assembly plant. An example for Ford Part Number DG9C-1508-B1A shipped the Hermosillo Assembly Plant in Mexico to be delivered on July 16, 2020 would be:
      - Example: HAP DG9C-B1A 07-16-2020
   4.2 The PDF file will include the data required and it can include single or multiple shipments for that day to the Assembly Plant. It is acceptable to use any existing documents, such as a bill of lading or excel spreadsheet, to create the PDF file as long as it includes the required information.
   4.3 In case more than one part number is placed in the trailer, all Ford part numbers will be listed in the file name.
      - Example, one shipment to DTP included 4 part numbers. The part numbers were 9L34-1508-KA, 4L34-1508-LA, 4L34-1508-BA, and 5C34-1508-CA.
      - This email subject and PDF file would be identified as:
        DTP 9L34-KA 4L34-LA 4L34-BA 5C34-CA 07-16-2020
5. Part Number Identification: (Mandatory for All Markets)
   5.1 Every unique TIN provided to Ford Motor Company must have a unique Ford part number assigned. Tires with different TINs may not share a Ford part number.
   5.2 Every unique tire construction, tire size, tire speed index, and tire load rating provided to Ford from a supplier must have a unique Ford part number assigned. Tires with the same size, speed index, and load rating from the same supplier or different suppliers must not have the same Ford part number assigned.
   5.3 Every unique tire in a Ford production facility (manufacturing plant, outside assembler, etc.) must have a unique Ford part number assigned.
      5.3.1 Example: Left and right-handed directional tires are considered unique even if the tires are of the exact same size, speed index, load rating, construction, and supplier. If the tread pattern shape requires that the tires are installed in a specific direction/orientation on the vehicle then the tires for the left-side and right-side vehicle usages must have different part numbers.

Table 1 - Ford Plant Codes for DOT Week Identification

Code	Ford Assembly Plant	
FRA	Flat Rock Assembly Plant	
CAP	Chicago	
CUP	Cuautitlan, Mexico	
DTP	Dearborn Truck	
HAP	Hermosillo, Mexico	
KCP	Kansas City	
KTP	Kentucky Truck	
LVL	Louisville	
MAP	Michigan Assembly	
OAK	Oakville	
OHP	Ohio Assy. (Avon Lake)	
DCP	Detroit Chassis Plant	
VBA	Valencia Body & Assembly	
CVB	Chennai Vehicle Assembly Plant	

K. TIRE STORAGE REQUIREMENTS

1. When storing tires prior to testing at all facilities, or when storing tires for production supply at a non-Ford location, the following must be followed:
   - 1.1 Tires will be stored dismounted, except for tires that are already mounted in preparation for testing.
     - 1.1.1 If test tires cannot be tested within 24 hours, the tire and wheel assembly inflation pressure must be removed. Remove valve stem core and allow tire to deflate to 0 (zero) psig (e.g. to atmospheric pressure).
     - 1.1.2 Tires may not be stored under pressure for more than 72 hours, less if specified in a specific test method.
   - 1.2 Tires must be stored in a laced configuration, stovepipe configuration, or on racks approved by Global Tire Manager. Tire storage must comply with all applicable fire code requirements.
   - 1.3 Future expectations are a controlled storage temperature of 21°C ± 3°C (70°F ± 5°F) for all test and production tires.
   - 1.4 Tire may not be stored in direct sunlight or outdoors.
   - 1.5 Tires may not be stored near an electric motor or other device that can produce ozone.

TABLE 2 : Current list of SC/CC items in the APQP - PPAP Evidence Workbook.

No.	Characteristic Description	DFMEA Classification	PFMEA Classification	Metric, P-Metric, LT, or EU Commercial	T-Type (Mini) Spare Tire	Commercial Steel Carcass Tire
	Below are the minimum requirements and could be extended to include other characteristics. (This is what's in FS4 Draft)					
1	Material identification / Materials Specification (raw mat'l spec., COA)	YC	CC	X	X	X
2	Laydown temperature of final batches	YC	CC	X	X	X
3	Compound content (recipe, mixing law)	YC	CC	X	X	X
4	Innerliner gauge	YC	CC	X	X	X
5	Innerliner width	YC	CC	X	X	X
6	Body ply gauge	YS	SC	X	X	X
7	Body ply width	YS	SC	X	X	X
8	Body ply cord spacing (EPI across sheet)	YS	SC	X	X	X
9	Body ply peel strength	YC	CC	X	X	X
10	Steel ply gauge	YS	SC	X	X	X
11	Steel ply cord spacing (EPI across sheet)	YS	SC	X	X	X
12	Creel room temperature	YC	CC	X	X	X
13	Creel room humidity	YC	CC	X	X	X
14	Body ply cord splice overlap	YS	SC	X	X	
15	Steel belt width	YC	CC	X	X	X
16	Steel belt angle	YC	CC	X	X	X
17	Belt edge gum width	YC	CC	X	X	X
18	Steel belt peel strength	YC	CC	X	X	X
19	Path of electrical conductivity (tread compound/machine set-up)	YS	SC	X		
20	Tread gauge	YC	CC	X	X	X
21	Tread width	YC	CC	X	X	X
22	Tread weight	YC	CC	X	X	X
23	Sidewall gauge	YC	CC	X	X	X
24	Sidewall width	YC	CC	X	X	X
25	Cooling water Ph value of extruder process	YS	SC	X	X	X
26	Bead inner diameter/circumference	YC	CC	X	X	X
27	Bead wire size/strand x turn	YC	CC	X	X	X
28	Apex profile	YC	CC	X	X	X
29	Tire building specification: component ID, spotting, tension	YS	SC	X	X	X
30	Nylon cover width	YC	CC	X	X	
31	Tread center to carcass center	YS	SC	X	X	X
32	Belt center to carcass center	YS	SC	X	X	X
33	Tread length	YS	SC	X	X	X
34	Curing time	YC	CC	X	X	X
35	Curing temperature	YC	CC	X	X	X
36	Curing pressure	YC	CC	X	X	X
37	Post cure inflation (PCI) (time and pressure)	YS	SC	X		
38	Green tire splice positioning (spotting-green tire relative to ID)	YS	SC	X	X	X
39	Sidewall marking	YC	CC	X	X	X
40	Radial force variation (cw)	YS	SC	X		
41	Radial force variation (ccw)	YS	SC	X		
42	Radial first harmonic (cw)	YS	SC	X		
43	Radial first harmonic (ccw)	YS	SC	X		
44	Radial second harmonic (cw) - region specific	YS	SC	X		
45	Radial second harmonic (ccw) - region specific	YS	SC	X		
46	Lateral force variation (cw)	YS	SC	X		
47	Lateral force variation (ccw)	YS	SC	X		
48	Lateral first harmonic (cw) - region specific	YS	SC	X		
49	Lateral first harmonic (ccw) - region specific	YS	SC	X		
50	Conicity sym/asym	YS	SC	X		
51	Static Balance or Dynamic Balance	YS	SC	X		X
52	Rolling Resistance	YC	CC	X		X
53	Radial run out	YS	SC	X		X
54	Lateral run out	YS	SC	X		X
55	Sidewall undulation (buldge & dent)	YS	SC	X	X	X
56	Conicity marking (quad or split)	YS	SC	X		
57	R1H high point mark location accuracy - region specific	YS	SC	X		X
58	Tire weight	YS	SC	X	X	X
59	Cured tire dimension check (e.g: OD, SW, CDS…)	YC	CC	X	X	X
60	Government regulatory lab testing	YC	CC	X	X	X
61	Component supply	YC	CC	X	X	X
62	Foam material (Material certification / uniform density)	YS	SC	X		
63	Foam Dimensions (Height, Width, Length)	YS	SC	X		
64	Foam Location on the tire (Centered on the innerliner)	YS	SC	X		
65	Foam Adhesive material (Material certification, Expiration date)	YS	SC	X		
66	Foam to Tire adhesion (Adhesive amount, Adhesive application location / pattern)	YS	SC	X		
67	Surface condition of the inner liner for adhesion	YS	SC	X		
67	Materials specification	YS	SC	X		
69	Sealant Package Thickness	YS	SC	X		
70	Sealant Package Width	YS	SC	X		
71	Sealant Package Profile	YS	SC	X		
72	Sealant Package Location	YS	SC	X		
73	Liner Preparation Specification	YS	SC	X		
74	Sealant Brand Sidewall Location	YC	CC	X		
75	Sealant Brand Sidewall Depth/Thickness	YC	CC	X		
76	Sealant Brand Sidewall Height	YS	SC	X		
77	Sealant Brand Sidewall Width	YS	SC	X		

Table 3 - Tire FS Test Summary

FS Section	SC & CC	Test Characteristics & Primary Test Responsibility Ford (F) / Supplier (S)	Lower Production Tolerance Limit Low value of the test char. Used to judge performance	Target Value Optimum value of test char.	Upper Production Tolerance Limit High value of test char.	Minimum Conformance to Tolerance	Prelim. Control Plan Parameter Sample Size
I.D.11.1	CC	Tire Supplier Compliance Letter (S)	N/A	N/A	N/A	N/A	N/A
I.D.18	CC	Tire Warranty Guide Approval (F)	N/A	N/A	N/A	N/A	N/A
II.BII.B	CC	High Speed P-Metric and Metric Tires (S)	N/A	N/A	N/A	All must pass	4 total
II.C	CC	High Speed LT-Metric, North American Truck/Bus, European Commercial (S)	N/A	N/A	N/A	All must pass	4
II.D	CC	Rim Roll Off (S & F)	N/A	No air loss	N/A	All must pass	2
II.E	CC	Tire Retention Test (S & F)	N/A	Retained on rim	N/A	All must pass	2
II.F	N/A	Ride, Handling, Braking (F)	N/A	Per TA	N/A	All must pass	1 vehicle set
0	N/A	NVH Requirements (S & F)	N/A	Per TA	N/A	All must pass	1 vehicle set
II.K	SC	Electrical Resistance (S)	N/A	N/A	10^8 Ω	All must pass	2
II.L	CC	Lab Durability (S)	N/A	~~DOT + 48 hours @ 130% of load~~ Per FS	N/A	All must pass	4
II.M	SC	PRAT (F)	Submission value – 1 Nm	Table 4 & Per TA	Submission value + 1 Nm	Avg must pass	4
II.N	SC	Low Speed Tire Uniformity (S)	N/A	Per FS	N/A	CPK ≥ 1.33	100 % Meas.
II.N.6	SC	Sidewall Undulation (S)	N/A	N/A	1.3 mm (.05 in.)	All must pass	4
II.O	N/A	Tire Measurements (S)	N/A	N/A	N/A	N/A	Per table
II.P.3	CC	Rolling Resistance SAE J2452 (F)	Submission value * .94	Per TA	DV value * 1.06	Avg must pass	As agreed by Tire Core
II.P.4	CC	Rolling Resistance ISO 28580 (S)	In accordance w/ ECE 117-02 EG 1222/2009	Per TA	In accordance w/ ECE 117-02 EG 1222/2009	Avg must pass	As agreed by Tire Core
II.Q.1	SC	Flat Spotting Short Term (F)	N/A	Per TA	N/A	All must pass	per Model
II.Q.2	SC	Flat Spotting Long Term (F)	N/A	Per TA	N/A	All must pass	per Model
II.Q.3	N/A	Heat Set Flat Spot (F)	N/A	N/A	63 degrees C	N/A	N/A
II.R	N/A	Tire Wear Projected Miles (S)	N/A	Per TA	N/A	All must pass	1 vehicle set or Model
II.R.7	N/A	Tire Wear Irregular Wear (S)	N/A	N/A	See Section II.N	N/A	1 vehicle set or Model
II.S	N/A	Gravel Test (S)	N/A	Per TA	N/A	N/A	2 or Model
II.U	CC	Tire Aging Durability (S)	N/A	Per FS	N/A	All must pass	4 tire sample
II.V	SC	Structural Accelerated Tire Durability (S)	N/A	Per FS	N/A	All must pass	2 tire sample

II. TEST PROCEDURES AND REQUIREMENTS

A. NOTES FOR ALL TESTING IN SECTION II

1. Test rim of the released minimum width application rim will be used. This will qualify the test tire for rim widths equal to or greater than the released width.
   - 1.1 For all tire component testing including, but not limited to, Ride, Handling, Braking and NVH, the test rim bead hump, offset, and other significant geometry should match that of the production intent rims. Deviations must be reviewed and approved by Global Tire Manager and Core Vehicle Dynamics (when applicable).
2. When tires are to be released in different sidewall configurations (e.g. BSW, OWL, WSW), only one sidewall configuration needs to be qualified. The sidewall configuration to be tested must be approved by Global Tire Manager.
3. Criteria for selecting tires prior to testing:
   - 3.1 Unless otherwise specified select tires to test that are more than 1 week old and not older than 20 weeks old based on the TIN week code.
     Tires outside of this range prior to testing must be approved by Global Tire Manager.
4. An **Incomplete Vehicle** is defined as follows:
   - 4.1 An assemblage consisting, as a minimum, of chassis (including the frame) structure, powertrain, steering system, suspension system, and braking system, to the state that those systems are to be part of the completed vehicle, but requires further manufacturing operations to become a completed vehicle.
5. For any test procedure or requirement with "Exceptions / Exclusions", the listed tire types are not required to perform this testing in order to satisfy DV sign off.
   This must be clearly documented on the DVP&R and approved by Global Tire Manager.
6. Instrumentation
   - 6.1 All test measurement equipment must be calibrated and maintained per FAP03-015, Control, Calibration, and Maintenance of Measurement and Test Equipment.
   - 6.2 All applicable safety guidelines and procedures must be followed.
   - 6.3 Tire pressure gauge accurate to ± 5 kpa (±1 psi)
   - 6.4 Thermometer accurate to ± 3°C (±5°F)

B. LAB HIGH SPEED – Metric and P-Metric Tires *(formerly TM-04.04-L-4150)*

1. Purpose / Goal of the Test
   1.1 This test evaluates the high-speed robustness of a loaded tire. The purpose of this test is to establish that the tire will not experience air loss, tread separation, cap ply separation, or other failures when evaluated at the defined test pressure(s), test load(s), and test speed(s).
   1.2 This test is defined for P-Metric tires, and Metric tires only.
   1.3 COMMONALITY. This is a CONTROL TEST and can be used to qualify vehicles or components throughout the world. The test may be conducted at any location having the necessary equipment and facilities. Proposed revisions to this procedure must be submitted per FAP03-179.
2. Equipment and Facilities
   2.1 Tire mounting equipment
   2.2 1.7 m (67 inch) drum in 40° C ± 3° C (104° F ± 5° F) regulated room with speed and load controls calibrated and run to NHTSA/ASTM standard conditions.
   2.3 Either a True Wheel (sometimes referred to as lab wheels or machine wheels) or a vehicle application wheel of the appropriate size and rim width as specified by the test requestor.
   2.4 Appropriate wheel adapters as necessary.
3. Sample Size: Four (4) tires
4. Sample Preparation
   4.1 Sample Conditioning: Set the tire/wheel assembly to the released cold inflation pressure while at ambient temperature [24 °C (75 °F)]. Place the tire/wheel assembly in the 40°C test room and allow a minimum of three (3) hours soak time for temperature and pressure to stabilize before adjusting the inflation pressure to the test pressure specified in II.B.5
   4.2 Each test must be run with new tires. Samples may not be cascaded to or used for additional test cycles of this procedure.
5. Test Conditions
   5.1 The test conditions are based on the placard pressure complexity as defined in the following table:
   Note: The tire maximum load capacity is defined as the maximum load corresponding to the tire load index per the appropriate standard (i.e. ETRTO, T&RA, etc…)

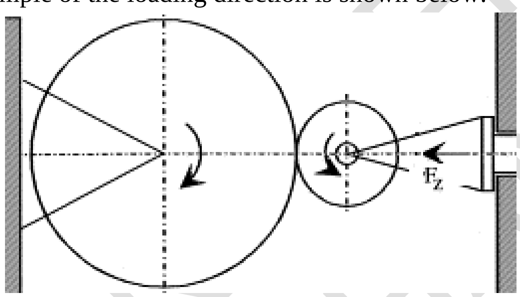
Metric and P-Metric Tires					
			Pressure	**Load***	**Speed**
Basic conditions One pressure (No split for load or speed)			Placard - 8 psi	Maximum of: 80% Load Capacity or GAWR / 2	Maximum vehicle speed
Dual pressure split for speed		Vehicles sold in European markets only	Released cold inflation pressure for speeds **greater** than 160 kph (100 mph) minus 8 psi (55 kPa)	Maximum of: 80% Load Capacity or GAWR / 2	Maximum vehicle speed
		Vehicles sold in all markets except Europe	Released cold inflation pressure for speeds **less** than 160 kph (100 mph) minus 8 psi (55 kPa)	Maximum of: 80% Load Capacity or GAWR / 2	Maximum vehicle speed
Dual pressure Front / Rear split	Determine the Maximum deflection case between: 1. FGAWR/2, FGAWR pressure 2. RGAWR/2, RGAWR pressure	Case 1: Worst case FGAWR	Front Placard - 8 psi (55 kPa)	Maximum of: 80% Load Capacity or FGAWR / 2	Maximum vehicle speed
		Case 2: Worst case RGAWR	Rear Placard - 8 psi (55 kPa)	Maximum of: 80% Load Capacity or RGAWR / 2	Maximum vehicle speed
Dual pressure Split for load (Normal vs GAWR)	Determine the Maximum deflection case between: 1. GAWR/2, GAWR pressure 2. normal load/2, normal load pressure	Case 1: Worst case GAWR	GAWR Placard - 8 psi (55 kPa)	Maximum of: 80% Load Capacity or GAWR / 2	Maximum vehicle speed
		Case 2. Worst case Normal load (Run both conditions)	GAWR Placard - 8 psi (55 kPa)	Maximum of: 80% Load Capacity or GAWR / 2	Maximum vehicle speed
			Normal Placard - 8 psi (55 kPa)	Normal Load / 2	Maximum vehicle speed
Winter Tires			Placard - 8 psi	GAWR / 2	based on the *Winter Tire Label* speed
		** For V, W, and Y rated tire, the load shall be reduced by an additional percentage as detailed*			

6. Procedure Steps
   - 6.1 Test Pressure: Inflate new tire/wheel test assembly to the pressure defined in II.B.5.
   - 6.2 Target Speed: Test to speed defined in II.B.5
   - 6.3 Test Load: Load defined in II.B.5. For V, W, and Y rated tires the load shall be reduced by an additional percentage identified in the table below.

Speed Rating	Load Index	Load Reduction
V	SL	10%
W	SL	17%
Y	SL	17%
V	XL	10%
W	XL	20%
Y	XL	20%

EXAMPLES:
H rating and below
Test Load = Max (80%* tire's maximum load<OR> Heaviest GAWR divided by two)
V rating (SL)
Test Load = 90% * Max (80% * tire's maximum load <OR> Heaviest GAWR divided by two)
W or Y rated Standard Load (SL)
Test Load = 83%*Max (80%* tire's maximum load <OR> Heaviest GAWR divided by two)
W or Y rated Extra Load (XL) or Reinforced
Test Load = 80% * Max (80% * tire's maximum load <OR> Heaviest GAWR divided by two)

   - 6.4 Mount the tire and wheel assembly onto the test axle and load the assembly normal to the outer face of the equipment drum. An example of the loading direction is shown below.



   - 6.5 Duration at each test speed: 30 minutes
     Note: All speed steps must be tested, in order, without interruption during or between steps
   - 6.6 Starting test speed is defined as Target Speed less 40 km/h
     Note: Run-up time less than 5 minutes from zero to Starting test speed
   - 6.7 Successive speed increments of + 10 km/h (6 mph)
     Note: Run-up time less than 15 second between steps defined in II.B.6.5
   - 6.8 Continue in 10 km/h (6 mph) steps every 30 minutes until tire structural breakdown is detected.
     The tire must be tested to failure, or to a MINIMUM of three speed steps beyond the target speed.

7. Specialty Vehicle Tire Performance
   Note: This procedure is to be run only for tires for Specialty Vehicle with a vehicle maximum speed in excess of 241 Km/h (150 mph) released for markets with restricted speed limits (such as North America). This procedure is not valid for markets with unrestricted speed limits.
   - 7.1 Inflate new tire/wheel test assembly to the pressure defined per II.B.5
   - 7.2 Repeat procedure II.B.6 with the target speed defined per II.B.5 reduced to 80% of the vehicle maximum speed.
   - 7.3 The test may be suspended after the completion of three speed steps beyond the target speed.

8. Acceptance Criteria:
   - 8.1 All tire samples must complete the 30 minute target speed step with zero failures per the defined failure modes in II.B.10.
   - 8.2 A tire experiencing structural break down at speed steps above the target speed constitutes an acceptable / passing result
   - 8.3 For tires with a failure which have pressures defined by II.B.5, the following must be conducted on new virgin tires.
     - 8.3.1 Increase the inflation pressure and re-run the test until all samples pass. Note the amount of additional pressure required above the value defined in II.B.5 and review the results with Global Tire Manager.
   - 8.4 A tire experiencing structural break down at speed steps above the target speed constitutes an acceptable/passing result.

9. Specialty Vehicle Tires Acceptance Criteria:

9.1 At or below 80% of vehicle max speed, zero of the tire samples may fail per the defined failure modes in II.B.10.

9.2 Benign failure after 1st speed step is a pass.

10. Failure Modes:
    a. Loss of air
    b. Tread separation
    c. Cap ply separation
    d. Belt ply to cap ply separation (if cap ply present)
    e. Belt ply to tread separation (if no cap ply present)
    f. Belt ply to belt ply separation
    g. Belt ply to body ply separation
    h. Cord Separation or Breakage
    i. Bead Separation
    j. Sidewall Separation
    k. Inner Liner Separation

11. Data Reporting:

11.1 Data should be prepared and present in the format defined in the Tire DVP&R.

11.2 Test termination example:

- The test(s) may be suspended after the completion of three speed steps above target. The following are examples of correct and incorrect testing to satisfy this statement.
- Assuming a V rated tire, the target speed is 210 km/h. The tire testing must complete; a full 30 minutes at 210 km/h and the start of the 220 km/h speed step, to be a valid passing end of test condition. A tire experiencing structural break down at speed steps above the target speed, in this example 220 km/h or higher, constitute an acceptable/passing result. A tire without structural breakdown must complete a full 30 minutes at three speed steps above the target, in this case 240 km/h (each speed step is an additional 10 km/h above target), to be a valid passing end of test condition.

12. Exceptions / Exclusions:

12.1 Specialty Vehicle Tires have a unique set of test conditions. The test is only to be run for Specialty Vehicle Tires with vehicle maximum speed in excess of 241 km/h (150 mph) intended for markets with restricted speed limits (such as North America) and not valid for markets with unrestricted speed limits.

12.2 Exceptions to the test conditions defined in II.B.5 must be reviewed and approved by the Global Tire Technical Specialist and Global Tire Manager, and the conditions must be detailed in the Tire DVP&R.

12.3 Exclusion: T-Type (mini spare) tires.

C. LAB HIGH SPEED - LT-Metric, North American Truck/Bus, and European Commercial Tires *(formerly TM-04.04-L-4151)*

1. Purpose / Goal of the Test
   - 1.1 This test evaluates the high-speed robustness of a loaded tire. The purpose of this test is to establish that the tire will not experience air loss, tread separation, cap ply separation, or other failures when evaluated at the defined test pressure(s), test load(s), and test speed(s).
   - 1.2 This test is defined for LT-Metric, North American Truck/Bus, and European Commercial Tires.
   - 1.3 COMMONALITY. This is a CONTROL TEST and can be used to qualify vehicles or components throughout the world. The test may be conducted at any location having the necessary equipment and facilities. Proposed revisions to this procedure must be submitted per FAP03-179.
2. Equipment and Facilities
   - 2.1 Tire mounting equipment
   - 2.2 1.7 m (67 inch) drum in 40 °C ± 3 °C (104 °F ± 5 °F) regulated room with speed and load controls calibrated and run to NHTSA/ASTM standard conditions.
   - 2.3 Either a True Wheel (sometimes referred to as lab wheels or machine wheels) or a vehicle application wheel of the appropriate size and rim width as specified by the test requestor.
   - 2.4 Appropriate wheel adapters as necessary.
3. Sample Size: Four (4) tires
4. Sample Preparation
   - 4.1 Sample Conditioning: Set the tire/wheel assembly to cold inflation pressure specified in II.C.4.3 while at ambient temperature [24 °C (75 °F)]. Place the tire wheel assembly in the 40 °C test room and allow a minimum of three (3) hours soak time for temperature and pressure to stabilize before adjusting the inflation pressure to the test pressure specified in II.C.4.3.
   - 4.2 Inflation Pressure: Test samples will be run at one test condition per test cycle. All pressures are capped during test. See II.C.4.1 for tire conditioning prior to test.
   - 4.3 Released cold inflation pressure. See the Tire Contract or Tire Functional Specification for alternative pressures required based on vehicle setup.
   - 4.4 Samples may not be cascaded to or used for additional test cycles of this procedure.
5. Test Conditions
   - 5.1 The test conditions are based on the placard pressure complexity defined in the following table:
     Note: The tire maximum load capacity is defined as the maximum load corresponding to the tire load index per the appropriate standard (i.e. ETRTO, T&RA, etc…)
   - 5.2 For vehicles with both front/rear and "normal load"/GAWR splits follow **both** sections as outlined in the table.

LT, NA Truck/Bus, EU Commercial					
			Pressure	**Load**	**Speed**
Basic conditions One pressure (No split for load or speed)			Placard	Maximum of: 80% Load Capacity or GAWR / 2	Maximum vehicle speed
Dual pressure Front / Rear split	Run the Max deflection case between 1. FGAWR/2, FGAWR pressure 2. RGAWR/2, RGAWR pressure	Case 1: Worst case FGAWR	Front Placard	Maximum of: 80% Load Capacity or FGAWR / 2	Maximum vehicle speed
		Case 2: Worst case RGAWR	Rear Placard	Maximum of: 80% Load Capacity or RGAWR / 2	Maximum vehicle speed
Dual pressure Split for load (Normal vs GAWR)	Run the Max deflection case between 1. GAWR/2, GAWR pressure 2. normal load/2, normal load pressure	Case 1: Worst case GAWR	GAWR Placard	Maximum of: 80% Load Capacity or GAWR / 2	Maximum vehicle speed
		Case 2. Worst case Normal load (Run both conditions)	GAWR Placard	Maximum of: 80% Load Capacity or GAWR / 2	Maximum vehicle speed
			Normal Placard	Normal Load / 2	Maximum vehicle speed

6. Procedure Steps

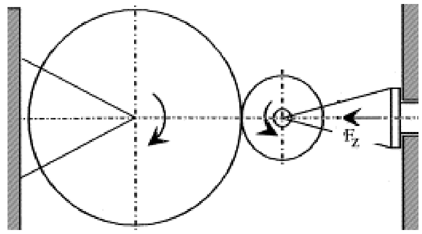
   Note: There is one test condition for this test as represented by II.C.5 with corresponding test steps. The test must be run with new tires, tires may not be cascaded from one test to another.

   6.1 Test Pressure: Inflate new tire/wheel test assembly to the pressure defined per II.C.4.2

   6.2 Target speed: The vehicle maximum speed (defined by the speed limiter).

   6.3 Test Load: Per table in II.C.5

   6.4 Mount the tire and wheel assembly onto the test axle and load the assembly normal to the outer face of the equipment drum. An example of the loading direction is shown below.



   6.5 Duration at each test speed: See table below.
   Note: All speed steps must be tested, in order, without interruption during or between steps

Speed Step	Duration (min)
1	10
2	10
3 or more	30

   6.6 Speed step 1 is defined as target speed (II.C.6.2) less 20 km/h.
   Note: Run-up time less than 10 minutes from zero to 0 (target less 20 km/h)

   6.7 Successive speed increments of + 10 km/h (6 mph)
   Note: Run-up time less than 15 second between steps defined in II.C.6.5

   6.8 Continue in 10 km/h (6 mph) steps every 30 min. until tire structural breakdown is detected. The test may be suspended after the completion of two speed steps beyond the target speed.

   6.9 Test termination example:

   - Test(s) may be suspended after the completion of two speed steps above target. The following are examples of correct and incorrect testing to satisfy this statement.
   - If we assume a Maximum Vehicle Speed is 170 km/h. The tire testing must complete; a full 30 minutes at 170 km/h and the start of the 180 km/h speed step, to be a valid passing end of test condition. A tire experiencing structural break down at speed steps above the target speed, in this example 170 km/h or higher, constitute an acceptable/passing result. A tire without structural breakdown must complete a full 30 minutes at two speed steps above the target, in this case 190 km/h (each speed step is an additional 10 km/h above target), to be a valid passing end of test condition.

7. Acceptance Criteria:

   7.1 Zero of the tire samples may fail, per the defined failure modes in II.C.7.3, at or below the vehicle max speed step.

   7.2 A tire experiencing structural break down at speed steps above the target speed constitutes an acceptable/passing result.

7.3 Failure Modes

- Loss of air
- Tread separation
- Cap ply separation
- Belt ply to cap ply separation (if cap ply present)
- Belt ply to tread separation (if no cap ply present)
- Belt ply to belt ply separation
- Belt ply to body ply separation
- Cord Separation or Breakage
- Bead Separation
- Sidewall Separation
- Inner Liner Separation

8. Data Generated & Formatting of Presentation

8.1 Record for each tire:

8.1.1.1 Test hours

8.1.1.2 Failure mode

8.1.1.3 Remarks

8.2 The data shall be presented in an excel format defined in the tire DVP&R which shall include, at a minimum; an individual line per tire tested, number of test hours each tire completed, test speed step last completed prior to end of test, and failure mode of each tire.

9. Exceptions / Exclusions:

9.1 Steel carcass tires used on Incomplete Vehicles must either run the Lab High Speed test, or the Tire Supplier must show compliance to applicable homologation requirements.

D. RIM ROLL OFF *(formerly TM-04.04-R-402)*

1. Purpose / Goal of the Test
   1.1 The goal of this test is to establish that the tire will not lose air or be unseated from the rim when subjected to lateral acceleration when the vehicle is at test axle GAWR and GVW loading.
   1.2 COMMONALITY. This is a CONTROL TEST and can be used to qualify vehicles or components throughout the world. The test may be conducted at any location having the necessary equipment and facilities. Proposed revisions to this procedure must be submitted per FAP03-179.
2. Instrumentation
   2.1 Calibrated speedometer (or engine tachometer).
   2.2 Calibrated tire pressure gauge.
   2.3 Vehicle weighing scales.
   2.4 Wheel bead seat diameter ball tape
   2.5 Torque Wrench (or Torque Stick, or equivalent, of correct level)
3. Equipment and Facilities
   3.1 Any facility with smooth dry asphalt or concrete surface area. The pad needs to have a marked circle 50 m (165 ft) in diameter (test circle) and a tangential approach road.
   3.2 A program vehicle or prototype for which the tire is to be fitted is recommended for this test. The use of a similar carryover vehicle or a representative surrogate vehicle and a wheel with the same width and flange type, as the selected wheel, is acceptable.
   NOTE: If a non-program vehicle is used, this alternate vehicle must have Front and Rear axle GAWRs and GVWR similar to the program vehicle for which the program tire is intended.
4. Sample Size:
   4.1 Vehicles with **same** tires front to rear – 2 samples total
   4.2 Vehicles with dissimilar sized tires – 2 samples front / 2 samples rear (4 samples total)
      - Samples to be run in their correct vehicle positions. For example, a tire that is only used in the rear position would only be tested in the rear position. That tire would not be tested in the front position.
   4.3 Vehicles with dual rear wheels - 2 samples front / 4 samples rear (6 samples total):
5. Sample Preparation
   5.1 Each tire construction should be tested on the minimum width road wheel, either steel or aluminum, for which that tire construction is released.
   5.2 Spare tires that are dissimilar from the road tires should be tested in combination with the spare wheel on which they are fitted.
   5.3 Measure the test wheel's individual bead seat diameters using an industry standard ball tape to confirm compliance to T&RA diameter standards.
   Note: Do not use any wheels that are not within T&RA or ETRTO tolerance for bead seat diameter.
   5.4 Mount the test tires on the wheels required for the test and inflate to the maximum pressure specified on the sidewall. Inspect to confirm the tire beads are fully seated.
   5.5 Adjust the air pressure of all samples to the program vehicle placard pressure for that tire wheel combination.
   Note: If a like size spare tire's recommended pressure is different from the front placard pressure, assume the front placard pressure for II.D.6.4.
   Note: If a dissimilar sized spare assembly is being tested, run the test at the 50% of spare tire placard pressure.
   5.6 Soak test tire/wheel assemblies at least 24 hours before testing or installing on vehicle.
6. Description of Test:
   6.1 The goal of this test is to establish that the tire will not lose air or be unseated from the rim when subjected to lateral loading when the vehicle is at test axle GAWR and GVW loading.
   6.2 This test will be done on a smooth dry asphalt or concrete surface per the FS.
   6.3 A prototype or representative vehicle may be used to conduct this testing per the FS.
   6.4 Inflation Pressure: 50% of placard inflation pressure for GAWR load.
   6.5 Load: Test axle GAWR and GVW loading per FS.
   6.6 Speed: Per the FS, depending on vehicle type and tire lateral mu.
7. Procedure Steps
   7.1 Install one (1) test sample at the RF position for LHD vehicles or LF position for RHD vehicles.

7.2 Install non-interest tires of same size and construction (road tires for that vehicle) at remaining three positions and adjust pressure to vehicle placard pressures.

7.3 For vehicles with dissimilar sized tires on front vs. rear axle:

- Two (2) samples of the test tire [one (1) at Left Rear and one (1) at Right Rear] shall be tested at the rear positions, four (4) for dual rear wheel vehicles [two (2) at Left Rear and two (2) at Right Rear];
- And two (2) samples of the correct size and type front tires [one (1) at Left Front and one (1) at Right Front] must be installed at the front positions.
- This should create a production representative tire setup on the test vehicle with tire samples at every axle position.

7.4 For Mini spares (T-type tires): When the spare tire is released for usage with more than one road tire, test with the highest lateral mu road tire. Place the mini-spare in either the RF position for LHD vehicles or LF position for RHD vehicles. Follow II.D.7.2 for the remaining vehicle positions.

7.5 Torque all wheel nuts to the recommended nominal torque level.

7.6 Load the vehicle to FGAWR (Front Gross Axle Weight Rating) and then try to achieve GVW (Gross Vehicle Weight) for the program vehicle. If the above described vehicle loading cannot be achieved, note this in the test report.

7.7 Calibrate the speedometer or engine tachometer at a road speed of 50 km/h (30 mph) on a straight course.

7.8 Mark with chalk possible areas of contact between test tire and vehicle during the test.

8. General Instruction / Supplemental Information

8.1 The steps in this procedure shall be run twice. For each new run, a new test tire/wheel assembly, as defined in section II.D.7 and prepared per section II.D.5, must be used. The same non-interest tires as defined in section II.D.7.2 can be used for both test runs.

8.2 Warm up the vehicle by driving for 10 km at 50 km/h (6 miles at 30 mph).

8.3 Run Test with Test Tire at Reduced Pressure
NOTE: It is recommended to disable any Stability Control system, if possible, since the intervention of these systems may make it difficult to achieve the Target Speed
NOTE: For vehicles with automatic transmissions, it is recommended to run the test with the gear selector in a lower gear (for example, "2") instead of Drive ("D") to prevent a gear shift while running the procedure
NOTE: For vehicles with manual transmissions, it is recommended to keep the vehicle in the same gear while the vehicle is on the test circle and running the procedure.

8.4 Reduce the pressure in the Test Tire (Right Front for LHD vehicles, and Left Front for RHD vehicles) to 50% of the placard inflation pressure

8.5 Position the vehicle on the approach road so that its longitudinal axis is tangent to the test circle, taking into consideration the direction of operation (counter-clockwise for LHD vehicles, clockwise for RHD vehicles).

8.6 Drive the vehicle along the tangential approach road onto the test circle at 25 km/h (15 mph), such that its longitudinal axis covers the test circle.

8.7 After the vehicle is established onto the test circle (approximately 90° around the test circle), gradually increase speed and adjust steering input to maintain the vehicles position on the test circle, until the target speed as defined in the table in II.D.8.8 is reached.

8.8 Complete one full circle at the target speed.

Vehicle type	Tire Lat Mu	Target Speed
Pass car, SUV, CUV - (Unibody)	< 1	42 km/h (26 mph)
Pass car, SUV, CUV - (Unibody)	≥ 1	48 km/h (30 mph)
Van, Car with Cargo Body, Pick-up, SUV	< .9	37 km/h (23 mph)
Van, Car with Cargo Body, Pick-up, SUV	≥ .9	43 km/h (27 mph)

8.9 To avoid or reduce damage to the test track, immediately terminate the test if any of the following occur:

- The test tire is clearly losing pressure (no inflation buildup).
- The test tire is rolled off the rim.
- The rim contacts the ground.
  CAUTION: When roll-off occurs do not brake at a high deceleration rate because damage to the test tire can result.

8.10 Remove the test wheel and tire assembly and replace as requested.

9. Acceptance Criteria: Acceptance is satisfying the following conditions after one (1) pass:
   9.1 The test tire must not lose pressure.
   9.2 The test tire must not roll off the rim.
   9.3 The test rim must not contact the ground.
10. Data Generated & Formatting of Presentation
   10.1 Record the following data:
      10.1.1 Tire: size, type, manufacturer, tread line (tire family), manufacturer's construction number (and individual serial number if it is a prototype level part), Ford part number, full DOT code, air pressure at the start and after completion of the test, tire position on vehicle
      10.1.2 Wheel: Size, Ford part number and contour.
      10.1.3 Vehicle: type and actual weight distribution.
      10.1.4 Test site: road surface type, day, time of the day, ambient temperature, test engineer and other pertinent information.
   10.2 Include the following observations made during the test:
      10.2.1 Vehicle behavior at test termination.
      10.2.2 Any areas of contact between the tire and vehicle during the test.
      10.2.3 Type of wheel damage, if any.
      10.2.4 Part of circle where roll-off or rim contact occurred.
      10.2.5 Type and degree of wear or damage to the tire after roll-off.
11. Exceptions / Exclusions:
   11.1 Steel carcass tires used on Incomplete Vehicles.
12. References
   12.1 FAP03-179, Developing Corporate Engineering Test Procedures.
   12.2 FAP03-015, Control, Calibration, and Maintenance of Measurement and Test Equipment.
13. Appendix / Attachment
   13.1 "Tire Lat Mu" is defined as the peak lateral mu of the tire as generated by the Test Method "TM-04.04-L-415_Vehicle Dynamics Tire Test, Flat Trac® Variable Lateral Sweep Test Procedure"

E. TIRE RETENTION (INSTANT AIR LOSS) *(formerly TM-04.04-R-802-US)*

1. Purpose / Goal of the Test

   1.1 The goal of this test is to establish tire retention requirements for all rims used on passenger cars. The test results may be used to support demonstration of compliance with Federal Motor Vehicle Standard No. 110, paragraph S4.4.1 (b), as September 11, 2016 in the Federal Register and as stated in Ford Engineering Requirements. *Reference: FMVSS110 – Tire Selection and Rims and Motor Home/Recreation Vehicle Trailer Load Carrying Capacity Information for Motor Vehicles with a GVWR of 4,536 kg (10,000 lb.) or Less*

   1.2 COMMONALITY. This is a CONTROL TEST and can be used to qualify vehicles or components throughout the world. The test may be conducted at any location having the necessary equipment and facilities. Proposed revisions to this procedure must be submitted per FAP03-179

2. Instrumentation

   Note - Equivalent instrumentation may be used if approved by test responsible management before test is run.

   2.1 U-Tube Accelerometer, 4 inch,

   2.2 A tire inflation pressure gage (0-100 psig with 1/2 psig graduation).

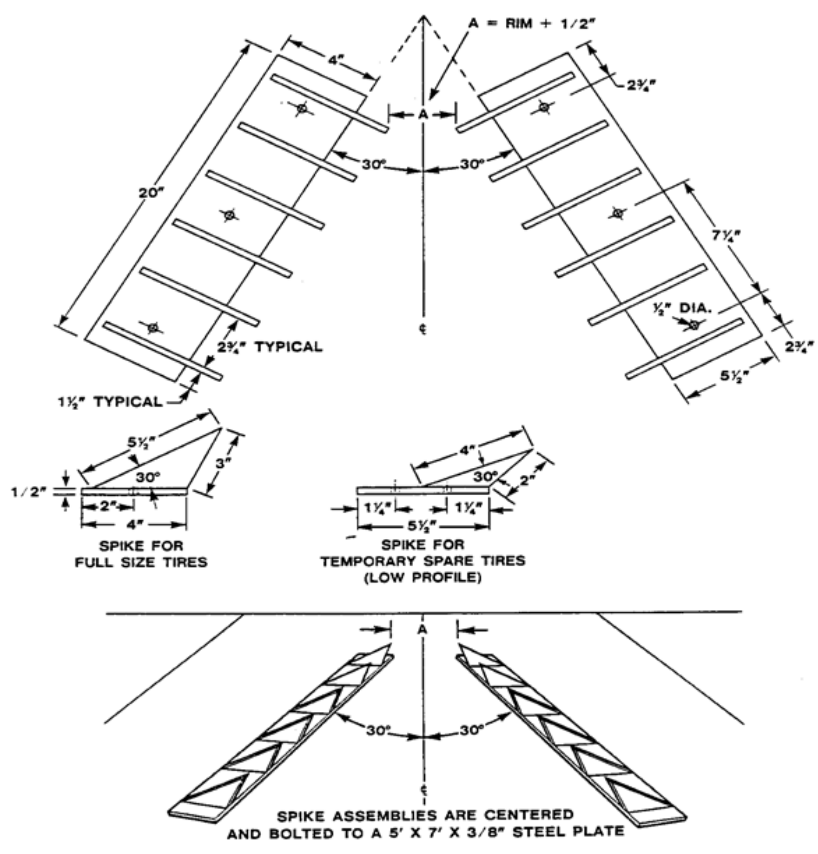
   2.3 Stopwatch or other timing device accurate to 30 seconds per hour.

3. Equipment and Facilities

   3.1 A straight portion of hard surface road sufficient to puncture tire and bring vehicle to a controlled stop.

   3.2 A puncture device for rapidly decreasing inflation pressure while the test vehicle is in motion. A device similar to that in Figure 1 may be used.

Figure 1.



4. Sample Size:
   4.1 Two (2) tires tested at the Right Front and Right Rear positions, or at the Left Front and Left Rear positions.
5. Sample Preparation
   5.1 Mount the accelerometer in the vehicle.
   5.2 Mount identical new unused tires on the two test rims and inflate to the maximum cold placard pressure + 20% (not to exceed the Maximum Sidewall Inflation Pressure, as marked on the tire).
   T-type (mini-spare) tires should be inflated to the cold placard pressure.
   5.3 Ballast the vehicle to its Maximum GVW at Maximum GAWR (the greater of the Front or Rear).
   5.4 Weigh vehicle and record individual tire reactions.
   5.5 To insure that both test rims are the same and are representative of the latest design level approved for production and are fully identified, a representative from the affected design responsible engineering department is required to inspect and provide documentation, including part numbers and pertinent Engineering Change Release Numbers, for all rims being tested and related parts which affect test results, and approve the rims for test. Record tire pressures and vehicle weight. Record the Tire Identification Number (TIN) and the construction number of each tire.
   5.6 A program vehicle or prototype for which the tire is to be fitted is recommended for this test. The use of a similar carryover vehicle or a representative surrogate vehicle and wheel with the same width as the selected wheel, is acceptable with approval from the Global Tire Technical Specialist.
6. Test Conditions
   6.1 Inflation Pressure: Max placard pressure + 20% (not to exceed the Maximum Sidewall Inflation Pressure, as marked on the tire).
   6.2 Load: Test loads should be set to the maximum application GVW the tire is released for usage on.
   6.3 Speed: 97 km/h (60 mph) per FMVSS procedure.
7. Procedure Steps
   The front and rear tires may be punctured simultaneously in one event or separately in two events.
   7.1 After mounting both test tires on one side of the vehicle, warm up the test tires by driving the vehicle for 10 minutes on a hard-surfaced road at 97 km/h (60 mph).
   7.2 Calibrate the speedometer over a measured mile during test tire warm-up.
   7.3 Approach the test location in a straight line at an actual speed of not less than 97 km/h (60 mph). Direct the test tires over the puncture device to cause rapid tire deflation. Immediately after deflation occurs, release the accelerator pedal, apply the brakes, and effect the necessary steering maneuvers to bring the vehicle to a controlled stop at a deceleration rate of 13 to 15 fpsps. Immediately after the stop, determine that rapid air loss has occurred; this means that there is no evidence of air escaping from either tire.
   7.4 Observe and record if either of the tire beads is pulled over the rim flange or if either tire bead slips into the rim well.
8. Procedure Steps / T-Type (Mini-Spare) Tires
   8.1 After mounting one test tire on the front of the vehicle, warm up the test tire by driving the vehicle for 10 minutes on a hard-surfaced road at 80 km/h (50 mph).
   8.2 Calibrate the speedometer over a measured mile during test tire warm-up.
   8.3 Approach the test location in a straight line at an actual speed of not less than 80 km/h (50 mph). Direct the test tire over the puncture device to cause rapid tire deflation. Immediately after deflation occurs, release the accelerator pedal, apply the brakes, and effect the necessary steering maneuvers to bring the vehicle to a controlled stop at a deceleration rate of 13 to 15 fpsps. Immediately after the stop, determine that rapid air loss has occurred; this means that there is no evidence of air escaping from either tire.
   8.4 Observe and record if either of the tire beads is pulled over the rim flange or if either tire bead slips into the rim well.
   8.5 Repeat steps II.E.8.1 through II.E.8.4, mounting the test tire on the rear of the vehicle.
9. Acceptance Criteria:
   9.1 During the controlled stop, each tire must be retained such that both tire beads remain between the inside and outside wheel flanges for the entire 360° of wheel rim.
   9.2 Rapid air loss must have occurred during the test; this means that there was no evidence of air escaping from the tire immediately after the stop.

10. Presentation of Data

10.1 The formal written report must include:

- A copy of the Test Authorization.
- A complete identification of the vehicle, tires, and rims used.
- A complete identification of the test fixture used.
- Total vehicle weight, vehicle weight distribution by tire, and tire inflation pressures.
- Documentation stating that the test samples are at the latest level approved for production.
- A statement signed by the test engineer that the samples were tested according to this tire FS , with a description of any variations from the procedure including required referenced material.
- A statement of whether or not the rim tested met the requirements of FMVSS 110. In the event of unsatisfactory test performance, describe the circumstances.

11. Exceptions/Exclusions:

11.1 Steel carcass tires used on Incomplete Vehicles.

11.2 T-Type (mini spare) tires are run to a maximum speed of 80 km/h (50 mph) for this test procedure.

12. References

12.1 TM-00.00-R-603, Tire Pressure Standards.

12.2 FMVSS 110 and the Ford Acceptance Criteria per the FS .

12.3 FAP03-015, Control, Calibration, and Maintenance of Measurement and Test Equipment.

12.4 FAP03-179, Developing Corporate Engineering Test Procedures.

F. RIDE, HANDLING, BRAKING

1. RIDE, HANDLING, BRAKING – LAB TESTING

1.1 Test Procedures / Descriptions:

1.1.1 TM-04.04-L-415: Flat Trac Variable Lateral Sweep test procedure (applies to all tire types)

1.1.2 TM-04.04-L-419: Flat Trac Brake Sweep test procedure

1.1.3 TM-04.04-E-4340; Tire Footprint Measurement (Ink Method / for reference only)

1.1.4 Snow Model Score

1.1.5 All-Terrain (A/T) Model Score (when applicable)

1.1.6 Stone Pecking Model Score

1.2 Sample Size: Defined in Tire Target Agreement and/or Tire FS

1.3 Acceptance Criteria:

1.3.1 Test tires will meet the targets specified on Tire Target Agreement, or as approved by the GAT Team, Program Vehicle Dynamics, NVH, and Global Tire Manager.

1.4 Exceptions / Exclusions:

1.4.1 Steel Carcass Tires used on Incomplete Vehicles.

2. RIDE, HANDLING, BRAKING – ON VEHICLE

2.1 Test Procedures / Descriptions:

2.1.1 TM-00.00-R-202: This procedure establishes standardized functional image attributes and the method to evaluate and rate the attributes.

2.1.2 TM-00.00-R-205: This procedure is used to subjectively evaluate base vehicle handling.

2.1.3 TM-00.00-R-216: This procedure is used to subjectively evaluate vehicle handling with a dissimilar spare tire/wheel.

2.1.4 TM-00.00-R-487: This procedure is used to subjectively evaluate wet handling

2.1.5 TM-00.00-R-488: This procedure is used to assess the vehicle dynamics performance on a snow-covered surface.

2.1.6 TM-00.00-R-448 Consumer Union lane change and stopping distance

2.1.7 TM-04.04-R-412 Tire performance screening (AMS brake performance test)

2.2 Sample Size: Defined in Tire TA and/or Tire FS

2.3 Description of Test(s):

2.3.1 On-vehicle requirements will be shown in the Tire Contract as delta VER rating difference from a control tire. The tire must meet acceptable handling under all loading conditions.

2.3.2 Functional test ratings will use the 1 - 10 rating scale as defined by TM-00.00-R-201.

2.3.3 All tire submissions for ride, handling and evaluations are to have known uniformity values (consistent with the BIC uniformity values for that particular model year), which are to be provided to the evaluator in a written format.

2.4 Acceptance Criteria:

2.4.1 Tires should meet the subjective VER targets from Tire Target Agreement, or as approved by the GAT Team, Program Vehicle Dynamics, NVH, and Global Tire Manager.

2.4.2 Non-conformance is prohibited and can result in withholding sign-off, definitions of non-conformance are:

a. Tire roll-off
b. Bead-chafing through the first layer of rubber
c. Fabric breaks
d. Tread separations and tread splice openings exposing the fabric
e. Bulges, bubbles, and separations
f. Tread chunking
g. Tread groove cracking
h. Inner liner cracking (no road hazard non-conformance)
i. Deterioration in operating smoothness

2.5 Repeated minor non-conformance will also be a basis for withholding sign-off. Definitions of minor non-conformance are:

a. Road hazard non-conformance's

b. Tread splice openings more than 3 mm (0.1 in.) deep

c. Weather checking

2.6 Exceptions / Exclusions:

2.6.1 Steel Carcass tires used on Incomplete Vehicles.

G. DYNAMIC MECHANICAL ANALYSIS *(formerly TM-04.04-E-4335)*

1. Purpose / Goal of the Test

1.1 Dynamic mechanical analysis (DMA) of elastomers is used to characterize the behavior of the elastomer as a function of temperature, frequency, and amplitude. DMA applies an oscillating force at a specific frequency to the elastomer while monitoring the viscoelastic response of the material through the stiffness and damping characteristics of the material. Storage modulus is the in-phase response to the force and is a measure of the sample's elastic behavior. Loss modulus is the out-of-phase response to the force and is a measure of the viscous behavior of the sample.

The DMA results can be used to predict the in-service performance of elastomeric materials, including tires. For tire materials a temperature sweep is run from -100ºC to 100ºC. Vehicle programs may use the physical properties to estimate attribute and field performance.

Currently the only approved instruments for use on tire materials used on Ford vehicles are the RSAII from Rheometrics and G2 from TA Instruments. Performance requirements for tire materials used on Ford vehicles are based upon output from temperature sweeps on samples tested in tension on those instruments. The approved DMA instruments for use on tire material submissions to Ford are located at Smithers in Akron, OH. Section 3

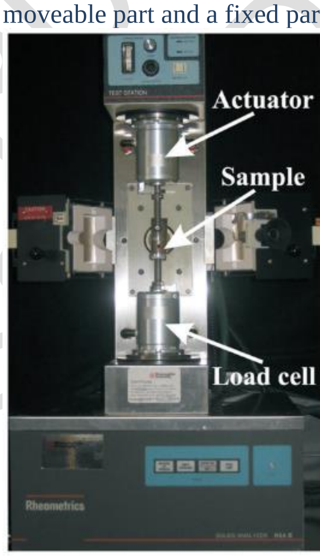
1.2 Instrumentation

1.2.1 The Ford approved instruments for use on tire materials are the RSAII from Rheometrics and G2 from TA Instruments.


Left: Rheometrics RSA II

Right: TA Instruments G2

1.2.2 A large tension clamping assembly is often used for elastomer samples. The clamp assembly consists of a moveable part and a fixed part. The drive motor oscillates the moveable part.



DMA Tension Clamps:

Left: Rheometrics RSA II

Right: TA Instruments G2

1.2.3 A liquid nitrogen dewar is necessary to obtain the temperature range necessary for DMA testing of elastomers. The liquid nitrogen dewar must have a pressure building valve and pressure controller to ensure a constant pressure of liquid nitrogen to the DMA equipment.

Liquid Nitrogen Dewar

1.3 Equipment and Facilities

1.3.1 Standard DMA test equipment will include a furnace capable of maintaining the sample temperature between -110°C and 120°C, a drive motor, a force sensor, and a displacement sensor. The drive motor generates a mechanical oscillation of the sample. The displacement sensor should be close to the sample so that only deformation of the sample is measured. The force sensor measures the dynamic force that is generated by the drive motor.

1.3.2 The DMA test equipment must be able to operate under the following conditions;

- Tension mode @ 0.10% strain
- Temperature Range: -100°C to 100°C
- Strain Frequency of Test: 1 Hz
- Temperature rise of about 0.05°C/sec

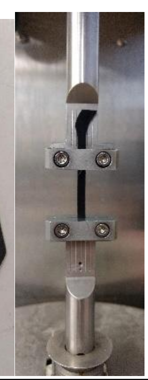
1.3.3 Currently approved equipment for tire materials are located at Smithers in Akron, OH and are the RSAII from Rheometrics and G2 from TA Instruments.

1.4 Sample Preparation

1.4.1 A tool and template must be used to properly cut the elastomer sample out of a cured plaque or part to ensure consistency within samples. Sample geometry is extremely important for DMA because the sample modulus is calculated from the sample stiffness and geometry. Sample stiffness is directly measured using the DMA. Samples for DMA analysis are created using a die punch, either from a plaque or from a slab of material removed from the tire.

Tire tread sample created at Smithers and the die used to create DMA samples

DMA sample created using the die



DMA sample in tension clamp

1.4.2 To accurately measure the mechanical properties using the DMA, the material must be deformed within the linear viscoelastic region of the material, meaning that the stress-strain relationship is linear. In the linear region the material response (modulus, tan delta) is independent of the deformation and the material structure is maintained or unbroken. This allows for an understanding of the impact of polymer system morphology on the material mechanical properties. At large deformations of the sample the modulus becomes a function of strain and decreases with strain as the material structure is disrupted.

Sample geometry impacts stress/strain behavior and stiffness. To ensure consistency in the testing of different samples, the geometry factor is held constant. The geometry factor for a rectangular tension sample is defined as GF = L/(W*T), where L is the length of the test region in the clamp, W is the sample width and T is the sample thickness. The target geometry factor for tension samples is 6-10.

Tension samples are typically 20mm long to ensure appropriate positioning in the clamp. The width and thickness of the sample can vary as long as the target geometry factor is obtained. Sample plaques are typically 2mm thick but samples from parts may be closer to 1mm thick.

It is important that the sample dimensions be consistent at all points across the sample. The width and thickness on a particular sample must be uniform. Samples can be punched out in a press or extracted utilizing other methods that result in a consistent sample size.

A tool to accurately measure sample thickness is also necessary. Any calibrated micrometer or digital caliper is sufficient.

	RSA II	**G2**
Total Sample Length		
Length (between clamps)	21-22 mm	21-22 mm
Width	3.31 mm	3.31 mm
Thickness	0.4 – 1 mm	0.4 – 1 mm

1.5 Procedure Steps

1.5.1 Load the sample into the large tension clamp following the procedure recommended by the DMA manufacturer.

1.5.2 Load the sample into the large clamping assembly in the DMA. Close the DMA furnace.

1.5.3 Run the experiment under the following conditions:

	RSA II	**G2**
Strain frequency	1 Hz	1 Hz
Tension strain	0.10%	0.10%
Heating rate	3°C/min	3°C/min
Min static force	0.01 N	0.01 N
Pretension/initial static force	0.5 N	0.5 N
Max force amplitude	6 N	9 N
Force tracking (axial force/dynamic force)	150%	150%

1.5.4 Additionally, the displacement amplitude must be ≤ 25µm

1.5.5 Electronically output time, E', E", tan delta, temperature, force, displacement

1.5.6 Cool the sample to between –100°C and -110°C. The sample may need to be re-tightened in the large tension clamp at the low temperature due to the glassy state of the sample. Once the temperature has stabilized at the low temperature (after reclamping, if necessary), measurement can begin.

1.5.7 At the conclusion of the test save the resulting curve and data, as well as sample dimensions.

1.6 Data Generated and Formatting of Presentation

1.6.1 Data is to be provided electronically. This includes the raw data file, graphs of E', E", and tan delta versus temperature from -100°C to 100°C, and sample dimensions of the volume of rubber between the clamps prior to testing.

1.7 References

1.7.1 FAP03-179, Developing Verification and Validation Methods

1.7.2 ASTM E1640-13 – Standard Test Method for Assignment of the Glass Transition Temperature by Dynamic Mechanical Analysis

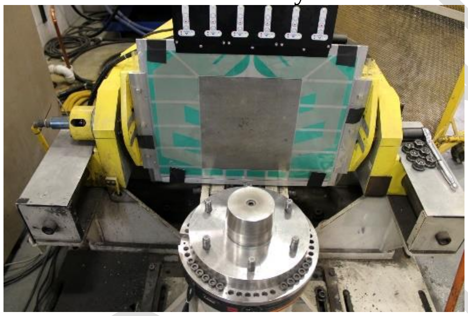
1.7.3 ASTM E2254-13 – Standard Test Method for Storage Modulus Calibration Using Dynamic Mechanical Analyzers

1.7.4 ASTM E2425-11 – Standard Test Method for Loss Modulus Conformance of Dynamic Mechanical Analyzers

1.7.5 ASTM D5026-06 – Standard Test Method for Plastics: Dynamic Mechanical Properties in Tension

H. TIRE FOOTPRINT MEASURMENT (ForceScan)

1. Purpose / Goal of the Test
   1.1 This test is used to capture a tire footprint and then to calculate metrics from this footprint used for attribute assessments. This procedure describes the method using a fixture and a pressure sensitive mat. This method is also called ForceScan.
2. Instrumentation
   2.1 Tread depth is no longer collected when this ForceScan method is used
3. Equipment and Facilities
   3.1 A fixture is required that can both apply the required load and camber angles. The fixture should have feedback for load, camber angle, and inflation pressure.
   3.2 A pressure sensitive mat capable of capturing the pressure from first contact to max load. The recommended system is a TekScan 8405 which has a 1.0 mm x 1.0 mm sense cell size over a 268 mm x 317 mm grid.
   3.3 The sample rate should be set such that the change in load between samples is between 10 N and 100 N.
   3.4 The pressure mat sensitivity should be set such that cells are not saturated during test and at least half of full scale is used during the test. A good approximation is full scan to be 4 times the inflation pressure.
   3.5 The applied load as measured by the fixture, camber angle, and pressure should also be recorded with the data.
4. Sample Preparation
   4.1 Mount tire on specified wheel size. Inflate and set pressure in accordance with TM-00.00-R-603.
   4.2 Allow assembly to stabilize at room conditions prior to test.
   4.3 Determine the test max (Fz_max) load by multiplying the tire max load (from either the tire sidewall or the associated tire industry Yearbook such as ETRTO) by 1.2 (max load will be 20% over the tire capacity).
5. Procedure Steps
   5.1 Mount tire / wheel assembly to ForceScan fixture



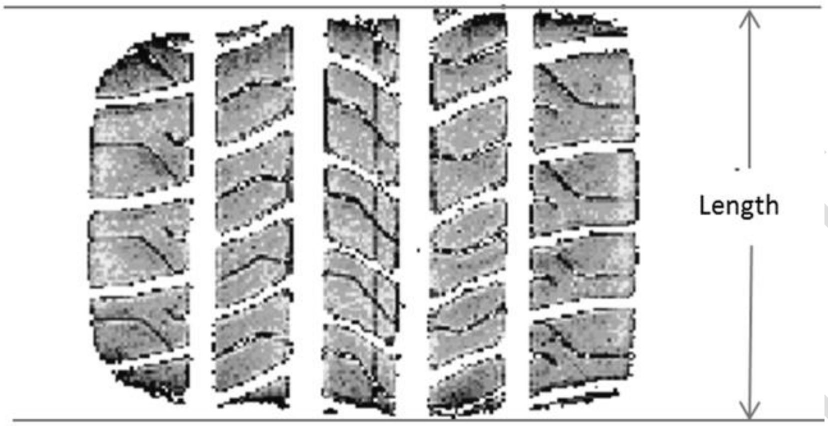
Ford Fixture showing pressure mat and spindle

   5.1.1 Recheck pressure once assembly is mounted on fixture
   5.1.2 Rotate tire such that footprint lines up with the DOT on the tire sidewall. Using a fixed position will allow to repeat the test on the same location of the tire.

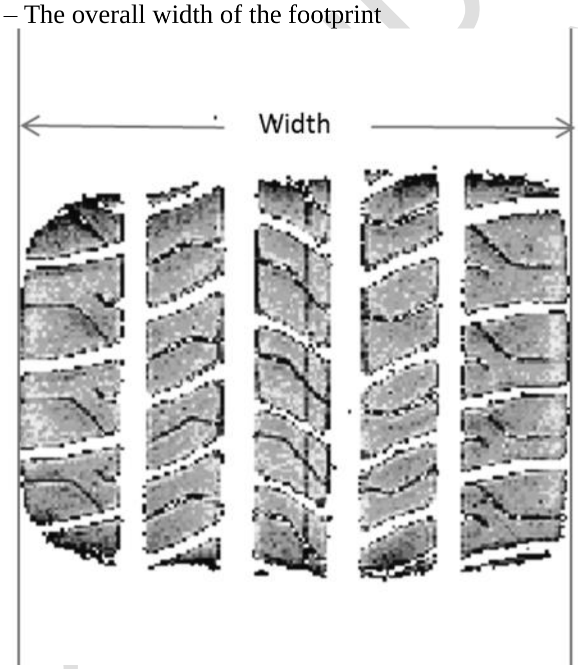
   5.2 Begin the test sequence. All steps should be recorded and included with the data.

Step	Camber	Start Fz	Max Fz
1	-6 deg	No contact	Fz_max
2	-3 deg	No contact	Fz_max
3	-1 deg	No contact	Fz_max
4	0 deg	No contact	Fz_max
5	+1 deg	No contact	Fz_max
6	+3 deg	No contact	Fz_max
7	+6 deg	No contact	Fz_max

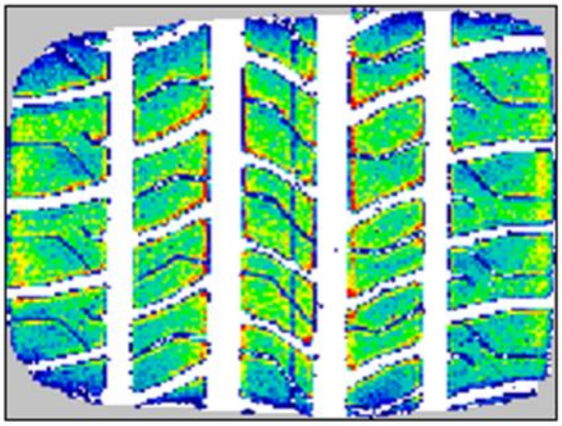
6. General Instruction / Supplemental Information
   - 6.1 None
7. Data Generated & Formatting of Presentation
   - 7.1 The following metrics should be determined from the footprints. These may be done by hand or with image processing software. In the steps above, two prints were taken. Either print may be used in the analysis described in this section.
   - 7.2 Footprint Length – The overall length of the footprint



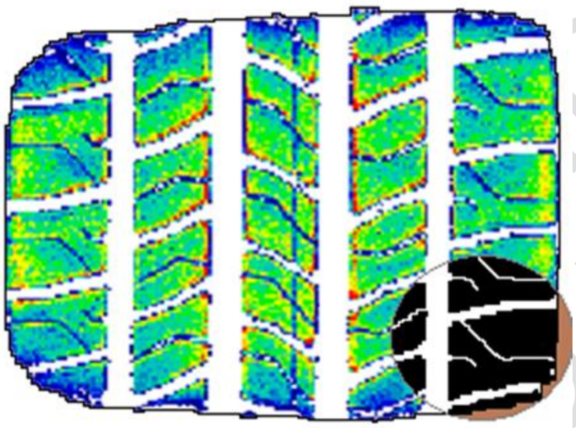
   - 7.3 Footprint Width – The overall width of the footprint



7.4 Corner Area – The total area of the box defined by the length and width minus the area defined by the outline of the footprint. It is the grey area in the picture below.



7.5 Void Area – The area within the outline of the footprint not in contact with the mat. It includes all circumferential and lateral grooves and sipes. In the example below, it is the sum of the white area (the circled area show the tread features in high contrast).



7.6 From the parameters determined in sections II.H.7.2 through II.H.7.5, the following metrics can be calculated:

- Contact Patch Aspect Ratio (CPAR) = Length / Width
- Roundness = Corner Area / (Length * Width)
- Void Ratio = Void Area / (Length * Width – Corner Area)
- Contact Surface Ratio (CSR) = 1 – Void Ratio

I. CTWIST MEASUREMENT (Laser Profilometry)

1. Purpose / Goal of the Test
   1.1 This procedure provides a method used to map the tread of a tire using a laser scan. This scan is conducted on the unloaded tire. The scan can be used to characterize tread features of the tire.
2. Applications
   2.1 Standard applications
      - Assessment of tread features for both attribute performance and error state estimation
      - Supporting production issues

   2.2 COMMONALITY. This is a CONTROL TEST and can be used to qualify components throughout the world. The test may be conducted at any location having the necessary equipment and facilities (see II.I.4). Proposed revisions to this procedure must be submitted per FAP03-179.
3. Instrumentation
   3.1 Laser scanning equipment capable of recording the tread features. The laser should have a spot size no greater than 0.25 mm, a distance resolution of 0.1 mm or better. The spindle should be rated to support tire/wheel assemblies up to 100 kg, and 1000 mm in diameter. The currently approved device is the Circumferential Tread Wear Imaging System (CTWIST) available from the L.S. Starrett Company. Any other device must be pre-approved by Ford engineering.
   3.2 Measuring tape to determine the diameter of the tire from a circumference measurement. A Pi Tape is the preferred device. The resolution should be 1 mm or better on the circumference.
4. Equipment and Facilities
   4.1 Laser scanner and associated computer for data acquisition.
   4.2 Rims in specified sizes.
   4.3 Tire and wheel mounting equipment.
   4.4 Temperature controlled test room capable of maintaining temperature of 20°C +/- 5°C (68°F +/- 8°F)
5. Sample Preparation
   5.1 Test Specimen
      5.1.1 Mount tire on wheel of the specified width. If release rim is not available, contact requesting engineer.
      5.1.2 Inflate to specified pressure.
      5.1.3 No balance required.
      5.1.4 Allow tire to soak at room temperature for 8 hours.
      5.1.5 Reset tire pressure to specified pressure.
6. Procedure Steps
   6.1 Measure and record the outside circumference of the tire along the centerline of the tread using flexible measuring tape. If a circumferential groove is located at the centerline, measure to one side as close to the centerline as possible.
   6.2 Scan tire using laser scanner
      6.2.1 Mount tire to scanner spindle
      6.2.2 Initiate and record scan using the following settings
         - Sampling frequency: 4096 samples per revolution
         - Scan line spacing: 0.5 mm preferred (1.0 mm is acceptable if limited by equipment)

      6.2.3 Check the recording for sample errors. If more than 3.0% of the samples have errors, rotate the laser 90 degrees and rescan the tire. Sample errors occur when reflected signal at bottom of groove is blocked by surrounding tread features. Use scan with few sample errors.
7. General Instruction / Supplemental Information
   7.1 General test information should be documented before and after the data acquisition and at other intermediate times if it is felt that the ambient conditions are significantly changing.

   For each test run, the following information should be documented and recorded on the test log sheet
      - Tire Manufacturer
      - Tire Model (Manufacturer tire line)
      - Tire Size (width, aspect ratio, diameter)
      - Tire pressure
      - Lab ID Number for tire sample

- DOT Code (with date stamp)
- Construction number if submission tire
- Test date
- Wheel size (diameter x width)
- Tire circumference

8. Data Generated & Formatting of Presentation

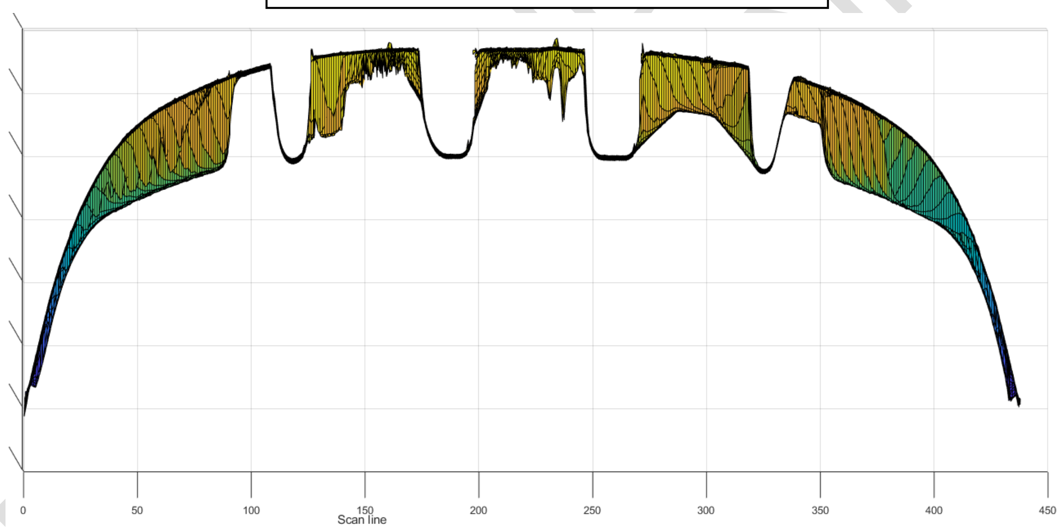
8.1 The following metrics should be determined from the scan.

8.1.1 Definitions

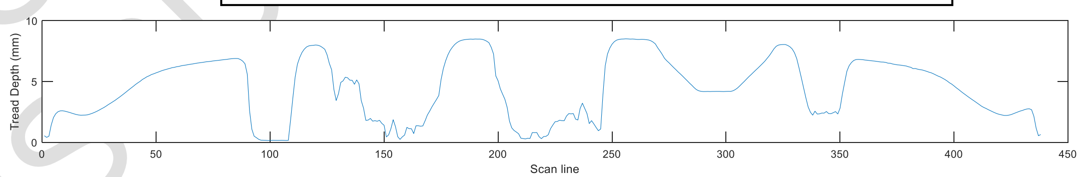
- Scan Point or Sample: A data point along a scan line. The distance between samples is dependent on the diameter of the tire and the number of samples per revolution. There are normally 4096 samples per revolution. With tire OD's between 600 and 900 mm, the sample size is between 0.45 and 0.7 mm.
- Sipe – A lateral tread feature 2 or less samples in size when there are 4096 samples per revolution
- Lateral Groove – A lateral tread feature 3 or more samples in size when there are 4096 samples per revolution
- Circumferential Groove – A groove that runs the circumference of the tire

8.2 Tread Depth is defined as the depth of the 90th percentile tread features. This includes all tread features both the circumferential and lateral grooves. Each scan line is processed and the 90th percentile depth is found for each scan line. This creates a tread depth profile across the tread.

Cross section of a sample tire tread



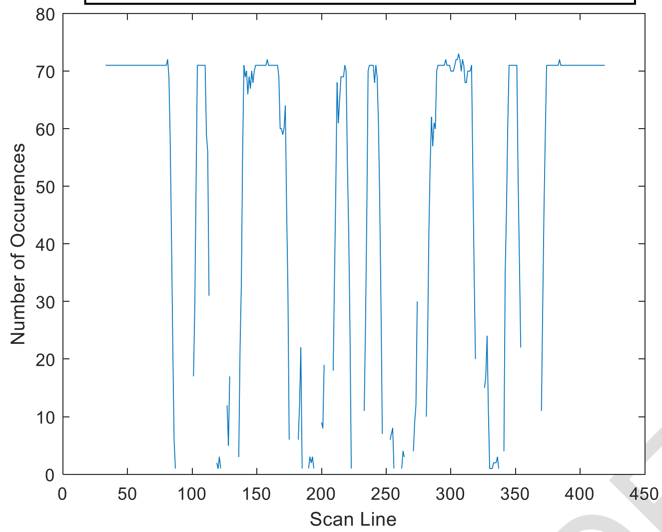
Tread depth by scan line across sample tire tread



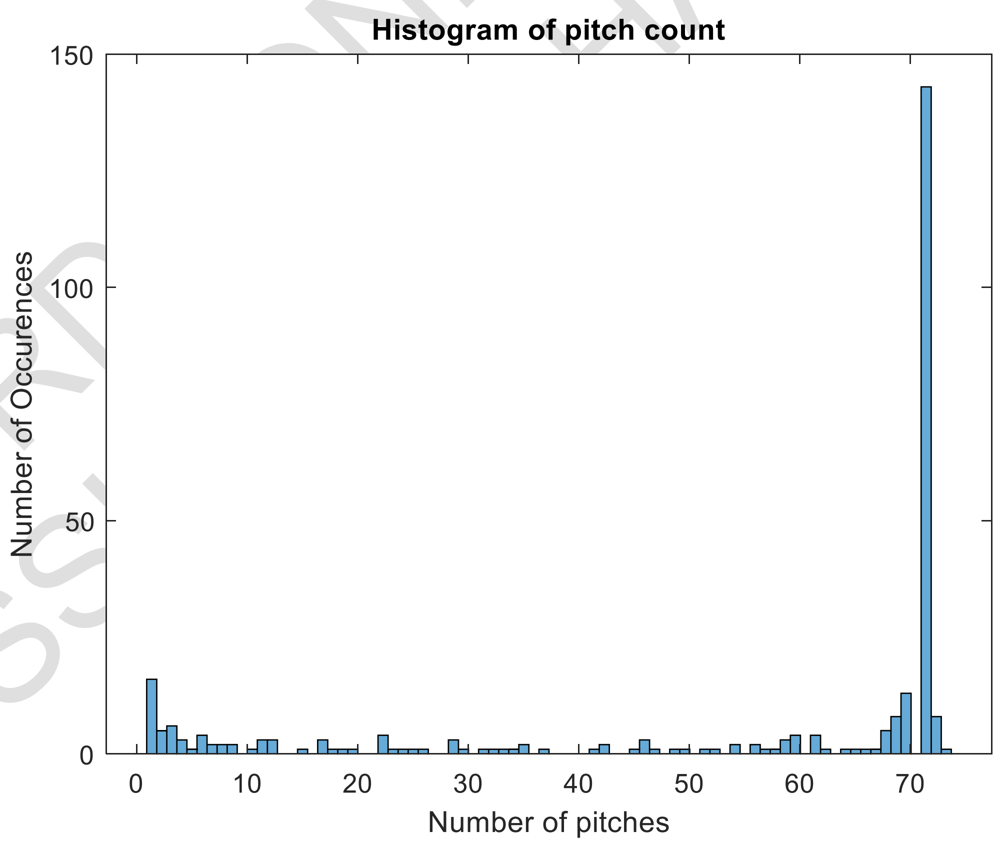
The tread depth is resolved into center and shoulder areas. This is defined by quarters across the tread; the outer quarters are the shoulders, and the center two quarters are the center. The reported tread depth is the 90th percentile across this area of the profile.

8.3 Pitch Count is the most frequent occurrence of lateral grooves across the tread. This is done by counting the number of lateral grooves in each scan line, then finding the Mode of this set. The Mode is the reported number.

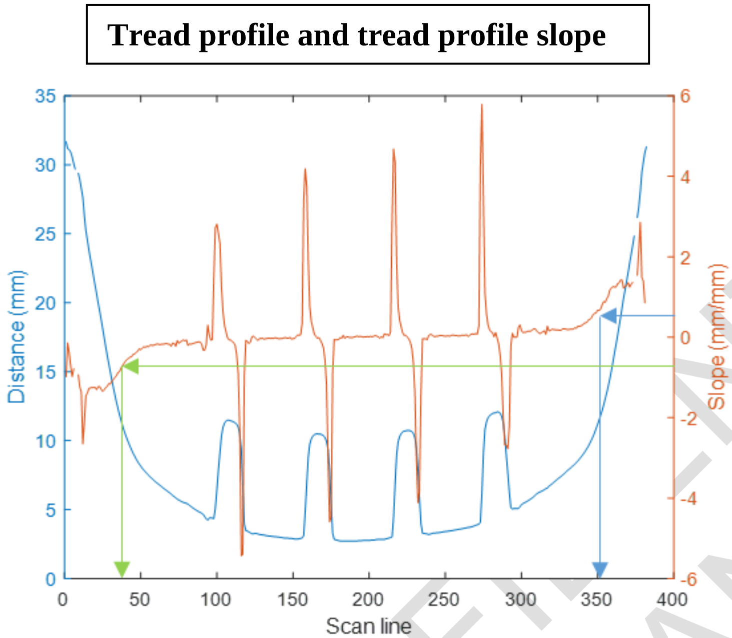
Tread profile and tread profile slope



Histogram of thread pitches

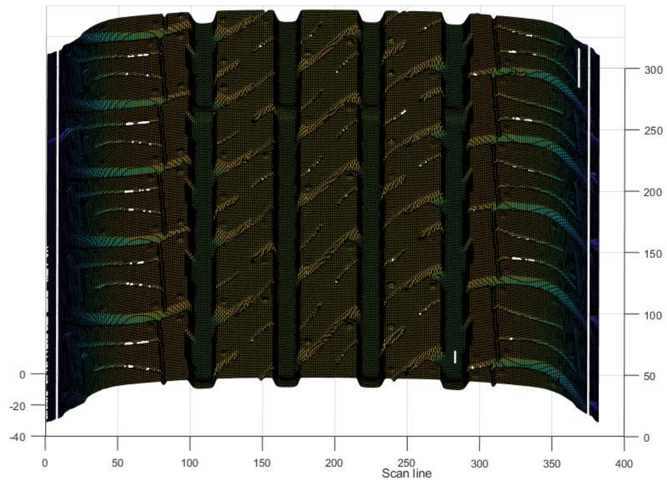


8.4 Rolling Tread Width (RTW) is the width of the portion of the tread that would likely be in contact with the road. The edges of the tread are where the slope crosses 50% (or 0.5 mm per mm) and the RTW is the distance between these edges.

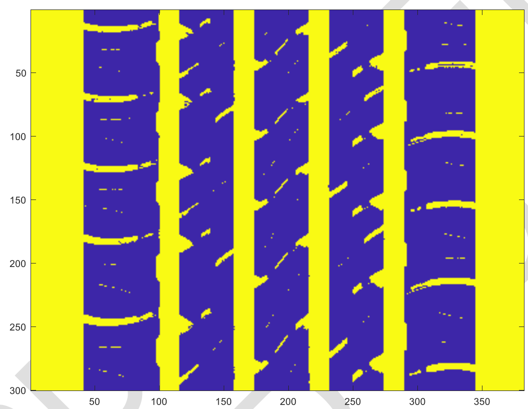


8.5 Median lateral groove width is the width of the lateral grooves 4 mm below the top surface of the tread. The median is defined as the middle in terms of frequency of groove size. A histogram can be used to represent the frequency of occurrence for groove sizes within the RTW. It is to represent the width of the grooves approximately midway down into the grooves. If the groove size is processed in terms of number of pixels (samples), it is converted to mm by multiplying by the circumference and dividing by the total number of samples per revolution (4096 in the case of the Ford machine).

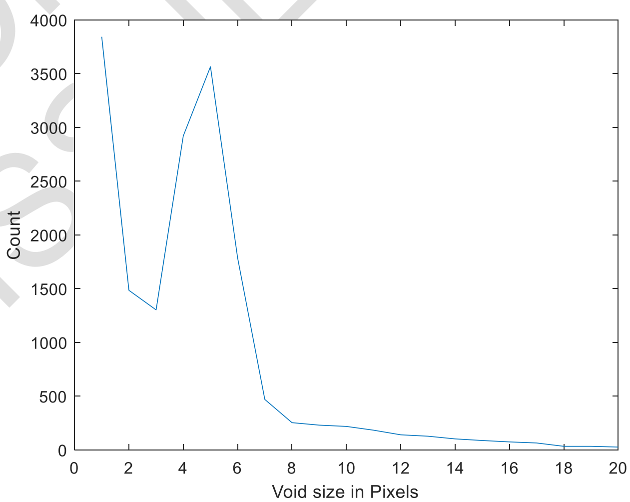
Example image of tread surface



Void map taken at 4 mm below surface

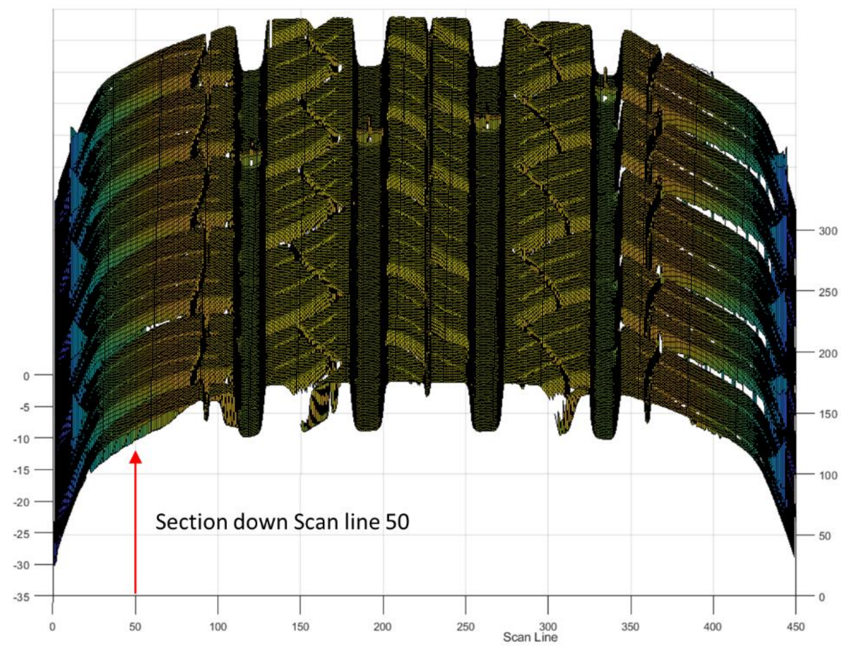


Histogram of void sizes along scan lines

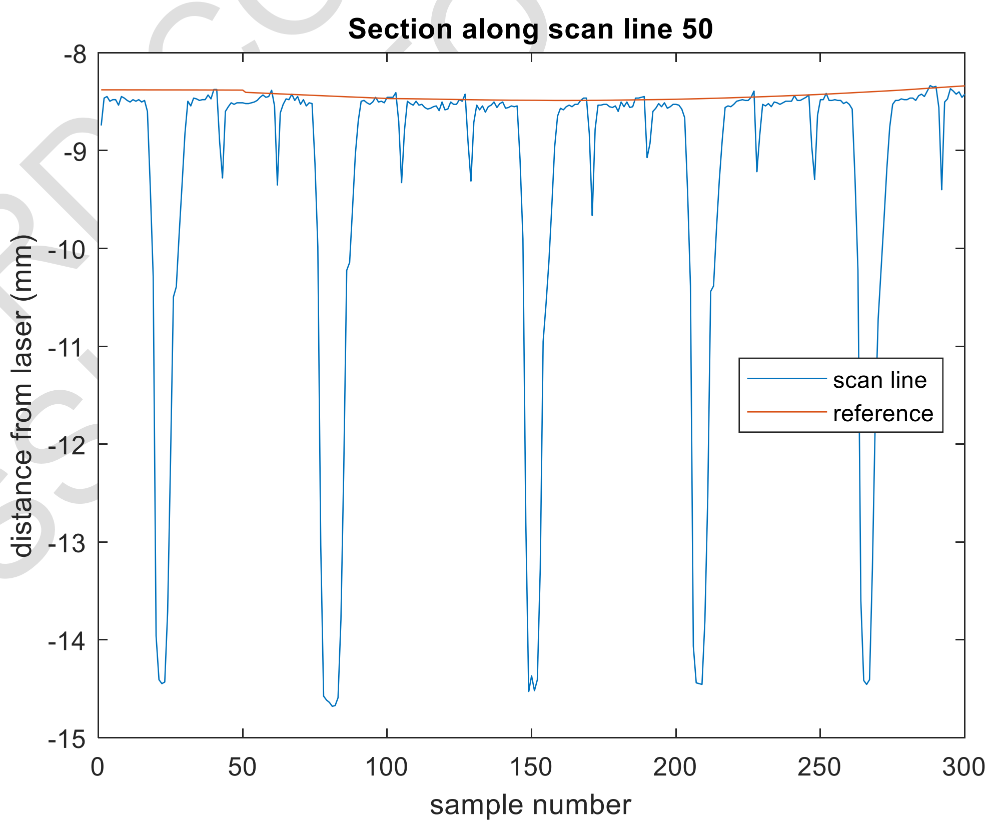


8.6 Void Cross Sectional Area. A lateral groove is a void in the tire tread. The void cross sectional size is used to characterize the frequency and size of the voids in the tread pattern. To find the cross-sectional area of a void , the area of a single void is the numeric integration from the distance between the reference surface and the laser sample from the first sample in the void to the last.

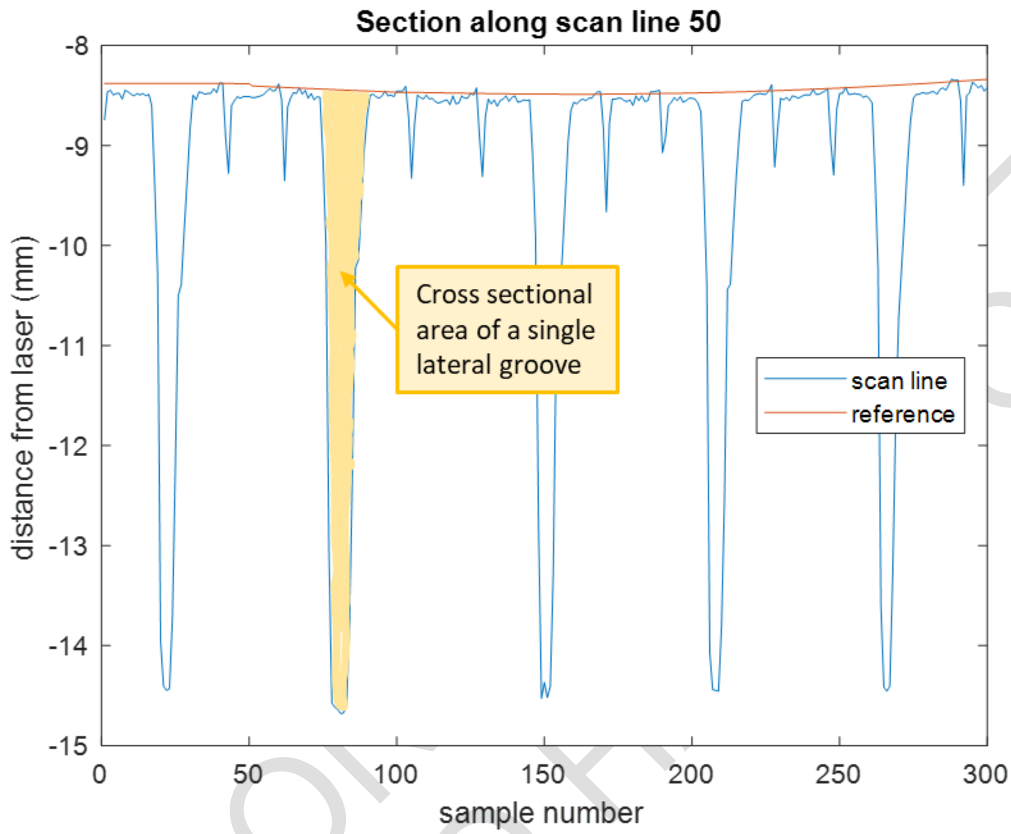
Image of tread arc and location of scan line



Distance from laser to tire along scan line



Cross-sectional area of a lateral groove



9. References

   9.1 SAE J670e, Vehicle Dynamics Terminology

J. NVH REQUIREMENTS

1. NVH LAB TESTING

1.1 Test procedures / description:

1.1.1 TM-04.04-E-10915: Laboratory Dynamometer Procedure for Tire Structure-borne Noise Measurement

1.1.2 TM-04.04-E-10914: Laboratory Dynamometer Procedure for Tire Impact Noise

1.1.3 TM-04.04-E-402: Laboratory Dynamometer Procedure for Tire Radiated Noise Measurement

1.1.4 TM-04.04-C-333: Getting a Modal Model of a Tire Wheel Assembly

1.1.5 TM-00.00-R-466: Vehicle NVH Test Procedure (VNVHTP C07) Vehicle Operation - Cruise - Coarse and Rough Road

1.1.6 TM-00.00-R-469: Vehicle NVH Test Procedure (VNVHTP C26) Vehicle Operation - Cruise - Smooth Road (Tire Noise)

1.2 All tire submissions for lab testing are to have known uniformity values (consistent with the BIC uniformity values for that particular model year), which are to be provided to the evaluator in a written format.

1.3 Acceptance Criteria:

1.3.1 Test results must be equal to or less than the targets specified on Tire TA or as approved by the GAT Team , Program Vehicle Dynamics, NVH, and Global Tire Manager.

1.4 Exceptions / Exclusions:

1.4.1 T-type (mini spares)

1.4.2 Steel Carcass tires used on Incomplete Vehicles.

2. NVH TIRE MODAL MODEL

2.1 Tire Supplier should provide the tire Modal Model for the approved tire construction upon request from the Ford Vehicle Engineering Team. The Tire Modal Model should include a note regarding the tire mass/inertia from FEA and the CAE-Test correlation plot for every model delivered. Ford quality metrics are that all modes below 100 Hz should match within +/-1 Hz on frequency and +/- 10% on amplitude.

K. ELECTRICAL RESISTANCE

1. Test Procedure: WDK Leitlinie 110/1 2 3
2. Sample Size: Two tires, minimum
3. Description of Test: The electrical resistance measurement of an inflated tire-wheel assembly as measured between the wheel and the conducting surface against which the tire is loaded.
4. Acceptance Criteria:
   4.1 Electrical Discharge: The average of two samples must have an electrical resistance less than: $1x10^8$ Ohms
5. Exceptions / Exclusions:
   5.1 T-Type (mini spare)
   5.2 Steel Carcass tires used on Incomplete Vehicles.

L. LAB DURABILITY *(formerly TM-04.04-E-4404)*

1. Purpose / Goal of the Test
   - 1.1 This test analyzes the durability of a loaded tire for potential failure modes. The conditions are based on FMVSS procedures (FMVSS 109 and FMVSS 119) with an additional 48 hours at a load of 130% above maximum load per the applicable tire standard (T&RA, ETRTO, etc…)
   - 1.2 T-Type, P-metric, Metric: reference FMVSS 109
     Light Truck (LT), North American Truck/Bus and EU Commercial: reference FMVSS 119
   - 1.3 COMMONALITY. This is a CONTROL TEST and can be used to qualify vehicles or components throughout the world. The roadwheel portion of the test must be run at a location with tire test equipment (1.7m roadwheel) and experience in tire testing (such as a tire supplier or independent lab that has extensive experience in tire testing). Proposed revisions to this procedure must be submitted per FAP03-179.
2. Equipment and Facilities
   - 2.1 Tire mounting equipment
   - 2.2 1.7m (67") roadwheel with speed and load controls calibrated and run to NHTSA/ASTM standard conditions.
3. Sample Size: Four (4) tires.
4. Sample Preparation
   - 4.1 All samples are prepared according to the following guidelines:
   - 4.2 Mounting:
     - Remove all tags, stickers, tape, stones and other foreign objects from tires
     - Mount tires per industry standard processes making sure no damage occurs to the tire, especially the bead area.
   - 4.3 Rim: Released minimum width rim (qualifies tire for rim widths equal to or greater than the released width)
   - 4.4 Inflation Pressure: Adjust the pressure to specified level (see Table 1) prior to test.
   - 4.5 Thermal Conditioning: Condition the tire/wheel assembly at test temperature (see Table 1 below) for not less than three hours.
5. Description of Test(s):
   - 5.1 Tire Durability to analyze the durability of a loaded tire for potential failure modes per conditions detailed in the TM.
6. Test Conditions: per the table below
7. Procedure Steps
   - 7.1 Readjust and record the tire pressure immediately before testing.
   - 7.2 Conduct the test without interruptions or pressure adjustments at speed, loads and pressure defined in Table 1.
   - 7.3 Test conditions for tires certified per FMVSS 119 that have bias plies or are used on speed-restricted service need to be reviewed with Global Tire Technical Specialist.
   - 7.4 Measure the tire pressure one hour after the procedure has completed.

Load Range	Ambient Temperature	Test Speed (kph)	Test Pressure (kPa)	Load (% of Maximum Load Rating) by Time Step (hours)			
T-type:							
-	**Per FMVSS 109 (no additional hours required)**						
P-metric, Metric:				0 – 4 hours	4 – 10 hours	10 – 34 hours	34 – 82 hours
Standard Load	35° +/- 3° C	80	180	85	90	100	130
Extra Load			220				
LT Metric / European Commercial:				0 – 7 hours	7 – 23 hours	23 – 47 hours	47 – 95 hours
LT Metric Load Range C EU Commercial with Reference Pressure < 400 kPa	35° +/- 3° C	80	Max Load Pressure*	75	97	114	130
LT Metric Load Range D EU Commercial with Reference Pressure 400-500 kPa				75	97	114	130
LT Metric Load Range E EU Commercial with Reference Pressure >500 kPa		64		70	88	106	130
North American Truck/Bus (Radial):				0 – 7 hours	7 – 23 hours	23 – 47 hours	47 – 95 hours
Load Range F	35° C	64	Max Load pressure*	66	84	101	130
Load Range G		56					
Load Range H		48					

**Inflation pressure corresponding to the maximum load rating marked on the tire. Use single maximum load value when the tire is marked with both single and dual maximum load.*

8. Acceptance Criteria:
   8.1 Tire samples must complete the test cycle with zero failures per the defined failure modes in II.L.9 as verified by visual inspection of the test samples.
9. Failure Modes
   - Loss of air
   - Tread separation
   - Cap ply separation
   - Belt ply to cap ply separation (if cap ply present)
   - Belt ply to tread separation (if no cap ply present)
   - Belt ply to belt ply separation
   - Belt ply to body ply separation
   - Cord Separation or Breakage
   - Bead Separation
   - Sidewall Separation
   - Inner Liner Separation
   - The tire pressure, when measured at least one hour after the end of the test, must not be less than the initial pressure.
10. References
    10.1 FAP03-179, Developing Corporate Engineering Test Procedures.
    10.2 FAP03-015, Control, Calibration, and Maintenance of Measurement and Test Equipment.
    10.3 FMVSS 109
    10.4 FMVSS 119
    10.5 FMVSS 139

M. PLYSTEER RESIDUAL ALIGNING TORQUE (PRAT - 0 Conicity)

1. Test Procedure: TM-04.04-L-414
2. Sample Size: 2 tires
3. Description of Test:

3.1 This TM describes recommended guidelines for the Pull Force and Residual Aligning Torque (PRAT) Tire Test performed for Ford Motor Company. The guidelines are intended for use when tire force and moment data is needed for computer simulation of vehicle handling. This tire force and moment test is sufficient for the simulation of vehicle drift/pull characteristics. This guideline applies the SAEJ670e Tire Axis System and associated definitions within its body.

3.2 Inflation Pressure: Specified in test request (placard cold inflation + 20 kpa)

3.3 Test Speed: 100 kph (62 mph)

3.4 Test Temperature: 24° C ± 3° C (75° F ± 5° F)

3.5 Test loads are defined in the Tire TA as defined by Vehicle Dynamics CAE. The primary reporting load for PRAT is at Curb + 2 Passengers (Curb + 2 Pass).

3.5.1 Three (3) test loads are required for load spans less than 4,000 N

3.5.2 Four (4) test loads are required for load spans greater than or equal to 4,000 N.

3.6 Test loads are calculated per the below method.

3.6.1 High load = based on front GAWR (chosen by Vehicle Dynamics CAE Engineer)

3.6.2 Mid load(s) = equally spaced midpoints between low and high

3.6.3 Low load = based on rear curb (chosen by Vehicle Dynamics CAE Engineer)

3.6.4 The curb + 2 passenger PRAT target will also be interpolated. The loads defined in section II.M.3.5 will be set at beginning of program and will remain the loads for the life of the vehicle for PRAT testing purposes.

3.7 Average PRAT must meet the specifications defined in Table 4 for LHD and RHD markets.

3.7.1 Target values defined in Table 4 of the Tire FS are defined at Curb + 2 Pass load. Acceptable design tolerance of target value is ± 1.5 Nm.

3.7.2 Vehicles sourced to LHD markets, as noted on the Tire TA, must use the first column of Table 4. Tires sourced to RHD markets, as noted on the Tire TA, must use the second column of Table 4. Tires sourced as shared for both LHD and RHD markets, as noted on the Tire TA, must use the high volume drive market PRAT.

3.7.3 The PRAT target set on the Tire TA is the PRAT target for a specific tire for the life of the tire, a change in PRAT target will require a new Tire TA to be issued.

3.7.4 The Tire Engineering Handbook must be referenced to assess the proper DV testing required for modified constructions of the same tire for LHD and RHD markets. (e.g. a RHD tire that is mirrored from a LHD tire)

3.8 Ford is responsible to provide the PRAT data for analysis; submission and DV samples are managed by the Global Analytical Tire (GAT) group.

3.9 Supplier Testing Process: Process for supplier research only, not for submission or DV data.

3.9.1 Approved Testing Supplier(s): Smithers Scientific Services, Inc. in Akron Ohio (PRAT@Smithers.com)

3.9.2 Tire Supplier requester will send, via email, a test request pro forma (External PRAT Tire Test Request.xls) to the approved testing location. The test request must contain the respective program Tire D&R Engineer contact information in order for the request to be processed. The subject line of email must state: PRAT TEST REQUEST / xxMY (PROGRAM) / (TIRE SIZE).

3.10 Acceptance Criteria:

3.10.1 Test samples must meet target as indicated on Tire TA.

- Example: PRAT Target as per Tire TA = -4.0 Nm, target range is then -2.5 Nm to -5.5 Nm
  PRAT @ curb + 2 passenger load = -4.4 Nm to a design specification of -4.0 Nm +/- 1.5 Nm
  Tire meets tire target and the audit is conducted at -4.4 +/- 1 Nm

3.11 Exceptions / Exclusions:

3.11.1 T-Type (mini spare)

3.11.2 Steel Carcass tires used on Incomplete Vehicles.

Table 4 – PRAT Specifications

Curb + 2 Pass Load Range (N)	PRAT Target by Market Usage (Nm)		
	LHD Unique	RHD Unique	LHD + RHD Shared
3500 - 4500	- 3	+ 3	- 1
4500 - 5500	- 3.5	+ 3.5	
5500 - 6500	- 4	+ 4	Follow High Volume Market Target if no unique alignment strategy for platform
6500 - 7500	- 4.5	+ 4.5	
7500 - 8500	- 5	+ 5	
8500 - 9500	- 5.5	+ 5.5	
9500 - 10500	- 6	+ 6	
	Assumes tolerance of + / - 1.5 Nm		

<u>Shared LHD & RHD markets:</u>

- -1 + / - 1.5 Nm for:
  - Vehicle lines < 5500 N avg front corner load
  - or > 5500 N loads IF unique alignment strategies are available by market
- Programs > 5500 N loads with no alignment provisions revert back to previous LHD dominant market strategy

N. LOW SPEED TIRE UNIFORMITY SPECIFICATIONS

1. For uniformity specifications, the Tire FS is the master unless superseded by other instruction included in the Tire TA for that model year vehicle line.
2. It is recommended that harmonics, up to and including the 16th harmonic, are measured and recorded (before and after grinding) for all developmental, pre-production, DV, and PPAP builds.
3. Test Procedure:

3.1 North America

3.1.1 P-metric and Metric Tires: SAE J332

3.1.2 LT-Metric, North American Truck / Bus

	Test Load	Inflation Pressure
Load Range C	85% 260 kPa (38 psi) Single Rated	260 kPa (38 psi)
Load Range D	85% 345 kPa (50 psi) Single Rated	345 kPa (50 psi)
Load Range E	85% 345 kPa (50 psi) Single Rated	345 kPa (50 psi)
Steel Carcass	85% 550 kPa (80 psi)* Single Rated	550 kPa (80 psi)*

**In cases where the supplier facility / test equipment is not able to perform the uniformity evaluation at these conditions, an exception must be reviewed and approved by the Global Tire Technical Specialist and Global Tire Manager and the conditions must be detailed in the Tire DVP&R*

3.2 All other regions

3.2.1 Passenger Car Tires: WDK 109/3

3.2.2 Commercial Vehicle Tires: WDK 109/4

4. Sample Size: Measure 100% or, if sampling, demonstrate capability cpk >1.33, ppk >1.67.

4.1 For any tires that use any post-cure process or material application (i.e. sealant, NVH foam, etc…) the uniformity measurement must be taken after the post cure process or material application. Optionally the tire supplier may provide data to Global Tire Manager to demonstrate capability of meeting the BIC uniformity requirements.

5. Description of Test:

5.1 Tires must be measured for radial first harmonic per procedure defined in section II.N.3 and the high point must be marked as indicated below.

5.2 All tire uniformity measurements are to be made in the clockwise and counterclockwise direction. The highest uniformity value, clockwise or counter-clockwise, is to be used to determine whether the tire meets the limits for tire uniformity and reported quarterly.

5.2.1 Exception: All tire uniformity measurements on directional tires may be made in only one direction, which is the intended direction of rotation, as reviewed and agreed by Global Tire Manager.

5.3 Wheel Width: As per procedure defined in section II.N.3.

5.4 Uniformity data including grind percent are to be reported upon request for assessing compliance to specifications, and special studies.

5.5 Uniformity measurements will include:

5.5.1 Radial force variation (RFV) N / (lb)

5.5.2 Radial first harmonic (R1H) N / (lb)

5.5.3 Radial second harmonic (R2H) N / (lb)

5.5.4 Radial third harmonic (R3H) N / (lb)

5.5.5 Lateral force variation (LFV) N / (lb)

5.5.6 Lateral first harmonic (L1H) N / (lb)

5.5.7 Conicity (Con) N / (lb)

5.5.7.1 No Split, Split, or Quad sort will be defined on the Tire TA. If no split is required, then use the split limit and provide one bin of tires. For example, if the split limit is +/- 53 N (12 lb) then tires would not be required to be split into 0 to -53 N and 0 to +53 N but rather one bin of +/- 53 N.

5.5.7.2 Split Sort Conicity Limits 0 to +/- conicity called out in Table 5A, 5B, and 5C

5.5.7.3 Quad Sort Conicity Limits for +/- conicity called out in Table 5A, 5B, and 5C:

a. For 15 lb quad limit: 0 to +/- 7 lb and +/- 7 to +/- 15 lb

b. For 18 lb quad limit: 0 to +/- 9 lb and +/- 9 to +/- 18 lb

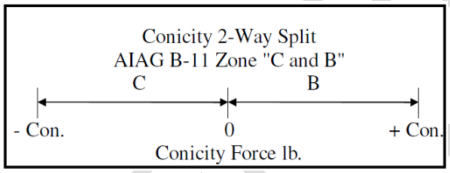
5.5.7.4 Conicity must be clearly marked by a red dot according to WDK Leitlinie 114. This is not applicable for tires with R1H high point marking.

5.5.7.5 Conicity (AIAG B-11) for "No Split" shipment is identified as "A".

5.5.7.6 Conicity markings for two-way (2) split shipment.

a. Negative conicity (-Con < 0): zone "C" tires are marked with one (1) white stripe across the tread surface in 2 locations 180 degrees (+/- 60 degrees) apart

b. Positive conicity (0 < +Con): zone "B" tires are marked with two (2) white stripes across the tread surface in 2 locations 180 degrees (+/- 60 degrees) apart



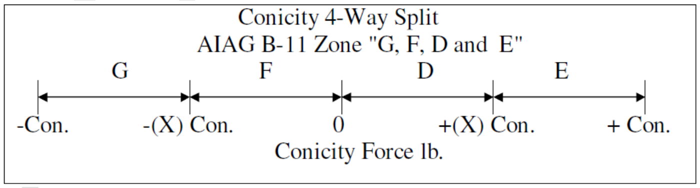
5.5.7.7 Conicity markings for four-way (4) split shipment. (Revised consistent with AIAG B-11 Specification)

a. Positive low conicity (0 < +X): zone "D" tires are marked with two (2) green stripes across the tire tread surface in 2 locations 180 degrees (+/- 60 degrees) apart

b. Positive high conicity (+X < +con): zone "E" tires are marked with two (2) bright red stripes across the tire tread surface in 2 locations, 180 degrees (+/- 60 degrees) apart

c. Negative low conicity (-X < 0): zone "F" tires are marked with one (1) green stripe across the tire tread surface in 2 locations 180 degrees (+/- 60 degrees) apart

d. Negative high conicity (-con < -X): zone "G" tires are marked with one (1) bright red stripe across the tire tread surface in 2 locations, 180 degrees (+/- 60 degrees) apart



5.5.7.8 Conicity AIAG B-11 Specification for "Not Available" is identified as "H"

a. Conicity for "Non-BIC Full-Sized Spares" is identified as "N".

5.5.7.9 Asymmetric tires:

a. Limitation for Conicity variation per Table 5A, 5B, and 5C. Red Spot for marking conicity direction according to WDK Leitlinie 114. The red dot must mark the outside of the tire in order to support tire and wheel assembly. The red dot used for this item must conform to the red dot specification noted in section II.N.5.6 and its subsections.

5.5.7.10 Directional tires:

a. Limitation for Conicity variation per Table 5A, 5B, and 5C. Red dot for marking conicity direction according to WDK Leitlinie 114. Two part numbers are required (left side – right side). A red dot or a green dot must mark the outside of the tire in order to support tire and wheel assembly. Tire production must be sorted for split shipping to the Ford plant. The dot used for this item must conform to the specification noted in section II.N.5.6 and its subsections.

5.5.8 Balance (Bal) gm-cm (in-oz) (No balance patches allowed).
Balance should be based on the Ford Assembly Plant balance method and the Tire Supplier capacity.

5.5.8.1 For dynamic balance, the specification is expressed in grams per flange allowed and is calculated by the following formulae: (.00225 * Tire Mass) * ((Tire radius) / (Rim Radius))

a. Where tire radius = Max Grown Tire O.D. from T&RA or ETRTO / 2

b. Where rim radius = Rim radius (mm) + Flange Height (mm) (Use J flange height of 17.3 mm per ETRTO for all tires)

c. Where tire mass = PPAP mass (g)

d. Example:

i. P235/70R16 tire that weighs 14 kg at PPAP (.00225 * 14000 g) * [(750 mm / 2) / ((405.6 mm / 2) + 17.3 mm)] = 53.7 g per flange allowed.

ii. For gm-cm specification use this formulae: ((.00225 * Tire Mass) * (Tire radius)) / 10

iii. For above example: ((.00225 * 14000 g) * (375 mm)) / 10 = 1181 gm-cm

5.5.8.2 For static balance, the specification is expressed in grams allowed and is calculated by the following formulae: (.0045 * Tire Mass) * (Tire radius / Rim radius)

i. For gm-cm specification use this formulae: ((.0045 * Tire Mass) * (Tire radius)) / 10

ii. For above example: ((.0045 * 14000 g) * (375 mm)) / 10 = 2363 gm-cm

5.5.8.3 Exception

i. In cases for light-weight tires where the calculation defines a balance specification that is below the supplier production capability, an exception must be reviewed and approved by the Global Tire Technical Specialist and Global Tire Manager and the specification must be detailed in the Tire DVP&R. The value of the approved tire balance specification must be consistent with nominal / typical values of the same tire size, or must be calculated using a weight value from a Ford approved tire weight model or the weight target from the approved initial Tire TA.

5.5.9 Center rib run-out (CRRO) mm/(in.). Radial runout must be checked at the center rib (or rib closest to the center) on the tread surface.

5.6 All tires must be marked for radial first harmonic with a Red Dot; this includes tires which also fall under sections II.N.5.5.7.9 and II.N.5.5.7.10. Additionally, directional tires will have a different color sidewall stripe or mark on the Left Hand Side tires than the Right Hand Side tires. Tape stripe is to be 35 mm (1.4 in.) to 80 mm (3.15 in.) long by 6 mm (.25 in.) to 20 mm (.8 in.) wide, {OPTIONAL : the length must include a non-adhesive pull-tab at 12 mm (.50 in.) to 30 mm (1.25 in.) long}, positioned radially 32 mm (1.3 in.) to 38 mm (1.5 in.) from the tire bead heel to the start of the tape stripe. Tape must reflect ultra violet light with a wave length of 530 to 570 nanometers. Tape must not be coated with any foreign material subsequent to installation and must remain affixed to the tire through the in-plant match mounting process. Feasibility of the sidewall stripe or mark color and size should be confirmed with the assembly plant. The marking(s) must be on the outside of the tire in order to support tire and wheel assembly.

5.7 Red Dot Specification: A paint dot diameter of 12 mm +/- 2 mm (Hot Foil) and 18 mm +/- 2 mm (Manual Marker). The red dot color will conform to PANTON® 1795 C (PANTONE® Formula Guide, Page 13.5 C)

5.7.1 A green dot must be used to indicate left side directional tires. The green dot size specification must be equal to the red dot size noted in the above section. The green dot color will conform to PANTONE® 363 C (PANTONE® Formula Guide, Page 40 C). Alternative colors may be used with written approval from Global Tire Manager.

5.7.2 Tolerance between the PANTONE® Standards and the Red/Green Dots to be validated using **CIELAB ΔE* =< 30.0** (= sqrt [$(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2$]) **CIELAB L* >= 40.0**

5.7.3 CIRCUMFERENCIAL LOCATION: Reference located at the maximum - R1H (High Point)

5.7.4 RADIAL PLACEMENT: The dot must be positioned with the bottom of the dot no lower than 12 mm (0.5 in.) up from the rim line and the top of the red dot must not extend beyond the tire sidewall equator. Exceptions must be reviewed with Global Tire Manager and respective plant PVT engineer for approval.

5.7.5 PAINT DOT QUALITY: The paint dot must have minimum 80% coverage at application (ie. Less than 20% black bleed through). The dot must not be coated with any foreign material and visual over spray from sidewall protective paint subsequent to application. The dot must maintain a color within spec and remain affixed to the tire without smearing / smudging / dissolving through the in-plant match mounting process. The preferred method of applying the dots is via hot stamp technology. However, the dots can be manually applied with paint sticks, pens, daubers, etc. as long as the above requirements are met.

5.8 Tire Match Mount Marking - Approved Product List

5.8.1.1 HOT STAMP APPLICATION

Company: Nakayama Trading Co. (Kyowa for US/Overseas, Hori for Japan) Location:Japan Products: Red Tape #TT-700 Green Tape #TT-169 Contact: 81-6-6245-4920	Company: ITW / Veneta Location: South Windsor, CT Products: Red Decal #TBD Green Decal #TBD Contact: (860) 647-5720	Company: Barta Location: Vienna, Austria Product: Red Decal #622.137 011-43-1-894689422 office.usa@barta.at

5.8.2 MANUAL APPLICATION (Oil-based Paint Only)

5.8.2.1 Recommended for repair and low volume application. Manual Red Dot application tool is available from Universal Technologies , Inc.

Company: APV Engineered Coatings Location: Akron, OH Products: Red Marker #D-2861-PDXL Green Marker #D-5865-PDXL (18 mm +/- 2 mm) Contacts: Mike Collart, Eric Rumley, Mike Couchie (330) 773-8911 mcouchie@apv-eng-coatings.com	Company: Matthews (Ink series – M156; System Type – Tampografic; Machine - F90) Location:Via Vittime delle Foibe 20/C; 10036 – Settimo Torinese (TO) Italy Product: I0007-RED-05 (corresponding to Pantone® 1795C) Contact: Dario Titotto, 011-39-011-800-55-70, dario.titotto@marking.it

- NOTE: The above products are approved by Ford for color only. It is the responsibility of the individual Tire Supplier to confirm a product can be used in its manufacturing processes to meet the Ford Tire Functional Specification. Other suppliers may be acceptable but must be approved by Global Tire Manager.

5.9 Acceptance Criteria:

5.9.1 P-Metric and Metric: Table 5A

5.9.2 LT-Metric, Floatation, and European Commercial: Table 5B

5.9.3 For European Commercial tires see Table 5D for ply rating versus load range association

5.9.4 North American Truck/Bus: Table 5C

5.9.5 Maximum tire grinding across the face of the tire must not exceed 1.0 mm (0.04 in.).

5.9.5.1 Grinding must not induce any heel / toe or other irregularities into the tire that may cause noise and / or NVH issues.

TABLE 5A (P-Metric and Metric) Tire Uniformity Specification

	13&14"	15"	16&17"	18-20"	21-24"
RFV	93 N / 21 lb	102 N / 23 lb	111 N / 25 lb	125 N / 28 lb	142 N / 32 lb
R1H	62 N / 14 lb	67 N / 15 lb	71 N / 16 lb	80 N / 18 lb	93 N / 21 lb
R2H	44 N / 10 lb	49 N / 11 lb	54 N / 12 lb	62 N / 14 lb	76 N / 17 lb
LFV	80 N / 18 lb	80 N / 18 lb	80 N / 18 lb	80 N / 18 lb	107 N / 24 lb
L1H	62 N / 14 lb	62 N / 14 lb	71 N / 16 lb	71 N / 16 lb	89 N / 20 lb
Conicity (no split)	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 53 N / ± 12 lb
Conicity (split)	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 53 N / ± 12 lb
Conicity (quad)	± 80 N / ± 18 lb	± 80 N / ± 18 lb	± 80 N / ± 18 lb	± 80 N / ± 18 lb	± 80 N / ± 18 lb
Balance	Calculated as Per Tire FS Section II.N.5.5.8				
CRRO	1.2 mm / 0.047 in	1.2 mm / 0.047 in	1.2 mm / 0.047 in	1.2 mm / 0.047 in	1.2 mm / 0.047 in

TABLE 5B (LT-Metric, Floatation, & European Commercial) Tire Uniformity Specification

	LRC	LRD	LRE & above	Floatation ≤ 33"	Floatation > 33"
RFV	187 N / 42 lb	205 N / 46 lb	222 N / 50 lb	222 N / 50 lb	267 N / 60 lb
R1H	80 N / 18 lb	98 N / 22 lb	116 N / 26 lb	133 N / 30 lb	178 N / 40 lb
R2H	80 N / 18 lb	98 N / 22 lb	116 N / 26 lb	133 N / 30 lb	178 N / 40 lb
LFV	160 N / 36 lb	160 N / 36 lb	160 N / 36 lb	133 N / 30 lb	157 N / 35 lb
L1H	98 N / 22 lb	98 N / 22 lb	98 N / 22 lb	89 N / 20 lb	111 N / 25 lb
Conicity (no split)	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 89 N / ± 20 lb	± 89 N / ± 20 lb
Conicity (split)	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 53 N / ± 12 lb	± 89 N / ± 20 lb	± 89 N / ± 20 lb
Conicity (quad)	± 67 N / ± 15 lb	± 67 N / ± 15 lb	± 67 N / ± 15 lb	NA	NA
Balance	Calculated as Per Tire FS Section II.N.5.5.8				
CRRO	1.3 mm / 0.051 in	1.3 mm / 0.051 in	1.3 mm / 0.051 in	1.3 mm / 0.051 in	1.3 mm / 0.051 in

TABLE 5C (Steel Carcass) Tire Uniformity Specification

	All Load Ranges
RFV	445 N / 100 lb
R1H	311N / 70 lb
R2H	311 N / 70 lb
LFV	222 N / 50 lb
L1H	222 N / 50 lb
Conicity (no split)	± 133 N / ± 30 lb
Conicity (split)	± 133 N / ± 30 lb
Conicity (quad)	NA
Balance	Calculated as Per Tire FS Section II.N.5.5.8
CRRO	2.5 mm / 0.098 in

Table 5D European Commercial Tires – Ply rating versus load range

Ply rating	2	4	6	8	10	12	14	16	18	20	22	24
Load range	A	B	C	D	E	F	G	H	J	L	M	N
ETRTO Pressure Range			325~375 kPa	425~475 kPa	525~575 kPa	625~675 kPa						

5.10 Exceptions / Exclusions:

5.10.1 Tires that are not match mounted to wheels do not have to be marked, but the tires must still meet the uniformity requirements.

5.10.2 Full size spare tires (including dedicated spare tires and reduced tread depth spare tires): Instead of meeting the specifications in tables 5A, 5B, or 5C, the tire must meet the OE Tire Manufacturer's Downstream Uniformity Specifications (aftermarket). Downstream uniformity limits to be provided when quoting.

5.10.2.1 All Non-BIC Full-Sized Spare Tires must be identified with a 1.25" or 2.5" Round Saturn Yellow Sticker high point sticker or other assembly plant approved sticker.

5.10.2.2 RADIAL PLACEMENT: The sticker must be positioned with the bottom of the sticker no lower than 12 mm (0.5 in) up from the rim line. Exceptions must be reviewed with Global Tire Manager and respective plant PVT engineer for approval.

5.10.3 T-type (mini-spare)

6. SIDEWALL UNDULATION

6.1 Test Procedure:

6.1.1 Measure perpendicular sidewall run-out (Figure A) with a dial indicator or equivalent device.

6.1.2 Figure A: Measure at approximate maximum section width.

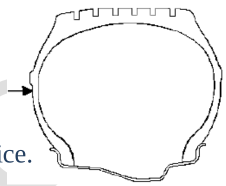


Figure A

6.2 Sample Size: Four (4) tires.

6.3 Description of Test:

6.3.1 The design intent of this specification is to prevent visually unacceptable undulations in the tire's sidewall.

6.3.2 Wheel: Approved wheels or true test wheels.

6.3.3 Pressure:

6.3.3.1 P-metric & metric 205 kPa (30 psi)

6.3.3.2 All other tires 345 kPa (50 psi)

- Note: these are minimum pressures, higher pressures can be used

6.4 The supplier is to measure undulation on a continuing basis, at their plant and provide data to Ford STA on request.

6.5 Acceptance Criteria:

6.5.1 Total undulation run-out in any 50 mm (2 in.) section of the circumference not to exceed: 1.3 mm (.050 in.).

O. TIRE MEASUREMENTS

1. See Table 6 for test methods, conditions, and sample sizes.
2. Tire Weight:

2.1 Test using certified scale with accuracy of ± 0.02 kg (0.05 lb.).

2.2 Sample Size: Four (4) tires.

2.3 Acceptance Criteria: Average tire weight will be equal to or less than the target value specified on the Tire TA.

2.3.1 Note - the weight value on the Tire DVP&R and supporting evidence should be the same as the value that was approved by the Program Vehicle Engineering. This will typically be the value that was reported from the submission tires.

Table 6 – Tire Measurement Standard Reference Conditions

	P-Metric and Metric	LT-Metric and North American Truck/Bus LR-A, B, C, D European Commercial ≤ 400 kPa Ref Pressure	T-Metric (Mini Spare)	LT-Metric and North American Truck/Bus LR-E and higher European Commercial > 400 kPa Ref Pressure
SSR: (one tire sample reported at two pressures on the CDS)				
Inflation Pressure	1 - 240 kPa 2 - 180 kPa	1 - Max per Load Index 2 - 250 kPa	420 kPa	1 - Max per Load Index 2 - 400 kPa
Load	1 - 240 kPa load 2 - 180 kPa load	1 - Max per Load Index 2 - 250 kPa load	420 kPa load	1 - Max per Load Index 2 - 400 kPa load
SSR is the slope of the line drawn from 70 to 100% of the respective tire type / application load and pressure in the above table. Example: 240 kPa SSR = (100% 240 kPa load - 70% 240 kPa load) / (deflection @ 100% 240 kPa load - deflection @ 70% 240 kPa load)				
Deflection curve coefficients are calculated from the set of deflection data developed under the following conditions: Per above table 180 kPa = Measured pressure is 180 kPa ; Load = 0 to 100% of the 180 kPa load Per above table 240 kPa = Measured pressure is 240 kPa ; Load = 0 to 100% of the 240 kPa load Per above table 350 kPa = Measured pressure is 350 kPa ; Load = 0 to 100% of the 350 kPa load Per above table 420 kPa = Measured pressure is 420 kPa ; Load = 0 to 100% of the 420 kPa load Per above table 450 kPa = Measured pressure is 450 kPa ; Load = 0 to 100% of the 450 kPa load Per above table 550 kPa = Measured pressure is 550 kPa ; Load = 0 to 100% of the 550 kPa load				
Solve for A1, A2, and A3 coefficients by minimizing the SSE (Sum Squared Errors) for the equation: Load = Deflection * A1 + A2 * (Deflection ^3) * Pressure [Units are: load = kg, deflection = mm, pressure = kPa]				
Note: Tire supplier to provide the load deflection curves				
SLR: (one tire sample reported at two pressures on the CDS)				
Inflation Pressure	1 - 240 kPa 2 - 180 kPa	1 - Max per Load Index 2 - 250 kPa	420 kPa	1 - Max per Load Index 2 - 400 kPa
Load	1 - 240 kPa load 2 - 180 kPa load	1 - Max per Load Index 2 - 250 kPa load	420 kPa load	1 - Max per Load Index 2 - 400 kPa load
SLR is obtained during the loading at a rate not to exceed 64 mm per minute				
Note: Tire supplier to provide the load vs radius curve where the radius is measured continuously as load is decreased to zero.				
RPK: (four tire samples)				
Inflation Pressure	260 kPa	Max per Load Index	460 kPa	580 kPa
Load	80% 180 kPa load	80% of Max Load	70% 420 kPa load	550 kPa load
Speed	72 kph	72 kph	72 kph	72 kph
RPK is determined per SAE J966 for P-Metric, Metric, or T-Metric tires. SAE J1025 for LT-Metric tires but using the above table for conditions				
Note: Conducting SAE J966 or J1025 on an indoor machine and correcting the results to a flat surface is acceptable if agreed to by Global Tire Manager.				
DLR: (one tire sample)				
Inflation Pressure	240 kPa	1 - Max per Load Index	420 kPa	1 - Max per Load Index
Load	90% 240 kPa load	1 - Max per Load Index	90% 420 kPa load	1 - Max per Load Index
Speed	80 kph	80 kph	80 kph	80 kph
Break In	80 kph for 10 min	80 kph for 10 min	80 kph for 10 min	80 kph for 10 min
Note:	-Break tire in at T&RA or ETRTO maximum load per “Break in” and “Inflation Pressure” conditions noted above -Measure the radius continuously as the tire load is gradually reduced to zero -Data should be plotted to produce a curve from which the DLR can be determined for any load -DLR must be reported out at the load and speed defined in this section			
Tire OD and Section Width: (one tire sample)				
Inflation Pressure	180 kPa (SL) / 220 kPa (XL)	1 - Max per Load Index	420 kPa	1 - Max per Load Index
Test Rim	Application Size & Width	Application Size & Width	Application Size & Width	Application Size & Width
Tire Footprint: (one tire sample / per TM-04.04-E-4340)				
Inflation Pressure	Application Pressure	Application Pressure	Application Pressure	Application Pressure
Test Rim	Application Size & Width	Application Size & Width	Application Size & Width	Application Size & Width
Load	Application Corner Curb	Application Corner Curb	Application Corner Curb	Application Corner Curb
Note	-For vehicles with the same tire at all corners, choose the front axle pressure and load for measurements -For vehicles with different tires on the front or rear axle choose the application axle load and pressure for measurements -For vehicles with split pressures based on speed assume pressures defined for speed less than 160 kph (100 mph) for measurements			

P. ROLLING RESISTANCE

1. Ford Motor Company will specify SAE J2452 for all programs and will use ISO 28580 as the test method for WLTP markets and / or other homologation requirements using ISO 28580.
2. All submission evaluations will use SAE J2452 through the final construction selection.
3. SAE J2452 (issued June 1999)
   - 3.1 Test Procedure: SAE J2452 issued June 1999
   - 3.2 Sample Size: Three (3) tires.
   - 3.3 Description of Test:
     - 3.3.1 Rolling resistance measurements for all P-metric, LT-metric, Metric, and Commercial tires will be conducted on a 1.7 m (67 in.) minimum diameter drum using the stepwise coast down method from SAE J2452 test procedures.
       North American Truck/Bus tires will not run this testing and will only run the ISO 28580 procedure.
   - 3.4 Approved Ford test vendors: Ford Machine in Building #4, testing from other locations is for reference only.
   - 3.5 Ford is responsible to provide the rolling resistance data for analysis; submission and DV samples are managed by the Global Analytical Tire (GAT) group.

Ford Test Conditions

Base Tire Load and Inflation Pressure

Tire Type	Base Inflation pressure, kPa (psi)	Base Load (lbs)
P-metric or metric (LL, SL, or XL)	240 (35)	P-metric / Metric: Load based on tire size at base inflation pressure using Standard Load (SL) condition. Use T&RA for P-metric and ETRTO for metric. Note: All P-metric and metric tires use the Standard Load (SL) index. Do not use the LL, XL, or Reinforced.
LT-metric, Floatation, and Commercial (does not include Truck and Bus	Max Load Pressure from T&RA for LT-Metric and Floatation ETRTO for Commercial	LT-metric / Floatation: T&RA load (based on tire size and load index) at base inflation pressure. Commercial: ETRTO load (based on tire size and load index) at base inflation pressure.

Ford SRC Condition

	P-Metric and Metric (LL, SL, XL, and Reinforced)	Floatation	LT-Metric	Commercial
Load	70% of 240 kPa (35 psi) load	70% of 240 kPa (35 psi) load	70% of 350 kPa (50 psi) load	70% of 350 kPa (50 psi) load
Pressure	262 kPa (38 psi)	262 kPa (38 psi)	372 kPa (54 psi)	372 kPa (54 psi)
Speed	80 kph (50 mph)	80 kph (50 mph)	80 kph (50 mph)	80 kph (50 mph)

BASE INFLATION DEVIATION FROM SAE - noted differences: For P-metric and metric tires base load is 35 psi instead of max load

   - 3.6 Acceptance Criteria:
     - 3.6.1 The average rolling resistance calculated for the engineering samples must be equal to or less than the rolling resistance target specified in the Tire TA.
   - 3.7 Exceptions / Exclusions:
     - 3.7.1 T-Type (mini spare)
     - 3.7.2 Steel Carcass tires used on Incomplete Vehicles do not require rolling resistance measurements unless required to show compliance to regulations.

4. ISO 28580
   4.1 Test Procedure: ISO 28580
   4.2 Sample size: refer to sections II.P.4.4.2 and II.P.4.4.3
   4.3 Description of Test:
      4.3.1 Rolling resistance measurements for all P-metric, Metric, LT-metric, Commercial, and North American Truck/Bus tires will be conducted using the ISO 28580 test method as required for markets where WLTP is in force and / or to satisfy other homologation requirements.
   4.4 WLTP (per Commission Regulation (EU) 2017/1151)
      4.4.1 The Tire Supplier must provide the RR data per the ISO 28580 test method and in accordance with guidelines of measurement and alignment procedures referenced per the WLTP regulation for each tire construction in the submission.
      4.4.2 Testing should be conducted at a facility correlated with the virtual lab per COMMISSION REGULATION (EU) No 1235/2011
      4.4.3 Conformity of the declared Rolling Resistance Class is required as per COMMISSION REGULATION (EU) No 1235/2011
5. Greenhouse Gas (GHG)
   5.1 Ford will require information from tire manufacturers for Greenhouse Gas (GHG) compliance for all LT and Commercial , and NA Truck/Bus tires used on heavy-duty pickups, vans and vocational vehicles. The supplier must run the ISO 28580 testing following the regulation of Part 1037 in the Federal Register
   5.2 The sample size is specified in Part 1037 in the Federal Register
   5.3 The Tire Supplier must provide the RR data for each construction in the submission in accordance to the GHG certification.
   5.4 At DVP&R sign off, and every year for the life of the tire on the program, the Tire Supplier must provide the Tire Attestation Letter that includes the RRC and SLR values obtained running the ISO 28580 test. The Tire Attestation Letter template is located in section II.DD “Appendix E - Tire Attestation Letter for Greenhouse Gas Emissions Regulation template”.

Q. FLAT SPOTTING

1. Short Term (overnight) parking.
   1.1 Acceptance Criteria
      1.1.1 Results from an approved Ford model are equal to or better (less) than the target specified on the Tire TA.
   1.2 Exceptions / Exclusions:
      1.2.1 T-Type (mini spare)
      1.2.2 Winter tires
      1.2.3 Steel Carcass tires used on Incomplete Vehicles
   1.3 For reference only / not required for DV sign-off
      1.3.1 TM-04.04-L-417 (Tire Two Hour Flatspotting Short Term Flat Spotting)

2. Long Term (elevated temperature) parking
   2.1 Acceptance Criteria
      2.1.1 Results from an approved Ford model are equal to or better (less) than the target specified on the Tire TA.
   2.2 Exceptions/Exclusions:
      2.2.1 T-Type (mini spare)
      2.2.2 Winter tires
      2.2.3 Steel Carcass tires used on Incomplete Vehicles
   2.3 For reference only / not required for DV sign-off
      2.3.1 TM-04.04-L-418 (Tire 7-Day Flatspotting Long Term Flat Spotting)

3. Heat-Set Flat Spot
   3.1 To prevent heat-set flat spot, the tire once mounted on the vehicle must not be left standing under any weight at temperatures in excess of 63° C during any manufacturing or repair process.
   3.2 The tires must be thermally protected during paint repair process.
   3.3 Requirement: Tire D&R must receive written confirmation from VO that the assembly plant will adhere to this requirement by keeping the tire under 63° C by limiting the tires exposure or by means of using heat blankets during repair process.

R. TIRE WEAR

1. Overview Information: Applicable to all wear testing noted below.
   - 1.1 Tread wear test is to be used as part of the overall tire DV when specified in the Tire TA. Optionally, Tire Supplier self-certification can be specified in the Tire TA.
   - 1.2 An on-vehicle wear test must be run for all tires that are on programs for the North American market.
     - 1.2.1 If a Tire Supplier has more than one tire on a vehicle program, the supplier need only test one of the tires. The assessment for all other tires will be based on the Ford model and data provided by the supplier indicated in II.R.7.2.4.
     - 1.2.2 On vehicle testing must follow the test procedures outlined in section II.R.2.
     - 1.2.3 Alternative test routes must be approved by Global Tire Manager and indicated in the DVP&R.
     - 1.2.4 Vehicle: Minimum of VP Level application vehicle required. Approval by Global Tire Manager is required for other vehicle or build level being used for testing. An approved Ford test vendor must be used as noted below.
       See Tire TA for instructions for vehicle type with control and test tires and if the lab wear is required.
     - 1.2.5 Approved Ford test vendors:
       - 1.2.5.1 High Q Incorporated, Pearsall, Texas
       - 1.2.5.2 Vehicular Testing Service, Lytle, Texas
       - 1.2.5.3 Texas Test Fleet, Devine, Texas
       - 1.2.5.4 Deutscher Kraft (DEKRA), Narbonne, France
2. Tire Wear Test Procedures for all Vehicles
   - 2.1 North America: per II.R.4
   - 2.2 Europe: per 0
     - 2.2.1 Optionally, Tire Supplier self-certification can be specified in the Tire TA.
   - 2.3 Markets other than Europe or North America:
     - 2.3.1 Tire Supplier self-certification can be specified in the Tire TA.
     - 2.3.2 Test results from either section II.R.2.1 or section II.R.2.2 may also be used to satisfy this section.
   - 2.4 All markets:
     - 2.4.1 An approved Ford Motor Company model will be run on all tires and produce a wear model rating. This model will assess the tires capability in markets approved for sale and be used to determine the tires performance relative to the Tire TA target.
3. Sample Size:
   - 3.1 Passenger Car / Same tire at all four positions: Eight (8) tires total; four (4) test tires, four (4) control tires.
   - 3.2 Passenger Car / Different tires front to rear: Eight (8) tires total: 2 (two) front test tires, 2 (two) rear test tires, 2 (two) front control tires, 2 (two) rear control tires
   - 3.3 Truck (tire used on SRW only, or used on ***both*** SRW and DRW applications): Eight (8) tires total; four (4) test tires, four (4) control tires
     SRW vehicle configuration
   - 3.4 Truck (tire used only on DRW applications): Twelve (12) tires total; six (6) test tires, six (6) control tires, DRW vehicle configuration
   - *3.5* Truck (DRW applications with Steer and Drive tires): 24 (24) tires total; six (6) test Steer tires, six (6) test Drive tires, six (6) control Steer tires, six (6) control Drive tires *, DRW vehicle configuration
     **Optionally, the test may be run using only the control Steer tire or control Drive tire, as agreed to by Global Tire Manager*
   - *3.6* Dedicated full-size spares or reduced tread depth full-size spares: one (1) test tire
     Test position is to be agreed by Tire D&R, Global Tire Technical Specialist, and the Supplier.

4. **Tire Wear Test for Passenger Car and Light Truck** *(formerly TM-04.04-E-406)*

4.1 Purpose / Goal of the Test

4.1.1 This tread wear test is part of the overall tire DV. This test enables tire-wear evaluation for North American passenger car and light trucks. The total duration of the test is approximately 16,800 km (10,500 miles). Output from this test includes tire life projection and irregular wear measurements. Both public highways and city streets are used in and around San Antonio, Texas.

4.1.2 Output from this test includes tire life projection and irregular wear measurement.

4.2 Instrumentation

4.2.1 Black box or Vbox to record fore-aft and lateral acceleration and vehicle speed using standard Ford set-up and usage criteria. The recorded data will be used to calculate the Driver Severity Number (DSN).

4.2.2 Ambient temperature gauge and tire pressure gauges

4.2.3 Electronic tire tread depth gauge

4.2.4 CTWIST (Circumferential Tread Wear Imaging System Two)

4.3 Facilities

4.3.1 IH-35, US083, US-90, SH127, SH173, city streets in San Antonio, Texas.

4.4 Vehicle Preparation

4.4.1 Use the following wheel, inflation, load and vehicle:

- Wheel: Test requester specified (Default: Production rims for control and candidate tires sizes)
- Inflation: Test requester specified (Default: Vehicle Placard Decal)
  Default for Dual Rear Wheel Vehicles (DRW vehicles) will still be Vehicle Placard Decal.
- Load: Test requestor specified (Default: Curb + 1 Driver + Instrumentation)
  Default for DRW vehicles will be 1/2 of vehicle GAWR.
- Vehicle: Test requester specified

4.4.2 The following vehicle preparation procedure should be used.

4.4.2.1 Vehicle is weighed at test initiation - individual wheel position loads are recorded.

4.4.2.2 Alignment is performed with vehicle at test weight.

a. OE tires are typically used for this condition.

b. Alignment is adjusted to center of vehicle manufacturer's specifications, using a tolerance of +/- 0.1 deg total toe and, +/- 0.2 deg Camber, +/- 0.5 deg Caster, if these are adjustable.

c. Alignment is checked every 2100 km (1300 miles). See section II.R.4.5.4 and II.R.4.5.5 for more information.

d. Realign vehicle only if alignment has changed by an amount greater than the vehicle manufacturer's specifications plus above tolerances.

e. Any testing with a toe setting (any wheel position) greater than 0.3° or less than -0.1° is to be considered a testing anomaly, testing should be halted, and the test requestor should be contacted so a plan for forward testing can be determined.

4.4.3 The break-in is the first ~1300 miles of the test, and this is included in the total test mileage.

4.4.4 Install instrumentation as specified in test authorization.

4.4.5 Install test tires and balance tires statically and/or dynamically consistent with vehicle assembly plant process.

4.4.6 Brake Inspection Prior to Wear Testing
To assure that wear test results are not adversely affected by brake concerns, the following procedures should be instituted for any wear test vehicle.

a. Inspect both front and rear for brake drag. Tire and wheel should rotate freely.

i. If evidence of front brake drag is found, this can, in many cases be corrected by greasing the rail slider. If this is not effective the caliper may have to be replaced.

ii. If evidence of rear brake drag is found, first check to assure the parking brake is properly adjusted. If the rear linings need adjusting, the adjusting lever can be accessed through the backing plate. These adjustment procedures are readily available in the applicable Ford Workshop Manual.

b. Inspect both front and rear brakes for excessive wear.

i. If wear is not even side to side, this is a sign that one side is hanging up.

ii. Replace front and/or rear linings as required. Use only approved Ford Motor Company linings. Replace both sides.

c. If either front or rear linings have been replaced, it is necessary to burnish the brakes before starting the wear test.

i. A sufficient burnish can be obtained by:
Running 50 stops from 40 mph to 20 MPH at 12 ft/sec/sec deceleration at 1 mile intervals.

ii. A brake pull or drift during this break-in is a sign that one side of either the front or rear is hanging up. Diagnose and fix.

d. Although front disk brakes are self-adjusting, rear brakes need periodic adjustment, which is generally accomplished by applying the brakes while backing up.

4.5 Operation

4.5.1 Administer the test route in section II.R.4.6 until approximately 16,736 km (10,464 miles) are completed.

4.5.2 Skid depth measurements are to be made approximately every 2100 km (1300 miles). See section II.R.4.5.4 for more information. Tires are measured at ten (10) places (in each groove) around the tire. Measurement locations are to be marked on the tread footprints. Shoulder rib measurements should be taken within 85 to 95 % of foot-print width, if tire design permits.

4.5.3 Tires are not to be rotated during the test, unless specified by the Ford test engineer.

4.5.4 If multiple/like vehicles are being tested, tires are to be moved from vehicle to vehicle per standard tire/vehicle/driver rotation schedule. For example, on a 2-vehicle test, move tires from vehicle to vehicle every 2 shifts and for 3 or more vehicles every shift. When every tire has seen every vehicle in the convoy, tread depths are measured. Thus, the # of vehicles in the convoy determines the exact measurement intervals, alignment check intervals, and test length.

4.5.5 Perform necessary vehicle maintenance in accordance with the appropriate shop manual or, as specified on the test authorization. Alignment settings are to be in accordance with available shop manual recommendations for the vehicle or as specified by the test requestor.

4.5.6 Mileage is accumulated at the rate of 1046 km (654 miles) per day. Mileage accumulation is suspended in the event of rain and/or wet roads.

4.5.7 At the beginning of each test day, all tires must be checked for air pressure and any abnormal conditions. If any tire lacks the remaining tread depth to perform the next segment of driving without reaching the tread indicator bars, the test will be halted. Advise test requestor if any concerns are found.

4.5.8 Operate vehicle at maximum legal speed limits, conditions permitting. The combined Driver Severity Number (DSN), lateral plus 20% fore-aft, should be in the range of 55 to 65. The DSN only needs to be determined for one vehicle in the convoy so long as the final report makes it apparent which tire was in operation for each determined DSN number.

4.5.9 General – Operation to be on dry roads only, no operation in the rain permitted (except to return to shop if caught in a rain storm). Must have 24 hour wait after rain subsides prior to resumption of mileage accumulation.

4.6 Test route

Subject to change due to road construction or closures; revised route to be reviewed with OEM Tire Suppliers and approved by Ford (NAE).

Broadway City Block Loop

INCR MILES		**DESCRIPTION**	**LEFT TURNS**	**RIGHT TURNS**	**STOP SIGN**	**STOP LIGHT**
	R	ON BROADWAY FROM IH35 EXIT				
0.4	R	ON JONES	0	1		1
	L	ON AVE B	1	0		
	L	ON 10TH	1	0		
	R	ON BROADWAY	0	1	1	
	R	ON 9TH	0	1		1
	L	ON AVE B	1	0	1	
	L	ON 8TH	1	0		
	R	ON BROADWAY	0	1		1
	R	ON BROOKLYN	0	1		1
	L	ON AVE B	1	0		
	L	ON 6TH	1	0		
	R	ON BROADWAY	0	1		1
	R	ON MC CULLOUGH	0	1		1
	L	ON AVE B	1	0		
	L	ON 4TH	1	0	1	
		CROSS BROADWAY	0	0		1
	L	ON ALAMO	1	0	1	
	L	ON MCCULLOUGH	1	0		1
	R	ON BROADWAY	0	1	1	
	R	ON 6TH	0	1		1
	L	ON ALAMO	1	0	1	
	L	ON BROOKLYN	1	0		1
	R	ON BROADWAY	0	1		1
	R	ON 8TH	0	1		1
	L	ON ALAMO	1	0	1	
	L	ON 9TH	1	0		
	R	ON BROADWAY	0	1		1
	R	ON 10TH	0	1		
	L	ON ALAMO	1	0	1	
	L	ON JONES	1	0	1	
		CROSS BROADWAY	0	0		1
	R	ON AVE B	0	1	1	
	R	ON 12TH	0	1		
	R	ON BROADWAY	0	1		
2.6	R	ON JONES	0	1		
Exit Path						
0.6		Jones to IH35/El Mira				1

	Left on IH35 S Access	1			1
0.3	Enter IH35 South				2
2.6	One Lap Totals	16	16	10	13
41.6	Sixteen Lap Totals	256	256	160	208
	PLUS Entry/Exit	1	1	0	5
	Totals	**257**	**257**	**160**	**213**

Step	Description		Incre Miles	Cumm Miles
	From IH35N at Exit 133:			
1	North on IH35 into San Antonio			
	Follow US281 North Signs			
	Exit US281 North/Broadway (Exit 158)			
2	Proceed to Broadway St		22.9	22.9
3	Right on Broadway to Jones St		0.4	23.3
4	16 laps 'City Blocks'		41.6	64.9
5	Exit City Blocks to IH35 South		0.9	65.8
6	IH35 South to SH 173		33.3	99.1
	FUEL			
7	North on SH 173 to US 90		21.9	121.0
8	West on US 90 to FM462		1.5	122.5
9	North on FM462 to FM470		26.1	148.6
10	Right on FM470 to SH 16		11.7	160.3
11	Left on SH 16 to FM337		10.8	171.1
12	Left on FM337 to US 83		36.5	207.6
13	Turn Around			
14	US 83 via FM337 to SH 16		36.5	244.1
15	Right on SH 16 to FM470		10.8	254.9
16	Right on FM470 to FM462		11.7	266.6
17	Left on FM462 to US 90		26.1	292.7
18	East on US 90 to SH 173		1.5	294.2
19	South on SH 173 to IH35N		22.0	316.2
20	North on IH35 to Exit 133		10.8	327.0
		Total	327.0	

Route Summary	**Miles**	**%**
Interstate Highway	**67.0**	**20.5%**
State Highway	**68.5**	**20.9%**
Secondary Roads	**148.6**	**45.4%**
City, Including City Blocks (41.6 miles above)	**42.9**	**13.1%**
Totals	**327.0**	**100%**

4.7 Presentation of Data

4.7.1 Test results are to be reported using a standardized tire wear report format containing at least the following sections:

4.7.1.1 General Test Information

i. Tire Brand, Tire ID (construction code or other identifying method), Tire Size, requested per-axle load, requested inflation pressures, requested vehicle (Make, Model Year, brakes, steering, transmission, a/c or not, engine), Route, Test start/end dates, Ambient Temperature range over test dates, test duration (miles and date range), any miles driven under wet conditions.

ii. From this point on, all points within the report which refer to a specific tire are to refer to the tire according to at-least the Tire ID plus identification of tire position. For example, if the Tire ID is "11AC15" for the test tire then it should be referred to in the document as "11AC15-LF", "11AC15-RF", "11AC15-LR", and "11AC15-RR" where "LF" is Left-Front, "RF" is "Right-Front", "LR" is "Left Rear", and "RR" is "Right-Rear".

iii. DRW vehicles will also include "outboard" and "inboard" in identifying tires based on which tires on the rear of the vehicle are towards the vehicles' outboard and which are towards the inboard.

4.7.1.2 Summary of Test Deviations

i. Detail here any deviations from standard test procedure such as, but not limited to: accidents, animal strikes, unusual acceleration/deceleration events, events that damaged the vehicle, events causing operation on un-planned road surfaces (off-road, wet road, detours, etc…)

ii. In the event of unusual acceleration/deceleration please include fore/aft/lateral g-force plots for the event.

iii. In the event of any replacement/repair/adjustment of the test vehicles/tires note the replacements/repairs/adjustments in the final report.

4.7.1.3 Tread Photos with Measurement Points

i. Photos, ink-blots, or equivalent, to show each tire's tread pattern and the points along it which were to be treated as groove 1, 2, 3, etc…

ii. "serial side" and "opposite serial side" is to be identified in this part of the report and kept consistent throughout.

4.7.1.4 Summary of Wear Results

i. Each tire will have its expected miles of tread life shown according to linear regression projection based on tread depth (total tread depth minus the wear bar height) measured at each interval versus the miles traveled. The first ~1300 miles of the test are included in the total test distance, but the initial tread depth measurement is not included in the projection.

ii. At this point and any other within this document, when a wear projection is performed the wear report must state the wear projection goes to the wear bars.

iii. Report will also show the % difference between the fastest wearing groove (any tire position) of the control tire versus the fastest wearing groove (any tire position) of the test tire.

4.7.1.5 Groove Depth Summary ("Smiley Plots")

i. Plots showcasing tread depth of each groove at each measurement interval, with each measurement interval uniquely color-coded. Tread wear indicator height to be shown as additional unique line. Grooves without tread wear indicators will still be shown.

4.7.1.6 Plots of Toe/Camber/Caster Measurements and Tolerances

i. At each alignment check interval, the alignment will be recorded, and the results will be plotted to show the values measured as well as the upper/lower limits for the alignment values.

4.7.1.7 Plots of Projected "Remaining Tread" versus "Miles Traveled" of Fastest Wear Groove

i. Plots will showcase the tread depth remaining before wear out (reaching the wear bar) versus the miles driven. This plot will showcase the linear regression which produces the "Fastest Wearing Groove" value for each tire.

4.7.1.8 Data Charts of Raw Tread Depth Measurements Per-Tire, Per-Groove

i. Charts will showcase all tread depth measurements taken on every tire for each groove.

ii. Charts will specify that the tread depth measurements provided are full-depth measurements (from top of tread to bottom of groove).

4.7.1.9 Data Charts for Ride Height History

i. Charts to display Ride Height of each tire at each measurement interval.

4.7.1.10 Data Charts for Ballast History

i. Charts to display Ballast History of each tire at each measurement interval.

4.7.1.11 Plots of Driver Severity

i. Fore/Aft g-Force Plot for entire test.
ii. Lateral g-Force Plot for entire test.
iii. Speed Profile Plot for entire test.
iv. Data Chart of Ford Driver Severity Number Per-Trip with identification information to know which tire was on the vehicle.

4.8 References

4.8.1 TM-00.00-R-602, Test Mass Standards.
4.8.2 TM-00.00-R-603, Tire Pressure Standards.
4.8.3 TM-00.00-R-605, Speedometer and Odometer Calibration.
4.8.4 Vehicle Driver Sheet; Form No. CEG 2227.
4.8.5 FAP03-015, Control, Calibration, and Maintenance of Measurement and Test Equipment.
4.8.6 FAP03-179, Developing Corporate Engineering Test Procedures.

5. Tire Wear Test for European Passenger Car (European Metric or P-Metric Tires) *(formerly TM-04.04-E-412)*

5.1 Purpose / Goal of the Test

5.1.1 This tread wear test is to be used as part of the overall tyre DV when specified in the tyre contract. Optionally, Tyre Supplier self-certification can be specified in the tyre contract. These tests will use appropriate control vehicles/control tyres. The test enables tire wear evaluation for passenger vehicles in Europe. Total duration of test is 15,000 km (9,321 miles). Output from this test includes tire life projection and irregular wear measurement. Both public highways and city streets are used in and around Narbonne, France. Deutscher Kraft (DEKRA) Test Center, Narbonne, France is the test vendor for performing this test.

5.2 Instrumentation

5.2.1 Black box (to record fore-aft and lateral acceleration and, vehicle speed using standard Ford set-up and usage criteria. This data will be used to calculate Driver Severity Number (DSN))

5.2.2 Temperature thermocouples and pressure gauges

5.2.3 Electronic tire tread depth gauge

5.2.4 CTWIST (Circumferential Tread Wear Imaging System Two)

5.3 Facilities

5.3.1 DEKRA Test Center, Narbonne, France

5.3.2 Highways and public roads in and around Narbonne, France

5.4 Vehicle Preparation

5.4.1 Use the following wheel, inflation and load:

- Wheel: Test requester specified (Default: Production rims for control and candidate tires sizes)
- Inflation: Test requester specified (Default: Vehicle Decal)
- Load: Curb + 1 Driver + Instrumentation + 1 front passenger of 68 kg (150 lb)

5.4.2 The following vehicle preparation procedure should be used:

5.4.2.1 Vehicle is weighed at test initiation - individual wheel positions are recorded.

5.4.2.2 Alignment is performed with vehicle at curb weight (curb with full tank of gas).

a. OE tires are typically used for this condition.

b. Alignment is adjusted to center of vehicle manufacturer's specifications, using a tolerance of +/- 0.1 deg. toe and, +/- 0.2 deg Camber, +/- 0.5 deg Caster, if these are adjustable.

c. Alignment is checked close to 3000 km (1864 mile) interval, dependent on shift cycles and vehicle count.

d. Realign vehicle only if alignment has changed by an amount greater than the OE in service tolerances.

5.4.3 Appropriate break-in procedure, if required, will be designated by the test requester. Break-in mileage must not be included in the test mileage.

5.4.4 Install instrumentation as specified in test authorization.

5.4.5 Install test tires and balance tires statically and/or dynamically consistent with vehicle assembly plant process.

5.5 Operation

5.5.1 Administer the test route in section II.R.5.5.9 until 15,000 km (9,321 miles) are completed.

5.5.2 Skid depth measurements are to be made every 3000 km (1864 miles). Tires are measured at ten (10) places (in each groove) around the tire. Measurement locations are to be marked on the tread foot-prints. Shoulder rib measurements should be taken within 85 to 95 % of foot-print width, if tire design permits.

5.5.3 Tires are not to be rotated during the test, unless specified by the test engineer.

5.5.4 If multiple/like vehicles are being tested, tires are to be moved from vehicle to vehicle at a maximum of 3200 km (1988 mile) intervals, per standard tire/vehicle/driver rotation schedule.

5.5.5 Perform necessary vehicle maintenance in accordance with the appropriate shop manual or, as specified on the test authorization.

5.5.6 Continue the test as well in wet conditions. The accumulated distance in wet conditions must be reported.

5.5.7 At the beginning of each test day, all tires must be checked for air pressure and any abnormal conditions. Advise test engineer if any discrepancies are found.

5.5.8 Operate vehicle at maximum legal speed limits, conditions permitting. Combined Driver Severity Number (DSN), lateral plus 20% fore-aft, should be in the range of 25 to 30.

5.5.9 Test route – DEKRA Circuit 45C 500 Narbonne

5.6 Presentation of Data

5.6.1 Test results are to be reported using the standardized tire wear report format.

5.6.2 Report the following on the vehicle driver sheet:

- All malfunctions and discrepancies during each shift
- Number of test circuits completed during each shift
- All components adjusted, repaired or replaced
- Any change in route, speed or procedure necessitated by weather or road conditions

5.7 References

5.7.1 TM-00.00-R-602, Test Mass Standards.

5.7.2 TM-00.00-R-603, Tire Pressure Standards.

5.7.3 TM-00.00-R-605, Speedometer and Odometer Calibration.

5.7.4 Vehicle Driver Sheet; Form No. CEG 2227.

5.7.5 FAP03-015, Control, Calibration, and Maintenance of Measurement and Test Equipment.

5.7.6 FAP03-179, Developing Corporate Engineering Test Procedures.

6. Markets other than Europe or North America:

6.1 This test will evaluate the on vehicle wear performance of tires in order to make a mileage assessment for expected customer usage on the application vehicle.

7. Acceptance Criteria:

7.1 Test samples will meet the target as indicated on the Tire TA.

7.2 Irregular wear specification:

7.2.1 Pass Car:

7.2.1.1 Max heel/toe 1 mm (.04 in.)

7.2.1.2 Max rib to rib 1 mm (.04 in.)

7.2.1.3 Max. across profile 1.5 mm (.06 in.)

7.2.2 Light Truck:

7.2.2.1 Max heel/toe 1 mm (.04 in.)

7.2.2.2 Max rib to rib 1.5 mm (.06 in.)

7.2.2.3 Max. across profile 2.0 mm (.08 in.)

7.2.3 Tire Worn noise:

7.2.3.1 To be evaluated if the irregular wear specification in II.R.7.2 is not met. Noise is to be evaluated by coasting down from 55 to 5 MPH, engine off, in neutral with the windows up. Evaluate on a 1 to 10 scale, new submission tires before test (new) and again after test (worn). Report only the rating difference for both evaluations on the tire data sheet.

7.2.4 When a model is used, an acceptable result must be from an approved Ford model

- If a model is used which does not contain an irregular wear assessment, then an assessment for irregular wear must be made by using Tire Supplier data. This data will include

a. tread compound analysis
b. footprint analysis
c. tread pattern analysis
d. supplier wear test data
e. field data assessed to the limits per section II.R.7.2

8. Exceptions / Exclusions:

8.1 Steel Carcass tires used on Incomplete Vehicles.

8.2 Winter tires

8.3 T-Type (mini spare): Run 3220 km (2000 miles) of P3-90A (or equivalent) tire wear test (front position only)

8.4 Dedicated Full Size spare or Reduced Tread Depth full size spare: Run 3220 km (2000 miles) of P3-90A (or equivalent) tire wear test. Test position to be determined by the Tire D&R and Global Tire Technical Specialist.

9. Accelerated, Irregular Tire Wear Test for High-Torque RWD Trucks (e.g. Super Duty)

9.1 For reference only. See Test Procedure: TM-04.04-E-4170

9.2 This test to be run as required by Global Tire Technical Specialist

S. GRAVEL ROAD CUT/CHIP

1. Ford Approved Model for Cut/Chip

1.1 Sample Size – per Tire TA

1.2 Description of Test

1.2.1 Uses footprint and compound properties to determine the cut/chip performance of a tire.

1.3 Acceptance Criteria

1.3.1 Cut/chip score equal to or less than the value specified in the Tire TA.

2. **On Vehicle Test: Tire Gravel Road Cut Chip Performance** *(formerly TM-04.04-E-4334)*
Other on-vehicle test procedure that shows equivalent results to TM-04.04-E-4334 can be used with approval from Global Tire Manager. Specific details of the test and correlation results with TM-04.04-E-4334 must be submitted to Global Tire Manager for review and approval.

2.1 Purpose / Goal of the Test

2.1.1 This test method is intended to outline a standard testing procedure to evaluate the on-vehicle gravel cut and chip performance of a tire. The subject tire performance will be compared to a control tire being run on the same test route at the same time. Overall evaluation of the tire performance is completed by visual jury evaluation as assessed by Global Tire Technical Specialist.

2.2 Equipment and Facilities

2.2.1 Uvalde Proving Grounds gravel "K" course, Uvalde, Texas, coordinated by Vehicular Testing Services (VTS) in Lytle, Texas

2.2.2 CEMA Testing Centre, Almeria, Spain, coordinated by Michelin Engineering and Services

2.2.3 Gravel Specification: Concrete Coarse Aggregate, 100% crushed limestone.

2.2.3.1 Source: Edwards Formation

2.2.3.2 Aggregate size per ASTM-33 #57 or TXDOT (1993) item 421 aggregate grade #4 which are equivalent

2.2.4 Two production or surrogate vehicles.

- If a surrogate vehicle is used it must have the same drive configuration as intended for the program and communicated via the test requestor. Example: If program is AWD, surrogate vehicle must be AWD.

2.3 Sample Size:

2.3.1 RWD: Four (4) tires; two (2) test and two (2) control tires tested at the rear position

2.3.2 FWD/AWD: Eight (8) tires; four (4) test and four (4) control tires tested at all vehicle positions.

2.4 Sample Preparation

2.4.1 Use the following wheel, inflation, load and vehicle:

- Wheel: Test requester specified (Default: Production rims for control and candidate tires sizes)
- Inflation Pressure: Test requester specified (Default: Tire placard pressure)
- Load: Driven axle at GAWR, non-driven axle at GVM. For AWD systems, "driven axle" is the one delivering most of the torque.
- Vehicle: Test requester specified

2.5 Vehicle Preparation

2.5.1 Vehicle is weighed at test initiation - individual wheel positions are recorded.

2.5.2 Alignment is performed with vehicle at curb weight.

- OE tires are typically used for this condition.
- Alignment is adjusted to center of vehicle manufacturer's specifications, using a tolerance of +/- 0.1 deg total toe and, +/- 0.2 deg Camber, +/- 0.5 deg Caster, if these are adjustable.

2.5.3 Install test tires and balance tires statically and/or dynamically consistent with vehicle assembly plant process.

2.5.4 Brake Inspection Prior to Wear Testing
To assure that wear test results are not adversely affected by brake concerns, the following procedures should be instituted for any wear test vehicle.

a. Inspect both front and rear for brake drag. Tire and wheel should rotate freely.

i. If evidence of front brake drag is found, this can, in many cases be corrected by greasing the rail slider. If this is not effective the caliper may have to be replaced.

ii. If evidence of rear brake drag is found, first check to assure the parking brake is properly adjusted. If the rear linings need adjusting, the adjusting lever can be accessed through the backing plate. These adjustment procedures are readily available in the applicable Ford Workshop Manual.

b. Inspect both front and rear brakes for excessive wear.

i. If wear is not even side to side, this is a sign that one side is hanging up.

ii. Replace front and/or rear linings as required. Use only approved Ford Motor Company linings. Replace both sides.

c. If either front or rear linings have been replaced, it is necessary to burnish the brakes before starting the wear test.

i. A sufficient burnish can be obtained using the following procedure:
Run 50 stops from 40 MPH to 20 MPH at 12 ft/sec/sec deceleration at 1 mile intervals.

ii. A brake pull or drift during this break-in is a sign that one side of either the front or rear is hanging up. Diagnose and fix.

d. Although front disk brakes are self adjusting, rear brakes need periodic adjustment, which is generally accomplished by applying the brakes while backing up.

2.6 Procedure Steps

2.6.1 Administer the test route in section II.S.2.6.9 until approximately 1600 km (1,000 miles) are completed. *Mileage: 1600 km (1,000 miles) on two vehicles driving concurrently. One vehicle will have specified control tire, the other, new candidate construction tires.*

2.6.2 Tires are not to be rotated during the test, unless specified by the test engineer.

2.6.3 If multiple/like vehicles are being tested, vehicle must be run in tandem on the course but tires will not be moved/rotated between vehicles.

2.6.4 Perform necessary vehicle maintenance in accordance with the appropriate shop manual or, as specified on the test authorization.

2.6.5 Mileage is accumulated at the rate of 270 km (168 miles) per shift. Mileage accumulation is suspended in the event of rain and / or wet surfaces. Course must be dry prior to running.
Note: Test and control tires are not to be driven on surfaces other than gravel. If the vehicle start location is not at the test location, then non-interest tires must be used on the vehicle until the vehicle is at the test location. The test and control tires shall then be installed onto the vehicle at the test location prior to the start of the test. See II.S.2.4 to verify test sample conditions.

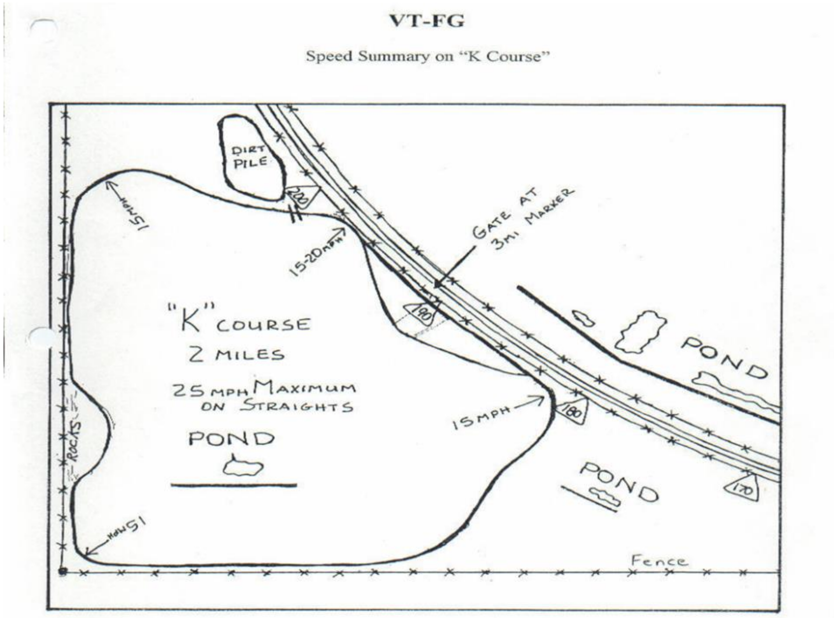
2.6.6 At the beginning of each test day, all tires must be checked for air pressure and any abnormal conditions. Advise test engineer if any discrepancies are found.

2.6.7 Operate vehicle between 24 & 40 kph (15 & 25 mph) per approved track conditions, conditions permitting.

2.6.8 When entering or exiting the test track surface accelerate or decelerate slowly to prevent excessive wheel spin or skidding.

2.6.9 Test route – Uvalde on the gravel “K” course.

Route Description



2.6.10 General Instructions / Supplemental Information

2.6.10.1 Operation to be on dry surface only, no operation in the rain permitted (except to return to shop if caught in a rain storm). Must have 24 hour wait after rain subsides prior to resumption of mileage accumulation.

2.7 Acceptance Criteria

2.7.1 After completion of test, new construction tires must not exhibit, to any greater extent, any chip, chunk, permanent set, tear or other severe deterioration compared to the control tires.

2.7.2 Approval will be based on the appearance of the test and control tires (extent of gravel damage), measurements are not required.

2.8 Data Generated and Formatting of Presentation

2.8.1 A summary of test data including; photos of control tires and test tires at beginning and end of test, mileage summary, vehicle data, control and test tire information, vehicle information, vehicle loading information, vehicle alignment information, and any other pertinent test data.

2.8.2 Disposition of tires will be identified by test requestor. Tires may need to be provided back to test requestor for analysis.

2.9 References

2.9.1 FAP03-179, Developing Corporate Engineering Test Procedures.

2.9.2 TM-00.00-R-602, Test Mass Standards.

2.9.3 TM-00.00-R-603, Tire Pressure Standards.

2.9.4 TM-00.00-R-605, Speedometer and Odometer Calibration.

2.10 Exceptions/Exclusions:

2.10.1 T-Type (mini spare)

2.10.2 Steel Carcass tires used on Incomplete Vehicles.

2.10.3 Winter tires

T. TIRE HYDROCARBON EMISSIONS

1. For reference only / not required for DV sign-off
   1.1 Test Procedure: TM-04.04-E-900

U. TIRE AGING DURABILITY

1. Test Procedure: TM-04.04-E-300 (Tire Aging Test Protocols)
2. Sample Size: Four (4) tires.
3. Description of Test:
   3.1 Test is designed to simulate the durability performance of a tire after 6 years of environmental aging in an Arizona-like climate.
   3.2 This test method includes:
      3.2.1 Mounting tires using a fixed or true wheel and inflating to the maximum sidewall inflation pressure, as marked on the tire, with a mixture of 50/50 N2/O2 gas (capped pressure)
      3.2.2 Aging tires in an air-circulating oven at 65 ° ± 3 ° C (149 ° ± 6 ° F) for 56 days
      3.2.3 Removing the tire inflation pressure every 2 weeks and re-inflating with 50/50 N2/O2 gas
      3.2.4 Removing tires from the oven, removing 50/50 N2/O2 gas mixture and re-inflating with air to the specified pressure
4. Acceptance Criteria:
   4.1 Zero of the tire samples may fail, per the defined failure modes in II.U.4.3, prior to the completion of 34 hours (47 hours for tires using FMVSS 119 criteria)
   4.2 A tire experiencing structural break down after 34 hours (47 hours for tire using FMVSS 119 criteria) constitutes an acceptable/passing result.
   4.3 Failure Modes
      4.3.1 Tread chunking
      4.3.2 Tread separation
      4.3.3 Cap ply separation
      4.3.4 Belt ply to cap ply separation (if cap ply present)
      4.3.5 Belt ply to tread separation (if no cap ply present)
      4.3.6 Belt ply to belt ply separation
      4.3.7 Belt ply to body ply separation
      4.3.8 Cord Separation, or Breakage
      4.3.9 Bead Separation
      4.3.10 Sidewall Separation
      4.3.11 Inner Liner Separation
      4.3.12 Open splices
   4.4 Exceptions/Exclusions:
      4.4.1 Steel Carcass tires used on Incomplete Vehicles

V. STRUCTURAL ACCELERATED TIRE DURABILITY

1. Test Procedure: TM-04.04-E-3281 (Tire Durability - Endurance with Cleat Test Protocols)
2. Sample Size: Two (2) new tires.
3. Description of Test:
   - 3.1 Lab test designed to measure and certify tire's durability robustness.
   - 3.2 Follow guidelines as per TIRE: LOSS OF TIRE FUNCTION DURING DURABILITY [RQT-040400-017807].
   - 3.3 This test method includes:
     - 3.3.1 Mounting tires using a fixed or true wheel and inflating to the maximum sidewall inflation pressure, as marked on the tire, with a mixture of 50/50 N2/O2 gas (capped pressure)
     - 3.3.2 Aging tires in an air-circulating oven at 65 º ± 3 º C (149 º ± 6 º F) for 10 days
     - 3.3.3 Removing tires from the oven and allow to cool at ambient conditions for 3 hours
     - 3.3.4 Re-inflating with air to the specified pressure and maintained with regulated air pressure and run tire on a 1.7 m (67 in) drum with cleat for required accumulations described below.
   - 3.4 Acceptance Criteria:
     - 3.4.1 All Tires: (except mini spares and tires that state on their sidewall they can be retread):
       - 3.4.1.1 4.8 days (115 hours) on the drum. Inspect tire after this test time and note conditions found on test report. If any failure modes defined in section II.V.3.5 are found, stop the test. If no failure modes are found, this is considered a pass. Continue running test until a failure defined in section II.V.3.5 occurs or the tire completes 6.4 days (153 hours) of testing, this will be considered the end of test.
     - 3.4.2 Mini spares:
       - 3.4.2.1 17 hours on the drum. Inspect tire after this test time and note conditions found on test report. If any failure modes defined in section II.V.3.5 are found, stop the test. If no failure modes are found, this is considered a pass. Continue running test until a failure defined in section II.V.3.5 occurs or the tire completes 1 days (24 hours) of testing, this will be considered the end of test.
     - 3.4.3 Tread cracking is allowed for this test but should be noted on the test report. Zero of the tire samples may fail, per the defined failure modes in II.V.3.5 after tire removal from oven and prior to completion of the test time defined in section II.V.3.5.
   - 3.5 Failure Modes
     - 3.5.1 Tread chunking
     - 3.5.2 Tread separation
     - 3.5.3 Cap ply separation
     - 3.5.4 Belt ply to cap ply separation (if cap ply present)
     - 3.5.5 Belt ply to tread separation (if no cap ply present)
     - 3.5.6 Belt ply to belt ply separation
     - 3.5.7 Belt ply to body ply separation
     - 3.5.8 Cord separation or breakage
     - 3.5.9 Bead separation
     - 3.5.10 Sidewall separation
     - 3.5.11 Inner Liner separation
     - 3.5.12 Open splices
   - 3.6 Exceptions/Exclusions:
     - 3.6.1 Steel Carcass tires used on Incomplete Vehicles
     - 3.6.2 Mini spare tires are not required to run TM-04.04-R-307 (Road Damage Robustness - Tire & Wheel Assembly / 100 Square Edge Pot Hole)

W. REVALIDATION REQUIREMENTS and SREA's

1. The manufacturing source, Global Tire Manager, Global Tire Technical Specialist, Tire D&R, and STA will jointly determine which potential changes to the process, components, materials, or material sources would have significant impact on the product's function, performance, durability or appearance. The supplier will describe these conditions in the FMEA, along with either (1) the revalidation plan that would be followed in each case, or (2) a provision to submit an amended FMEA for approval if any of those process changes are planned. No change to processing may be allowed without prior engineering approval of the process changes and the attendant Control Plan changes.
2. If a Tire Supplier desires to either move the production of a previously approved tire construction to another manufacturing location, or add additional manufacturing locations to a previously approved tire construction, a completed Supplier Request for Engineering Approval (SREA) must be submitted to the affected Ford responsible engineer.
3. Changes that require revalidation of the tire including, but not limited to, process, components, materials, or material sources, or for a move / addition of a production location, must be presented to the Global Tire Technical Specialist and Tire D&R using the latest version of the Tire Submission Summary (TSS). The TSS and Ford Tire Engineering Handbook will be used along with an SREA to define the revalidation requirements. The Ford Tire Engineering Handbook shows the recommended minimum requirements for revalidation, however DVP&R items may be added or removed with agreement from the Global Tire Manager or Global Tire Technical Specialist.
   - 3.1 Additionally, the TSS must show a comparison of the new tire and previously approved tire. It should demonstrate that the new tire is equivalent to the previously approved tire for all attributes, both lab and on-vehicle.
   - 3.2 Changes to a tire should follow the standard development and approval process. This includes assessment and approval of the tire by Ford Vehicle Engineering / Vehicle Dynamics (both on-vehicle and lab testing, as required), and a completed DVP&R with all of the required signatures.
4. All Federal safety requirements (e.g. FMVSS, ECE R30, and other local market regulations) must be met.
5. The Tire Supplier cannot begin shipping the revalidated tire or tires from the new location until the Global Tire Manager, Global Tire Technical Specialist, Tire D&R Engineer, and Tire Supplier have
   - 5.1 Reviewed the TSS requested in section II.W.3
   - 5.2 Reviewed the Tire DFMEA and control plan
   - 5.3 Signed the SREA
   - 5.4 Completed the re-DV and re-PPAP requirements.
   - 5.5 Upon these signatures, the new plant code will be added to the Tire CDS and re-signed by all designated to approve.
6. The tire supplier may request changes to a tire with an SREA and must follow the Global SREA Process. In addition to any approvals required per the Global SREA Process, the SREA must always be approved by Ford STA, Ford PD (Tire D&R Engineer), Global Tire Technical Specialist, and Global Tire Manager. The SREA must be fully approved prior to the supplier making any changes to the tire.

X. REQUIREMENTS FOR POLICE VEHICLES (Pursuit rated in North America)

1. Tires released for use on Police programs must meet all applicable FS requirements
2. Police tires will also be evaluated by two different Police agencies on three tracks:
   - 2.1 Auto Club Raceway (Fontana, CA)
   - 2.2 Grattan Raceway (Belding, MI)
   - 2.3 FCA Chelsea Proving Grounds (Chelsea, MI)
3. Police tires are expected to pass the criteria established by the Michigan State Police (MSP) and the Los Angeles County Sheriff's Department (LASD)
4. MSP testing is conducted at the FCA Chelsea Proving Grounds and Grattan Raceway.
   - 4.1 Evaluations at the FCA Chelsea Proving Grounds include:
     - 4.1.1 Acceleration performance
     - 4.1.2 Maximum vehicle speed
     - 4.1.3 60 mph – 0 mph stopping distance (using 20 stops)
5. Evaluations at the Grattan Raceway include:
   - 5.1 Vehicle Dynamics (High Speed Handling) - This consists of 4 drivers each completing 8 hot laps for a total of 32 laps.
   - 5.2 Failure occurs if a tire is found to be chunking, has a loss of air or exposed belts, or has the tread worn off the tire during or at the conclusion of testing.
6. LASD testing is conducted at the Auto Club Speedway.
7. Evaluations at the Auto Club Speedway include:
   - 7.1 Acceleration performance
   - 7.2 Vehicle Dynamics
   - 7.3 Braking
   - 7.4 Pursuit course
   - 7.5 Each tire must pass through the vehicle dynamics test and recorded brake test in order to be certified for high-speed handling law enforcement use. Currently the vehicle dynamics testing involves successfully completing 32 laps around a high-speed performance track, with 4 drivers each completing 8 laps. The brake test is conducted IMMEDIATELY after the 32 lap session, which means that the tire is worn and hot, and the brakes are hot. Tires and brakes can then be changed for the City Pursuit test (two drivers, two laps each).
   - 7.6 Failure occurs if a tire is found to be chunking, has a loss of air or exposed belts, or has the tread worn off the tire during or at the conclusion of testing.
8. Evaluations of submission tires will be completed by Ford, jointly with the tire supplier, at the Police facilities prior to any evaluations conducted by either of the Police agencies.

Y. COMPILATION OF REFERENCE DOCUMENTS

Item	Title	Date
TM-00.00-R-201	Vehicle and Attribute Rating System	Latest Release Level
TM-00.00-R-202	Functional Attribute Rating of Vehicles	Latest Release Level
TM-00.00-R-205	Base Vehicle Handling Safety	Latest Release Level
TM-00.00-R-216	Dissimilar Spare Tire/Wheel Handling Safety	Latest Release Level
TM-00.00-R-448	Consumer Union Lane Change and Stopping Distance	Latest Release Level
TM-00.00-R-466	Vehicle NVH Test Procedure (VNVHTP C07) Vehicle Operation - Cruise - Coarse & Rough Road	Latest Release Level
TM-00.00-R-469	Vehicle NVH Test Procedure (VNVHTP #C26) Vehicle Operation - Cruise - Smooth Road (Tire Noise)	Latest Release Level
TM-00.00-R-487	Wet Handling Procedure	Latest Release Level
TM-00.00-R-488	Snow Handling Procedure	Latest Release Level
TM-04.04-C-333	Getting a Modal Model of a Tire_Wheel Assembly	Latest Release Level
TM-04.04-E-300	Tire Aging Test Protocols	Latest Release Level
TM-04.04-E-402	Laboratory Dynamometer Procedure for Tire Radiated Noise Measurement	Latest Release Level
TM-04.04-E-406	Tire Wear Test for Passenger Car and Light Truck	Latest Release Level
TM-04.04-E-412	Tire Wear Test for European Passenger Car (European Metric or P-Metric Tires)	Latest Release Level
TM-04.04-E-414	Gage R&R for Tire Uniformity Measurement at Low Speed	Latest Release Level
TM-04.04-E-900	Tire Hydrocarbon Evaporative Emissions Determination (Mini-SHED Evaporative Emissions Test)	Latest Release Level
TM-04.04-E-3281	Tire Durability – Endurance with Cleat Test Protocols	Latest Release Level
TM-04.04-E-4170	Accelerated, Irregular Tire Wear Test for High-Torque RWD Trucks	Latest Release Level
TM-04.04-E-4334	Tire Gravel Road Cut & Chip Performance ~~Gravel Road Cut/Chip (1K)~~	Latest Release Level
TM-04.04-E-4335	Dynamic Mechanical Analysis of Elastomers	
TM-04.04-E-4340	Tire Footprint Measurement	Latest Release Level
TM-04.04-E-4404	Tire Lab Durability (FMVSS plus 48 hours)	Latest Release Level
TM-04.04-E-10914	Laboratory Dynamometer Procedure for Tire Impact Noise	
TM-04.04-E-10915	Laboratory Dynamometer Procedure for Tire Structure-borne Noise Measurement	
TM-04.04-L-414	Vehicle Dynamics Tire Test, Pull Force and Residual Aligning Torque Test Procedure	Latest Release Level
TM-04.04-L-415	Vehicle Dynamics Tire Test, Flat Trac® Variable Lateral Sweep Test Procedure	Latest Release Level
TM-04.04-L-417	Tire Two Hour Flatspotting Short Term Flat Spotting	Latest Release Level
TM-04.04-L-418	Tire 7-Day Flatspotting Long Term Flat Spotting	Latest Release Level
TM-04.04-L-419	Vehicle Dynamics Tire Test, Flat Trac Brake Sweep Test Procedure	Latest Release Level
TM-04.04-L-4150	Lab High Speed – P-metric & Metric Tires	Latest Release Level
TM-04.04-L-4151	Lab High Speed – LT-Metric, North American Truck/Bus, European Commercial, and European Industrial Tires	Latest Release Level
TM-04.04-R-402	Tire Rim Roll-Off Test	Latest Release Level
TM-04.04-R-412	Tire Performance Screening (AMS Brake Performance Test)	Latest Release Level
TM-04.04-R-802-US	Passenger Car Tire Retention Test	Latest Release Level
VDT-LOP 23.13	FoE procedure (ON VEHICLE TEST) *Not referenced anywhere in the document*	Latest Release Level
SAE J332	Testing Machines for Measuring the Uniformity of Passenger Car and Light Truck Tires	2002-11-01
SAE J670	Vehicle Dynamics Terminology	2008-01-24
SAE J2452	Stepwise Coastdown Methodology for Measuring Tire Rolling Resistance	1999-06-01
ISO 28580	Passenger Car Rolling Resistance	8/1/1992
WDK 109-2 / 3 / 4 [2]	Uniformity Measurements of Pneumatic Tires	Latest Release Level
WDK Leitlinie 110/1 2 3	Test Procedure for Determining the Electrical Resistance of Pneumatic Tires	Feb-97
FMVSS 109	Federal Motor Vehicle Safety Standard. (49 CFR PART 571) New Pneumatic and Certain Specialty Tires	Vol. 78 No. 12 – 17.01.2013

FMVSS 110	Federal Motor Vehicle Safety Standard. (49 CFR PART 571) Tire Selection and Rims and Motor Home / Recreation Vehicle Trailer Load Carrying Capacity Information for Motor Vehicles with a GVWR of 4,536 Kilograms (10,000 Pounds) or Less	Vol. 81 No.217 - 09.11.2016
FMVSS 119	Federal Motor Vehicle Safety Standard. (49 CFR PART 571) New Pneumatic Tires for Motor Vehicles with a GVWR of More Than 4,536 Kilograms (10,000 Pounds) and Motorcycles	Vol. 77 No.173 – 06.09.2012
FMVSS 139	Federal Motor Vehicle Safety Standard. (49 CFR PART 571) New Pneumatic Radial Tires For Light Vehicles	Vol. 77 No. 4 – 06.01.2012
Tire Identification and Recordkeeping	NHTSA 49 CFR Part 574 – Tire Identification and Recordkeeping	13-April-2015
ECE R30	Uniform Provisions Concerning the Approval of Pneumatic Tyres for Motor Vehicles and their Trailers	22-February-2017
B-11	Automotive Industry Action Group (AIAG) for Tire & Wheel Identification Label Standard	08-May-2001

[2] For WDK procedures, contact: Kautschuk-Wirtschaftsfoerderungs-GmbH, Postfach 90 03 60, 60443 Frankfurt, GERMANY

Z. Appendix A - Change Log (Note: Change log from prior versions have been removed, only current changes are shown.)

Date	Rev Level	Revision Description	Section #	Proposed Implementation*
	FS Version 5	Added abbreviations for RQT and TSS	Abbreviations	1
		Updated signatures for the TA	I.B	1
		Added "Requiremens (RQT's)"	I.C.9	1
		Added Exception for directional tires	I.E.3	1
		Updated SC/CC list. Revised Rolling Resistance to YC / CC	Table 2 – SC/CC Items	1
		Added "(S)" to item II.C. Revised II.Q.1 and II.Q.2 to "per Model" (was "2")	Table 3 –FS Summ	1
		Clarified inflation pressure for Tire OD and SW (180 kPa SL / 220 kPa XL)	Table 6	1
		Added section II.A.6	II.A.6	1
		Added all details of the test procedure, in its entirety. Updated Pressure/Load/Speed chart to reflect elimination of "Part A" Clarified Load Reduction for V-rated / XL tires Update to require minimum of three speed steps Added requirements for an exception to the test criteria	II.B	1
		Added all details of the test procedure, in its entirety.	II.C	1
		Added all details of the test procedure, in its entirety. Corrected the Vehicle Type in the chart (deleted "body on frame" for Pass Car, SUV, CUV	II.D II.D.8.8	1
		Added all details of the test procedure, in its entirety.	II.E	1
		Added "…or as approved by the GAT Team, Program Vehicle Dynamics…"	II.F.2.4.1	1
		Added the DMA test procedure, in its entirety	II.G	1
		Added the Tire Footprint Measurement (ForceScan) test procedure, in its entirety	II.H	1
		Added "…or as approved by the GAT Team, Program Vehicle Dynamics…"	II.J.1.3.1	1
		Added all details of the test procedure, in its entirety.	II.L	1
		Revised bullet point to "…front corner load" (was "front axle load")	Table 4 PRAT Spec	1
		Revised test load and inflation pressure for LR-E to 50 psi (was 80 psi) Added the exception for Steel Carcass tires	II.N.3.1	1
		Corrected the Table references to 5A, 5B, and 5C (was 2A, 2B, 2C)	II.N.5.5.7.9	1
		Corrected the Table references to 5A, 5B, and 5C (was 2A, 2B, 2C)	II.N.5.5.7.10	1
		Added Exception for balance spec of light weight tires	II.N.5.5.8.3	1
		Added statement for reduced tread depth spare tires	II.N.5.10.2	1
		Revised to use the Ford Model as the primary method of measurement	0	1
		Added all details of the test procedure, in its entirety.	II.R	1
		Added	II.R.3.6	1
		Added all details of the test procedure, in its entirety.	II.R.4	1
		Added	II.R.8.4	1
		Revised to be for reference only	II.R.9	1
		Added all details of the test procedure, in its entirety.	II.S	1
		Revised to be for reference only	II.T	1
		Added	II.V.3.6.2	1
		Revised section	II.W	1

*** Note: Implementation: 1—Immediate; 2---Time Stamped (Date Specified in Tire FS); 3---TA Driven (per Target Agreement)**

AA. Appendix B - Tire Construction Detail Sheet (CDS)

Tire Construction Detail Sheet (CDS) - Version 8.1
Design Verification Sign-Off

Identification Detail

Supplier	0	**UTQG Ratings (Wear / Traction / Temp)**	
Size / Description	0	**Load Rating / Speed Index**	#VALUE!
Brand	0	**Rim Size(s)**	0
Trade Name	0	**Tire Label Values (RRC / Noise / Wet)**	
Tire Type	0	**ECE Certification Number**	
Sidewall / Additional Features	0	**Other Sidewall Markings**	
Part Number	0	**Production Plant Code (2 or 3 Characters)**	
Construction Number	0	**Plant Name, State, Country**	
Tire Functional Specification	0	**Plant Code(s) for additional plant(s)**	
Tire & Wheel SDS	0	**Tire ID Number (TIN / DOT)**	0

Cured Measurements

Component	**Compound Detail**	**Cross-Section Detail**							
		# of Plies and / or Configuration	**Cured Gauge (mm)**			**Material**	**Strands / Denier**	**Cured EPI Nominal**	**Angles (deg) Nominal**
			Measurement reference point	**Location from reference point (mm)**	**Measurement (mm) (reference / anti-reference)**				
Tread		total tread gauge					N/A	N/A	N/A
Sidewall		gauge at equator					N/A	N/A	N/A
Inner Liner		gauge at shoulder					N/A	N/A	N/A
Wedge		maximum gauge					N/A	N/A	N/A
Apex		gauge above bead					N/A	N/A	N/A
Bead		gauge outside bundle					N/A	N/A	N/A
Body Ply			N/A	N/A	N/A				
Belt Ply			N/A	N/A	N/A				
Cap Ply			N/A	N/A	N/A				

Stone Pecking Info	Rolling Tread Width (mm)	
	Number of Grooves	
	Standoff Distance - tread edge to center of 1st groove (mm)	

							Measurement Reference Point	**Location from Reference Point (mm)**
Belt Width (mm)	1st		2nd		3rd			
Body Ply Ending (mm)	1st		2nd		3rd			
Nominal Cap Ply Width (mm)								
Wedge Width (mm)								
Apex Height (mm)								
Non-Skid Depth (mm)	Center		Shoulder					

Footprint Dimensions (mm)	Length	0	Width	0	Inflation (kpa)	0	Load (kg)	0	PCI'd? (Y/N)	0
Tire OD and SW (mm)	OD	0	SW	0	SW w/ Rim Flange Protector			Inflation (kpa)	#N/A	

Performance / Dimensional Details

SSR (Condition 1)	0	(N/mm)	Inflation (kpa)	#N/A	Load (kg)	0	Rolling Resistance	
SSR (Condition 2)	0	(N/mm)	Inflation (kpa)	#N/A	Load (kg)	0	RRC	#DIV/0!
SLR (Condition 1)	0	(mm)	Inflation (kpa)	#N/A	Load (kg)	0	SRC (lb)	#DIV/0!
SLR (Condition 2)	0	(mm)	Inflation (kpa)	#N/A	Load (kg)	0	SRC (N)	#DIV/0!
DLR	0	(mm)	Inflation (kpa)	#N/A	Load (kg)	0	SRC Load (lb)	0
Lateral Mu			Inflation (kpa)		Load (kN)		Aligned ISO RRC (kg/tonne)	
Longitudinal Mu			Inflation (kpa)		Load (kN)		Aligned RRC Class	
Cornering Stiffness		(N/deg)	Inflation (kpa)		Load (kN)		Tire Class (ECE R-117-02)	
PRAT (Curb + 2 Pass)	0.00	(N-m)	Inflation (kpa)	20	Load (N)	0		
Tire Average Weight	#DIV/0!	kg	#DIV/0!	lb				
Revs per Kilometer (RPK)	#DIV/0!	Revs per Mile (RPM)	#DIV/0!					

**Aligned RRC information must be per the certification and done on supplier equipment.*
RRC value must be aligned and correlated to virtual lab

Based on the above information provided by the tire supplier, the tire is approved for use as released

Vehicle Program(s)			
Tire D&R Engineer	0	**Date:**	
Tire D&R Supervisor	0	**Date:**	
Core Tire Engineering	0	**Date:**	
Tire Supplier	0	**Date:**	

BB. Appendix C - Tire Drawing Template

Tire Drawing Template

NO.			INVERTED DELTA			
1	SUPPLIER					
2	TIRE SIZE / DESCRIPTION					
3	TIRE BRAND					
4	TRADE NAME					
5	TIRE TYPE					
6	SIDEWALL / ADDITIONAL FEATURES					
7	PART NUMBER		YES			
8	CONSTRUCTION NUMBER					
9	TIRE FUNCTIONAL SPECIFICATION					
10	TIRE & WHEEL SDS					
11	UTQG RATINGS (WEAR / TRACTION / TEMP)					
12	LOAD RATING / SPEED INDEX					
13	RIM SIZE(S)					
14	TIRE LABEL VALUES (RRC / NOISE / WET)					
15	ECE CERTIFICATION NUMBER		NO			
16	OTHER SIDEWALL MARKINGS		NO			
17	PRODUCTION PLANT CODE (2 OR 3 CHARACTERS)					
18	PLANT NAME, STATE, COUNTRY					
19	PLANT CODE(S) FOR ADDITIONAL PLANT(S)					
20	TIRE ID NUMBER (TIN / DOT)		YES			
PERFORMANCE/ DIMENSIONAL DETAIL						
21	TIRE OD (mm)					
22	TIRE SECTION WIDTH (mm)					
23	LATERAL Mu		INFLATION (kPa)		LOAD (kN)	
24	LONGITUDINAL Mu		INFLATION (kPa)		LOAD (kN)	
25	PRAT @ CURB + 2 PASS LOAD (N-m)		INFLATION (kPa)		LOAD (N)	
26	TIRE AVERAGE WEIGHT (kg)					
27	ROLLING RESISTANCE SRC (lb)		RRC		SRC LOAD (lb)	
28	ROLLING RESISTANCE SRC (N)					
29	REVS PER KILOMETER (RPK)					

CDS Data Import
(using CDS v8.1)

Note: Obtain file from Ford to populate new data and submit. The above data is only an example. Tire Suppliers may not submit alternative drawings to satisfy this request (e.g. 2-D drawings); only content consistent with the Tire Drawing Template is permitted.

Tire Suppliers will also be required to supply an image of both the inboard and outboard tire sidewalls. Both sidewall images must be in one file in a *.CGM, *.DWG or *.DXF format.

The preferred format is *.CGM

CC. Appendix D - Tire Submission Summary Sheet (TSS)

Appendix "D"

	Vehicle Program:	0	Model Year:	0	Part #:	0	Wheel Size/Width/Profile:	0	0	0
Submission Tire Information:	Placard Front/Rear kPa:	0	0	DOT:	0	Brand:	0	Model:	0
	Size, LR, SI, LT/LR:	0	0	0	0	0	0	0	Tread: 0
PRAT Loads:	Fz1 [N]:	0	Fz2 [N]:	0	Fz3 [N]:	0	Fz4 [N]:	0	

Item	Rd Feedback	Value	Load [kN]	Pres. [kPa]	Submission 1						Submission 2					
Submission Number					1						2					
Test Vehicle Tag/VIN and Test Track																
Test Vehicle Config.																
Submission Report Date					Control Tire						Control Tire					
Full Tire Construction																
Vehicle Evaluation Information:	Rd Feedback	TA Values			Control Tire											
Ride (Subjective) (00.00-R-202)																
Noise (Subjective) (00.00-R-202)																
Steering (On-Center) (00.00-R-202)																
Steering (Cornering) (00.00-R-202)																
Wet Handling (00.00-R-202/R205)																
Dry Handling, Sub-Limit, Curb - R202 (Subjective, VER)																
Dry Handling, Sub-Limit, Loaded - R202 (Subjective, VER)																
Snow (Subjective) (00.00-R-202, R-488)																
Dry Handling, Limit, Curb - R202, R205 (Subjective, VER)																
Dry Handling, Limit, Loaded - R202, R205 (Subjective, VER)																
Airborne Noise (630-3150Hz) Delta (Objective) [dBa]																
Structural Noise (80-300 Hz) Delta (Objective) [dBa]																
Dry Brake Delta (00.00-R-448) (m)																
Wet Brake Delta (00.00-R-487) (m)																
Stability																
Lab Evaluation Information: Ford Lab Data I Suppli	Rd Feedback	Value	Load [kN]	Pres. [kPa]	Control Tire											
RRC [SAE J2452] (Tire FS Section III.R)																
Weight [3 Tires For SAE J2452] (for reference only) (kg)																
PRAT @ X.XX N [Nm] (CETP 04.04-L-414, Curb + 2 Pass)																
Long. Mu @ XX kN (CETP 04.04-L-419)																
Lat. Mu @ XX kN (CETP 04.04-L-415)																
Vertical Stiffness [N/mm] (CETP 04.04-L-415)																
Lateral Compliance [mm/kN] (CETP 04.04-L-415)																
Cornering Stiffness @ F Curb + 2 Pass [N/deg] (CETP 04.04-L-415)																
Cornering Stiffness @ R Curb + 2 Pass [N/deg] (CETP 04.04-L-415)																
Cornering Stiffness @ RGAWR Curb + 2 Pass [N/deg] (04.04-L-415)																
Airborne Noise (Primary) 800, 1000Hz, 1/3 Octave [dB A] CETP E-402																
Airborne Noise (Secondary) 1600, 2000, 2500 Hz 1/3 Octave [dB A] E-402																
Structural Noise (Rumble) (40-150Hz) (N) (CETP 04.04-L-10915)																
Structural Noise (Cavity) Cavity Mode +/-20Hz [N] CETP 04.04-L-10915																
Impact Noise (Impulse) (N*s) (CETP 04.04-L-10914)																
STFS @ XX kPa (CETP 04.04-L-417) (N)																
STFS - Model 2.0 (N)																
LTFS @ XX kPa (CETP 04.04-L-418) (N)																
LTFS - Model 2.0 (N)																
Snow Model Score																
Cool/Wet Storage Modulus (E' @ 0° CETP(04.04-E-4335) (MPa)																
Glass Transition Temperature (Tg) (CETP(04.04-E-4335)																
Cut/Chip Model Rating																
Wear Model Rating (PAW Model) (@ Driven Axle Curb load) [km]																
Wear Model Rating (PAW Model 2.1) (@ Driven Axle Curb load) [km]																
Footprint Length (mm)																
Footprint Width (mm)																
Footprint Roundness																
Footprint Contact Surface Ratio																
RRC - ISO 28580 - per WLTP regulation																
WLTP Class																
Weight [kg] (Tire FS Section III.Q)																
OD [mm] @ XX [kPa] (Tire FS Section III.Q)																
Section Width [mm] @ XX [kPa] (Tire FS Section III.Q)																
SSR [kg/mm] @ XXXX kg, XXX kPa (Tire FS Section III.Q)																
SLR [mm] @ XXX kg, XXX kPa (Tire FS Section III.Q)																
DLR [mm] @ XXX kg, XXX kPa (Tire FS Section III.Q)																
RPK (Tire FS Section III.Q)																
Electrical Resistance [Ohms] (Tire FS Section III.K)		1.E+08														
High Speed A @ target KPH (Tire FS Section III.A1)		MR														
High Speed B @ target KPH (Tire FS Section III.A1)		MR														
Lab Durability (Tire FS Section III.L)		MR														
Rim Roll Off (Tire FS Section III.B)		MR														
Instant Air Loss (Tire FS Section III.C)		MR														
Tire Aging Durability (Tire FS Section III.AI)		MR														
Structural Accelerated Durability (Tire FS Section III.AJ)		MR														
Wear [% vs. control] (Tire FS Section III.V)																
Full Tire Construction	Rd Feedback				Control Tire	0	0	0	0	0	Control Tire	0	0	0	0	0

Notes: **Obtain newest file from Ford to populate new data and submit.** **The image above is only an example. Add all pertinent data needed to explain differences in constructions and performances. The Vehicle Evaluation Information section of the Tire Submission Summary Sheet (TSS) is color coded to assess delta versus target.**

TSS Responsibilities:

1. Ford Automatic:
   a. Lab Evaluation Information (blue section)
   b. Tire System and Quality Measurements (purple section)
2. Ford Manual:
   a. Test Request Tab
3. Supplier (Master and Predictions tabs):
   a. Subjective Vehicle Evaluation Information (brown section)
   b. Lab Evaluation Information (green section)
   c. Tire Design Information ("tannish orange" section)

d. On the Test Request tab there are tannish-orange cells under each submission to fill out with the expected arrival date for each shipment of tires to the EVB.

DD. Appendix E - Tire Attestation Letter for Greenhouse Gas Emissions Regulation template

SUPPLIER HEADER

DATE

MFG Rolling Resistance Coefficient (RRc) – Signed Statement in accordance with 40 C.F.R. Part 1037.520(c) of the EPA/NHTSA Final Rule dated 9/15/2011 (Greenhouse Gas Emissions Standards and Fuel Efficiency Standards for Medium- and Heavy-Duty Engines and Vehicles)

______SUPPLIER NAME________________________confirms that the tire rolling resistance levels ("TRRL") provided in the table below are based on testing conducted in accordance with EPA 40 C.F.R. Part 1037.520(c).

This letter satisfies the requirement of the regulation for the tire manufacturer to supply the vehicle manufacturer with a signed statement verifying that the testing performed to achieve the results were conducted according to EPA 40 C.F.R. Part 1037.520(c).

Supplier	Supplier Product Code	Tire Size	Load Rating	Vehicle line	Ford part number	RRC	SLR (mm)